

STRATEGIC OPINION-WRITING ON APPELLATE COURTS

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ABSTRACT. Ruling on thousands of cases each year, U.S. federal courts of appeals make some of the most impactful decisions in modern society. I study how polarized partisan environments affect consensus-building using quasi-random three-judge panels on these courts from 1970–2013, documenting a novel pattern in dissenting opinions. Compared to party-unanimous panels, party-mixed panels cause all judges to dissent more often, and at equal rates: median and majority-party judges are just as driven to dissent as their more politically extreme colleagues. This result is incompatible with classical models of judicial politics and is unique to partisanship. To explain my results, I introduce a theoretical framework where judges’ favored coalitions are more homogeneous along both partisan and non-partisan dimensions. Using judge metadata, I find suggestive evidence for the model’s result that polarization increases dissents by judges of panel-minority law school or gender. With state-of-the-art machine learning tools from natural language processing, I generalize beyond dissents, showing that those same features drive differences in opinion text even when rulings are unanimous. My findings show that partisanship has a powerful and complex effect on agreement and illustrate the need for new tools to capture its consequences in this opaque yet high-stakes environment.

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1. INTRODUCTION

Partisan behavior by judges has been spotlighted in the U.S. following recent high-profile, politically impactful decisions by the Supreme Court. Early in his tenure, current Chief Justice John Roberts famously claimed that judges are like baseball umpires who “[only] call balls and strikes,” i.e., clarify rules without being political. However, polling from the Supreme Court’s most recent term in summer 2024 finds that a majority of Americans believe those justices decide cases based on their personal ideology rather than impartial interpretations of the law ([Associated Press, 2024](#); [Copeland, 2024](#)).

Less remarked upon, but far more influential in terms of volume, is the next level of the U.S. judicial system: thirteen federal courts of appeals which serve as the final word on about 40,000 cases each year. A decades-old empirical literature starting with [Sunstein et al. \(2006\)](#) has studied individual judge behavior on these courts using quasi-random assignment of judges to three-judge panels, and of those panels to cases. The interactions *between* judges on panels have been less carefully analyzed, with many prior works treating within-panel decision-making as a black box and focusing directly on the consequences of panel characteristics for outcomes. However, appellate panels provide a rich environment in which to study classical questions from political economy about how partisanship affects consensus-building and coalition formation. In this work, I combine insights from the theory of judicial politics with new tools from computer science to show how partisanship increases overall disagreement among judges, but does not do so along party lines.

To capture the high-stakes consequences of how judges interact on a panel, I focus on *dissents*—the decision to cast a vote against the majority’s preferred ruling and the most visible form of disagreement among judges. A key result from prior work is that bipartisan panels—with, e.g., two Democratic judges and one Republican judge—cause a sharp increase in dissenting opinions compared to party-unanimous panels ([Lu and Chen, 2024](#)). I first verify this pattern in the full sample of published appellate court opinions from 1970 to 2013, finding a causal increase of about 25% more dissents on bipartisan panels compared to party-unanimous panels. This effect is quantitatively meaningful, accounting for about 1,000 more cases with dissents each year. I also contribute novel evidence that the distance

between the panel median and the most extreme judge—not a judge’s individual distance to the median—drives this increase.

The importance of the extremal judge suggests that predictions from classical median voter models might explain the increase in dissents on bipartisan panels, i.e., a minority partisan or more polarized extremal judge will dissent more often and raise the total dissent rate. However, I provide robust evidence of a previously unexplored pattern in dissenter ideology: minority partisans are no more likely to dissent than majority partisans. More strikingly, the median judge dissents just as often as the extremal judges; on an average of over 1,300 cases each year, the leftmost and rightmost judges form a majority coalition which excludes the center judge. Thus, judge partisanship causes the *existence* of more dissents, but it does not predict the *identity* of the dissenter. Even taking the bounds of my estimates which are most favorable to partisan differences in dissent rates, 75% of the additional disagreements caused by partisanship do not split the judges along partisan lines.

Combining theoretical and simulated results, I show that these two empirical facts are incompatible with both a model of sincere dissents with noise and a median-voter-like approach to strategic dissents. I also examine non-partisan judge characteristics—law school training, gender, and age—and show that partisanship is unique among those characteristics in its effect on dissents. The presence of an outlier judge along those non-partisan dimensions is associated with an increase in total dissents; however, that increase is smaller than the one caused by bipartisan panels, and concentrated on the relevant outlier judge. For example, switching a panel where all judges share a law school to one where only two judges share a law school mainly increases dissents by the minority law school judge. Unlike the effect of party, these new results are explained by classical approaches to judicial politics.

To partially address the puzzle of partisan dissents, I describe a framework where judges can differ on both partisan and non-partisan dimensions, and gain more utility from coalitions with more similar judges. The two most similar judges form a potential minimal majority coalition, while the least similar judge will dissent whenever they are so dissimilar that their share of the coalition surplus is less than their outside option. This model provides a link between partisanship and non-partisan characteristics and comes close to explaining the puzzle of partisan dissents. As a simple example, consider a setting where judges’ partisan

and non-partisan characteristics are orthogonal, so that, e.g., a Republican and a Democrat are equally likely to have gone to Law School X. The increase in dissents on bipartisan panels is then straightforward—party-unanimous panels contain only judges who may or may not share a law school, while bipartisan panels are mechanically more divided along party lines and equally likely to be divided along law school lines. Thus, since reduced similarity increases the chances of dissent, bipartisan panels have a higher overall dissent rate. However, focusing exclusively on bipartisan panels, the full vector of judge characteristics—not just partisanship—determines which judge is least similar and therefore at risk of dissenting. Consider a panel with one Democrat from Law School X, one Democrat from Law School Y, and one Republican from Law School X. Both the partisan and the law school minority judges are tempted to dissent; which one actually does depends on whether partisanship or law school background contributes more to similarity.

The interactions between partisan and non-partisan characteristics are challenging to measure because of reduced sample size. However, I provide suggestive evidence that partisanship activates disagreements along non-partisan dimensions. I compare judges who are outliers in some characteristic to those in the panel majority for that characteristic, and test whether bipartisan panels cause differential increases in those judges’ dissent rates. I find a quantitatively meaningful difference in the point-estimated dissent rates of judges in the law school or gender minority on a panel, though statistical power is reduced due to the rarity of shared-law-school and majority-women panels. I then compare these effects to the total increase in dissents caused by partisanship, scaling them using the share of bipartisan panels where they apply. For example, about one-quarter of bipartisan panels have two unique law schools, and on those panels law-school-minority judges dissent 10.9 percentage points more often. If partisanship’s only effect was to provoke disagreements along law school lines, then the increase in dissents on bipartisan panels would be about 3%, one-eighth of the observed increase of 23.5%. The combined effects of law school and gender could account for about one-third of observed disagreements on bipartisan panels.

To provide further evidence of latent disagreements along non-partisan dimensions, I use state-of-the-art natural language processing (NLP) tools from the machine learning community to study opinions even in the absence of observed dissents. While the ruling on a case

settles that particular issue, the content of the majority opinion sets legal precedent and decides the consequences of that case for future appellate decisions. An outlier judge who knows majority judges are likely to write a different opinion is thus likely to disagree with the majority. That disagreement may be insufficient to prompt a dissent, or the outlier judge may even strategically join the majority with the hope of swaying opinion content; either of these factors would prevent standard outcome-based measures from detecting disagreement. However, by directly analyzing similarity in opinion text, I can quantify which judge characteristics drive the largest differences in opinion content, and are thus most likely to produce disagreements.

Specifically, I fine-tune a large transformer model (Chalkidis et al., 2020) to predict legal topics in each opinion, resulting in learned semantic text embeddings: high-dimensional vector representations of opinion text which capture its full semantic context. My first similarity measure follows a recent approach in applied machine learning and uses predicted probabilities as an estimate of the share of opinion text devoted to each topic. I show that opinion text has only minor differences across party; in particular, the topic distribution on bipartisan panels is essentially the same regardless of which party holds the majority. However, the topic distribution is markedly different by gender, complementing recent results which mapped measures of gender bias in opinion text to more conservative rulings on gender-related cases (Ash et al., 2024).

My second measure of opinion similarity directly leverages the semantic embeddings. Using an approach established in machine learning but largely new to economics, I directly compute the similarities of embedding vectors and look at the neighborhood of each opinion to assess co-location of opinions along characteristics of interest. This approach allows me to directly study the effect of law school, which—since it takes over 150 possible values—could not be easily interpreted as an explanatory variable in previous specifications. Benchmarking against the tendency of opinions to cluster by primary legal topic, I find that neighboring opinions are more likely to be written by judges of the same gender or law school than random chance would predict, while they are if anything slightly *less* likely to be written by judges of the same party than under random chance. Together, these results provide clear

evidence that even on panels where a dissent does not occur, non-partisan variables capture a larger and more influential share of legal disagreement than partisan ones.

This work is part of, and contributes to, a growing modern literature studying the effect of political polarization on judicial decision-making. Driven by increased perceptions of court politicization in the modern era, as well as the greater amount of available administrative data on courts and judges, recent work (discussed in detail in Section 2) has shown that the effects of partisanship may be subtle, but are detectable and quantitatively impactful. Many of these works have focused on the outcome of an appellate court hearing; for example, which litigants are favored, or whether a lower court’s decision is overturned. I augment this analysis by introducing more fine-grained measures of opinion content via additional judge characteristics and the novel use of NLP to study opinion text. Dissents and legal topics are not as immediately related to appellate outcomes, but they play a vital role in shaping the reach and longevity of rulings. Appellate courts are constantly in dialogue with legal precedent, the other levels of the court system, and potential future cases; understanding when and why rulings feature consensus or disagreement is key to understanding whether those rulings are likely to be influential. I also provide a demonstration of the power of modern machine learning tools in political economy. Applications of transformer models and text embeddings are novel but likely to grow, with large language models (LLMs) making embedding-based measures and their reduced-form predictions or generative output more easily accessible. By studying opinion text even when observable outcomes like dissents or partisan splits are not available, this work shows the power of these tools for extending standard approaches to richer domains.

The remainder of the paper proceeds as follows: I begin by describing related literature on judicial partisanship in economics and law, as well as applications of NLP to legal domains both within and beyond economics, in Section 2. Sections 3 and 4 describe key institutional details, features of the appellate court data, and my empirical approach. Section 5 applies classical models of judicial politics to my setting and introduces my coalition formation framework, providing key testable implications of both. I then present evidence for my two central empirical facts—partisanship causes dissents, and politically-extreme judges do not

dissent more often—in Section 6. In Section 7, I show that this pattern is unique to partisanship and provide suggestive evidence that disagreements along non-partisan dimensions are activated on bipartisan panels, increasing dissents by outlier judges. Section 8 supplements that analysis with NLP measures that apply even to unanimous decisions, showing that non-partisan characteristics have more influence on opinion content than partisan ones. Finally, Section 9 concludes with a discussion of possibilities for future research, both on the methodological side and on real-world questions around judicial politics. Appendices A–F contain additional visualizations and robustness checks for the data and results in this work; each begins with a short paragraph describing its contents in more detail.

2. RELATED LITERATURE

My work contributes to the modern literature using quasi-random assignment to study the effect of appellate judge and panel characteristics on outcomes. This literature includes work using opinions hand-coded as liberal or conservative (Sunstein et al., 2006; Kastellec, 2011), number of Democratic-appointed judges (Cohen, 2024; Cohen and Dehejia, 2024), and more modern machine learning tools (Lu and Chen, 2024) to measure panel and outcome partisanship. In general, these works find that judge or panel partisanship causes different outcomes, such as partisan-aligned opinions and reversals. Rather than comparing Democratic- and Republican-dominated environments, this work focuses instead on the difference between party-unanimous and bipartisan ones.

Closely related to this work’s analysis of partisanship are Lu and Chen (2024) and Chen et al. (2023). The former uses machine learning tools similar to those in Section 8 to construct an opinion polarization measure based on prediction accuracy. I instead study legal topics and use both predictions and the full semantic embeddings of opinions. As an auxiliary result, Lu and Chen (2024) documents an increase of about one percentage point in the dissent rate of partisan-minority judges compared to partisan-majority judges. This effect does not appear in my sample, likely due to small differences in data cleaning, and is not sufficient to account for the large increase in total dissents on bipartisan panels.¹ I also use judge ideology scores

¹I document the full data preparation and provide code to produce all results and tables in this work as part of the replication archive available on my [website](#).

to show that the median judge contributes approximately one-third of the dissenting votes in my sample, which would remain a puzzle for standard models even if minority partisans were more likely to dissent.

The latter work, [Chen et al. \(2023\)](#), studies a judge’s individual distance from the panel and circuit median. The authors document a U-shaped pattern of dissents relative to the panel median and an M-shaped pattern relative to the circuit median, which they explain with a model where judges trade off their personal disagreement with an opinion against social disapproval for dissenting. I complement this across-panel explanation with richer within-panel analysis. In Section 6 I show that the most extreme judge on a panel is responsible for increased dissents distributed evenly across judges. Using the model of Section 5.3 and the data in Section 7.3, I provide suggestive evidence of how partisanship and other judge characteristics may interact to drive dissents.

Turning to non-partisan judge characteristics, prior work has studied the causal effect of race ([Kastellec, 2013](#)) and gender ([Ash et al., 2024](#)) on rulings in the subset of cases where those features are particularly salient. I contribute to this literature by providing a general theoretical framework for similarity between judges, encompassing partisanship, demographics, and other characteristics such as law school training. Empirically, the success of my NLP approach to topic modeling introduces the possibility of extending these works by testing whether the effects of judge identity generalizes to cases where that identity is less salient.

The theoretical literature on appellate decisions has largely elided any separation between partisan and non-partisan judge characteristics by focusing on a unidimensional policy space ([Carrubba et al., 2012](#); [Parameswaran et al., 2021](#)). Predictions about the degree to which the median judge controls opinion content vary, but a general result is that less polarized judges are more likely to vote strategically and thus less likely to dissent. In my sample, the substantial share of dissents by the median judge suggests that a richer policy space or different preferences over coalitions may be important components of strategic behavior. The former approach is investigated empirically in [Lauderdale and Clark \(2014\)](#), which uses word-level statistical data from the U.S. Supreme Court to show that the identity of the median justice varies depending on which topics are most important to a given case. The latter approach is the focus of my framework in Section 5.3, which directly considers judge

preferences over possible majority coalitions and describes both the settings in which dissents are more frequent and the relevant traits of dissenting judges.

Looking beyond this work’s focus on partisanship and dissenting opinions, text data and NLP have seen growing applications in economics (Ash and Hansen, 2023), political science (Rodriguez and Spirling, 2024), and law (Katz et al., 2023). My work uses NLP tools as a supplement to extensive metadata on cases, judges, and panels. Given the rarity of observable features like dissents, as well as the limited number of hand-coded case topics and outcomes, novel applications of these tools are vital for generalizing existing conclusions about judge behavior and are often the only way to measure properties like opinion similarity at scale. Adapting existing machine learning approaches for the judicial setting, developing new NLP tools to tackle problems of real-world interest, and ensuring that those tools are compatible with existing frameworks for causal inference and estimation, are important technical challenges which I hope to address in future work.

3. SETTING

3.1. Structure of US Appellate Courts. There are thirteen circuit courts of appeals. Eleven cover geographically contiguous sets of states, jointly containing all fifty; they hear appeals from one or more district courts (the next level down in the federal court system) located in their area. There is also a circuit court covering Washington, D.C. only and a Federal Circuit which hears cases not tied to a particular district.² The total modern caseload across all circuit courts combined is about 40,000 cases per year; larger circuits, like the Ninth, hear more cases than small ones, like the Seventh. About 7,500 of these cases are filed for further appeal with the Supreme Court, but less than 200 of those are actually reviewed. Thus, for about 99.5% of cases, a circuit court provides the final word.

Circuit court judges are appointed by the president, with approval historically requiring a two-thirds majority vote in the Senate.³ Seats on the court are assigned to particular districts; each judge will have a “duty station” located in a given district, and the court will

²See Figure 4 in Appendix A for a visualization of circuits and districts.

³The threshold is now a simple majority following procedural changes during President Obama’s second term. This work’s data ends before that change was made.

convene there for part of the year to hear appeals. The senators for the state containing that district, called “home-state senators,” have significant influence on the appointment of the corresponding appellate judge (Giles et al., 2001). This influence is reflected in some commonly-used measures of judicial ideology, which take into account the political leanings of home-state senators (see Appendix A.2 for details).

3.2. The Opinion-Writing Process. Appellate courts decide whether to uphold or reverse a lower court’s ruling; they do not make independent determinations of, e.g., guilt or innocence. Their cases cover a broad range of topics—criminal sentencing, constitutional law, antitrust, etc.—and the vast majority are heard by a panel of three judges chosen quasi-randomly (see Section 3.3 for a thorough discussion). A select subset of cases deemed particularly challenging are heard by the entire circuit, which is referred to as an *en banc* case. As is common in work on appellate courts, my sample omits *en banc* cases.

Some cases, which are routine or have procedural issues, are decided without oral arguments after reading the submitted legal briefs. My work, and almost all other work outside law itself, considers cases where appellate judges hear oral arguments and discuss them in order to decide on a *disposition*: whether to uphold or reverse a lower court’s ruling. If a judge votes against the majority disposition, they are *dissenting*, and write a separate non-binding opinion. If a judge agrees with the majority disposition but disagrees with the reasoning, they are *concurring*, and often write a separate non-binding opinion. The senior judge among those who agree on both disposition and reasoning assigns the writing of the *lead opinion* to a member of that group.⁴ This assignment is opaque and may be strategic, driven by personal considerations, or affected by availability and other bureaucratic reasons.

In cases where an opinion sets legal precedent, its text is published; unpublished opinions are more routine and feature a lower share of dissents and concurrences. My sample contains

⁴The term “lead opinion” accounts for the possibility that only one judge remains in the group, i.e., one judge dissents and the remaining two have different reasoning or all three judges agree but with different reasoning. In these cases, the senior judge’s opinion is the lead and other judges who agree with the disposition are considered concurring. Other than these two (rare) special cases, the colloquial label of “majority opinion” is equivalent to that of lead opinion.

only published opinions, which are generally believed to be more affected by political and strategic behavior (Cohen, 2024).

3.3. Quasi-Random Assignment on Appellate Courts. Quasi-random assignment of appellate judges to three-judge panels has been treated as an established fact in social science for decades. However, the details of the assignment process are complex. Several law review publications, in particular Levy and Chilton (2015), provide a thorough discussion of the history and current practice of appellate panel formations; I summarize key facts here.

Circuits make schedules in advance for several months’ worth of cases, and often for the entire year. There is no centralized scheduling procedure; each circuit has its own guidelines, procedures, and institutional norms. Assignment of judges to panels, and panels to cases, may be the responsibility of the chief judge, the clerk of court, the circuit executive, or another member of the court staff; it may be done by computer (e.g., shuffling an Excel spreadsheet), by hand (e.g., drawing names from a hat), or both. The Fourth, Fifth, Sixth, Seventh, Ninth, Tenth, and Eleventh Circuits explicitly state in official procedure documents that they randomly assign judges to panels. Several judges, lawyers, and legal scholars have stated that panels are randomly drawn on other circuits as well (Levy and Chilton, 2015).

There are several reasons why panel assignment may not be truly random despite these statements. First, some circuits explicitly note that they attempt to implement what amounts to stratified random assignment. For example, the Seventh Circuit states that panels are adjusted so that judges sit “approximately the same number of times with every other judge of the court” over a two-year period (Levy and Chilton, 2015). Additionally, across all circuits schedules may be adjusted in advance so that, e.g., a judge is not forced to hear cases for two months straight despite their name appearing in all of the first 50 panels drawn. Finally, schedules are also necessarily adjusted after being finalized, e.g., if a judge is ill or if a case is withdrawn. However, the consensus among both legal scholars and practitioners is that these deviations from randomness are orthogonal to my variables of interest: case topics, ideology, and most judge demographics.

Due to the idiosyncrasies of scheduling, testing random assignment by compiling a full calendar of panels and the cases they heard is a monumental task even in modern data. Instead,

two practical tests are used in prior literature. The first is to perform balance checks on observed panels—e.g., in the observed sample of cases from the Ninth Circuit in 1975, do panels of three Republicans hear First Amendment cases about as often as panels of three Democrats (assuming both types of panel are equally common)? The second is to simulate panels and check whether the observed distribution of panel characteristics is an outlier in the simulation—e.g., using the actual judges serving on the Ninth Circuit in 1975, randomly generate the observed number of panels and count the share which have two Democratic judges and one Republican judge; repeat this simulated circuit $\times$ year 1,000 times and compare the distribution of shares to the observed share. I present results from both methods alongside detailed descriptions of the exact procedures used in Appendix B.

In the appellate court data used here, the observed distributions of case topics, judge characteristics, and panel characteristics are extremely close to what is observed in the “truly random” simulated panels. Depending on the method used, some statistically significant deviations from randomness are observed, but the magnitude of those deviations is generally smaller than half a percentage point. This result accords with the recent literature studying political behavior by appellate judges.⁵

4. DATA AND EMPIRICAL STRATEGY

4.1. Opinion Data. My main data is provided by the [Data Science Justice Collaboratory](#); more details are in [Ash et al. \(2024\)](#). Text is originally compiled by Bloomberg Law and contains the universe of published opinions from the 11 geographic circuits, the DC Circuit, and the Federal Circuit from 1890–2013. Each opinion is associated with the following metadata:

- The circuit and year where the case producing the opinion was heard.
- Whether the opinion upheld or reversed the lower court’s ruling.
- The names of the three judges on the panel producing the opinion.

⁵For further results confirming random assignment in similar data, see [Cohen \(2024\)](#), [Ash et al. \(2024\)](#), and [Chen and Yeh \(2020\)](#).

- The votes of the three judges (i.e., whether each judge supported the lead opinion, dissented, or concurred).
- The author(s) of the lead, dissenting and concurring opinions (if any).
- The text(s) of the lead, dissenting and concurring opinions (if any).
- Case and opinion metadata; of particular interest is a hand-coded primary topic from Bloomberg Law in one of eight categories: criminal, civil rights, First Amendment, due process, privacy, labor relations, economic activity and regulation, miscellaneous.

In Table 1, I present summary statistics for opinion text; full distributions of relevant variables are in Appendix A.1. Both dissents and concurrences are relatively rare. For opinions where the lead/majority has a known author and contains some text, 9.9% have a dissent with a known author; 99.4% of those dissents have some text, and 73.4% have at least one single-spaced page (500 words) of text. Again restricting to opinions where the lead/majority has a known author and contains some text, 6.8% have a concurrence with a known author; 74.9% of those concurrences have some text, and 38.7% have at least one single-spaced page of text.

4.2. Judge Data. Judge characteristics are originally from Appeals Court Attribute Data, the Federal Judicial Center, data collection by [Chen and Yeh \(2020\)](#), and [Bonica \(2014\)](#). They include age, gender, region of birth, law school attended, and prior experience. Full distributions of relevant variables are in Appendix A.2.

Critical to my analysis are measures of judge ideology, which is challenging to quantify because appellate judges are not elected and vote in panels too small for structural estimation methods like DW-Nominate ([Poole and Rosenthal, 1985](#)) to apply. Still, I can rely on three existing measures. The first is self-reported party, which is almost always the same as party of appointing president, but may have slight variation over time. The second is Judicial Common Space (JCS) scores from [Epstein et al. \(2007\)](#), which are based on the ideology of senators involved in judicial appointment. JCS scores are the gold standard measure of judicial ideology and are the most commonly used measure across economics, political science, and law. The third measure is Database on Ideology, Money in Politics, and Elections—Campaign Finance (DIME-CF) scores, from [Bonica and Sen \(2017\)](#), which

measure judge ideology using campaign contributions. In many cases JCS and DIME-CF scores are highly correlated, but the authors of the latter claim it captures preferences over a broader set of issues. More details of these ideology measures are in Appendix A.3.

While party affiliation is available for over 99% of judges, ideology scores are generally only available for judges sitting on panels from 1970–2013. This time period also coincides with a period of greater stability in judges’ partisan stances following the passage of major civil rights legislation in the 1960s; as such, I focus primarily on the shorter time period and supplement my analysis with the full range of panels when possible.

4.3. Learned Semantic Embeddings. To quantify properties of opinion text, I follow the state-of-the-art approach in natural language processing (NLP) and use machine learning to generate *embeddings*: high-dimensional vectors which capture the full semantic content of the text and are then used for downstream prediction, classification, and clustering tasks. These embeddings can be generated off-the-shelf from existing large transformer models such as BERT (Devlin et al., 2019), or from large language models (LLMs) such as GPT-4o from OpenAI and LLaMA from Meta. However, their performance for particular tasks can be improved by fine-tuning those models on a supervised learning task. In Appendix D, I describe a typical fine-tuning pipeline as deployed in this work, which I use to generate the embeddings used in Section 8. I also present performance metrics for the main model used (tables 15 and 16 in Appendix D), which accurately classifies approximately 75% of opinions into the eight Bloomberg Law topics described in Section 4.1.

Text embeddings can be used directly as numerical representations of qualitative data. For example, I can compute measures of similarity between opinions by computing distances between embeddings, or assess clustering of opinions by plotting them (after reducing their dimension using an appropriate algorithm). They can also be used to generate reduced-form data, such as predictions of author party or topic distributions. The latter is more common in applied work (e.g., Lu and Chen 2024). In Section 8, I work with both reduced-form predictions and full embedding vectors, and discuss my approach in more detail there.

4.4. Empirical Strategy: Quasi-Random Assignment. To identify the causal effect of partisanship on dissents, I leverage quasi-random assignment of judges within each circuit

and year to three-judge panels. If assignment of judges to panels were truly random, then I could use specifications of the form

$$\text{Outcome}_p = \alpha \text{panel type}_p + \gamma_{ct} + \varepsilon_{pct}, \quad (1)$$

where each observation is a panel p on circuit c in year t . The circuit $\times$ year fixed effects γ_{ct} ensure that the panel type on the left-hand side (e.g., whether the panel is bipartisan or party-unanimous) is randomly assigned and is not influenced by omitted variables (e.g., the case topic) or reverse causality from the outcome. They also account for the varying composition of each circuit from year to year, which affects the probability of each panel type, e.g., there may be a greater share of Democrats on the First Circuit in 1975 than on the Seventh Circuit in 2000, and thus the chances of a panel with 3 Democrats is higher on the former than the latter. After including fixed effects, the estimated coefficient $\hat{\alpha}$ represents the causal effect on outcomes of a change in panel characteristics.

As discussed in Section 3.3, assignment is not strictly random due to institutional features, but is quasi-random with respect to characteristics of interest in my sample. Still, I generally include a control for legal topic to increase the precision of my estimates and sometimes include judge fixed effects, controls for panel characteristics (e.g., number of unique law schools represented), or controls for author characteristics.

5. MODELING DISSENTS ON BIPARTISAN PANELS

Before presenting my empirical results, I describe results from a model of sincere dissents with noise and show that it cannot plausibly explain the equal dissent rates of median and extremal judges. In Appendix C.2, I enrich this model by allowing judges to make strategic choices considering both case dispositions and opinion locations in policy space, and show that the structure of the cost of dissent determines the effect of increased distance to the mean on dissent rates by different judges (see Proposition 2 in Appendix C.2 and the discussion at the end of that appendix). However, even in this richer model, the inability to fully describe empirically observed patterns in dissents remains.

As a partial explanation of those equal dissent rates, I propose a framework where judges are multidimensional and have preferences over coalitions rather than opinions. This approach

comes closer to explaining the patterns in partisan dissents, but my ability to test its predictions about the effect of non-partisan judge characteristics is limited by sample size issues (see Section 7). However, it provides both a direction for theoretical progress and important guidance as to what data may be most useful for reconciling the puzzle of partisan dissents.

5.1. Sincere Dissents with Noise. I begin by presenting a simple model in the style of Carrubba et al. (2012) where judges vote sincerely on whether to uphold or reverse a lower court’s decision. Consider three judges $i \in \{1, 2, 3\}$, who have cutoffs $x_i \in [0, 1]$ below which they would like to uphold a ruling and above which they prefer to overturn it. Without loss of generality, let $x_1 < x_2 < x_3$, and let $x_2 - x_1 < x_3 - x_2$, so that judge 2 is the median and judge 1 is closer to the median than judge 3. I will refer to judge 3 as “extremal” and judge 2 as “second-extremal.”

A case $c \in [0, 1]$ is a value drawn from some distribution $F(c)$; for simplicity, let $F(c) \sim U[0, 1]$. Then, under sincere dispositional voting, judge 1 dissents if $c \in (x_1, x_2)$ and judge 3 dissents if $c \in (x_2, x_3)$; judge 2 never dissents. By uniformity of $F(c)$, judge 3 dissents more than judge 2, and increasing either judge’s distance from the median by one unit increases their probability of dissent by one unit.

Adding case-specific noise is analytically complex, even when the noise is mean-zero noise with a uniform or normal distribution. However, if noise is sufficiently small (e.g., i.i.d. uniform with sufficiently small variance), I can show that increasing the extremal judge’s distance has more of an effect on total dissents than increasing the second-extremal judge’s distance (Proposition 1 in Appendix C.1). Intuitively, noise may create any of the six possible permutations in judge order, but with sufficiently small noise I can focus only on three: the original ordering, the order $x_2 < x_1 < x_3$ with judge 1 as the new median, and the order $x_1 < x_3 < x_2$ with judge 3 as the new median. In the first permutation, increasing judge 1 or judge 3’s distance from judge 2 has the same positive effect on total dissents. However, in the second permutation, only increasing judge 3’s distance affects total dissents; in the third permutation, only increasing judge 1’s distance affects total dissents. The second permutation is more likely than the third, so the dampening effect caused by permuted judge orders will be stronger for second-extremal distance than for extremal distance.

5.2. Simulating Sincere Dissents. Noise produces two attractive conclusions which, as I will show in Section 6, are present in the data: the median judge’s dissent rate is greater than zero (see Table 3), and increasing extremal distance has a stronger effect than increasing second-extremal distance (see Table 2). However, simulations show that neither effect is sufficiently strong to account for the observed results, and a better fit on one implies a worse fit on the other. To conduct this simulation, I define a grid of variances σ^2 between 0 and 0.5; I choose this range because the sample variance of JCS ideology scores is 0.158 (Figure 11, Appendix A.3). For each variance, I draw 100,000 random panels from the data and add i.i.d. $N(0, \sigma^2)$ noise to each JCS score, then compute the share of post-noise panels for which the original median, second-extremal, and extremal judges are now the post-noise median judge. Figure 1 shows the results.

The median judge’s share of dissents is upper-bounded by the share of permuted panels where they are no longer the median; however, reaching this upper bound requires that only the median judge dissents on those panels. A more reasonable assumption is that the median judge and the remaining non-permuted judge are equally likely to dissent, and thus the median judge accounts for half of those dissents. In Section 6.2, I show that the upper bound of a 95% confidence interval for the difference in dissent rates across judges produces a median dissent share 9% lower than that of non-median judges (27% compared to 36%).⁶ This rate thus requires at least 27% of panels where the median judge is permuted to another role, and realistically requires about 54% of panels. The left image of Figure 1 shows that the latter is achieved when the noise has variance approximately equal to the sample variance of JCS scores. Thus, even using the most model-compatible difference in dissents, the required variance makes noise just as informative about judge partisanship as well-documented, historically predictive ideology scores.⁷ Since the growth in permuted panels is approximately logarithmic, even increasing noise variance to triple that of JCS

⁶Of course, about half of the confidence interval is composed of median dissent shares *higher* than those of non-median judges, which this model can never explain.

⁷In Section 6.1, I provide strong evidence that JCS scores are predictive of dissents, especially on bipartisan panels, making it even less likely that they are weakly less informative about judge preferences than an arbitrary noise term.

scores only delivers a median dissent share of 30.5%, still 2.7% short of the point estimate in Table 3 (Section 6.2).

The greater effect of increased extremal distance compared to increased second-extremal distance comes from the lower probability of permutations making the original extremal judge into the post-noise median. However, for σ^2 large enough, the difference in the probabilities of becoming the post-noise median judge for the original extremal and second-extremal judges must decrease.⁸ The right image of Figure 1 shows that in my simulations, the difference monotonically decreases in σ^2 for all $\sigma^2 > 0.03$. For $\sigma^2 = 0.158$, the value needed to fit the 95% confidence interval of median dissent shares, the difference is about 15%. In Table 2 (Section 6.1), I find that the lower bound of the 95% confidence interval for the difference between the effect of increased extremal distance and that of increased second-extremal distance is about 25%, best fit by the difference-maximizing variance $\sigma^2 = 0.03$. Thus, this fact is even more challenging to fit, and the best approximation dramatically reduces the fit with median dissent rates.

5.3. A Framework for Coalition Formation. I now step back from the unidimensional model of cases and propose first steps towards a model of multidimensional judges who directly have preferences over the majority coalitions that can be formed. Full details of this framework are in Appendix C.3; I work with a simple example here for ease of exposition. Let judges be two-dimensional, with a type $t_i = (x_i, y_i)$ consisting of their party $x_i \in \{D, R\}$ and law school $y_i \in \{0, 1\}$. The value of a majority coalition is given by the average weighted similarity of its members; two judges who share a party get similarity weight w_x while two judges who share a law school get similarity weight w_y , and the overall similarity is the sum of those weights. Additionally, unanimous coalitions are scaled by $\alpha > 1$, representing the additional value of unanimity even if judges are more dissimilar, and dissenting judges receive some payoff $d \geq 0$. A majority coalition is stable if its value can be divided among member judges so that no subset of judges can form a more valuable coalition.

⁸Proposition 1 (Appendix C.1) shows that the difference is strictly positive for some σ^2 , but it is necessarily 0 for arbitrarily large σ^2 , where the original ideology score is irrelevant.

Consider a panel with Democratic judges and judges from Law School Zero in the majority. If there is only one dimension of difference, then the two majority judges may want to exclude the dissimilar minority judge from the unanimous opinion.⁹ More interesting is the case where there is a 2-1 split along both dimensions. Then there are two possibilities:

- (1) With probability $1/3$, there is one double-minority judge with type $(R, 1)$.
- (2) With probability $2/3$, the lone R judge has characteristic type 0.

In the first case, the unanimous opinion is stable if and only if it beats the 2-1 split with both identical judges in the majority and the “double-minority” $(R, 1)$ judge dissenting. In the second case, with $w_x = w_y$, the 2-1 splits with the $(R, 0)$ and $(D, 1)$ judges in the minority deliver equal value to their members. (The majority coalition with the $(D, 0)$ judge in the minority may be stable for low d and α , but can be ruled out by increasing either of those parameters.) If instead $w_x < w_y$, the 2-1 split with the $(D, 1)$ judge in the minority will deliver higher value and thus be the main challenge to the unanimous opinion.

This simple setup can largely account for the two key empirical facts of Section 6. First, bipartisan panels will have more dissents than party-unanimous panels, since on a party-unanimous panel there is only one possible dimension of disagreement while on a bipartisan panel there are two. Second, if $w_x \ll w_y$, then I can approximate the observed result that the minority partisan makes up a $1/3$ share of dissents. Under that assumption, the $(R, 1)$ judge always dissents when present; otherwise, the $(D, 1)$ judge dissents much more often than the $(R, 0)$ judge. Since the $(D, 1)$ judge is present on $2/3$ of panels and the $(R, 1)$ judge is present on only $1/3$ of panels, slightly less than $2/3$ of dissents are attributed to D judges and slightly more than $1/3$ to R judges. As w_x increases to equal w_y , the share of R dissents judge increases to $1/2$. However, this result can be offset by adding party-correlated noise to the partisan component of judge types. As seen in the sincere dissents model, this noise reduces dissents by R judges on net by occasionally making a D judge the post-noise partisan minority.

⁹There is insufficient data to test that, e.g., bipartisan panels with only men from the same law school have higher dissents by minority partisans than majority partisans. In the sample of men-only panels, the minority partisan is no more likely to dissent, but this sample is large and includes many other potential dimensions of judge heterogeneity.

This framework is only a first step towards explaining judges’ preferences for unanimous or split decisions. Placing more structure on preferences at the judge level and explicitly specifying a choice rule among stable opinions would allow me to directly estimate a model of this style in the data and compare its accuracy to the noisy sincere dissents approach in the previous setting. However, these first results already suggest an important direction for empirical analysis: measuring the interaction between partisanship and non-partisan outlier status. In Section 7.3, I do so (to the extent feasible given available data) and find suggestive evidence supporting two predictions: double-minority judges are most likely to dissent, and characteristic-minority judges dissent more than minority partisans when both are present. Before turning to the joint analysis of party and non-partisan characteristics, though, in the next section I first robustly establish the effect of partisanship alone on dissents.

6. RELATING PARTISANSHIP AND DISSENTS

6.1. Partisanship Causes More Dissents. Prior literature has found that judges’ votes on an appellate panel can be partially attributed to partisanship—both of an individual judge, and of the panel as a whole (Sunstein et al., 2006; Kastellec, 2011; Cohen, 2024). In this section I establish the degree to which partisanship is causally relevant for dissents in my sample, which contains all published appellate opinions from 1970 until 2013.¹⁰ My first specification, at the panel level, follows existing approaches and compares bipartisan panels to party-unanimous ones under the setup of Equation (1):

$$\text{dissent}_p = \alpha \text{bipartisan}_p + \text{controls}_p + \gamma_{ct} + \varepsilon_{pct}. \quad (2)$$

“dissent” on the left-hand side is an indicator for the presence of a dissenting vote on panel p (observed on circuit c in year t). On the right-hand side, “bipartisan” is an indicator for whether panel p has two judges of one party and one of the other;¹¹ γ_{ct} is the circuit $\times$ year fixed effect ensuring quasi-random assignment. To correct for minor deviations from exact

¹⁰As discussed in Section 4.2, when using ideology scores I focus on the time period 1970–2013, where those scores are nearly complete. All results using party only (and thus the full sample, 1890–2013) are robust to restricting the sample to include only 1970–2013.

¹¹Results are robust to separating Democratic- and Republican-majority panels; see tables 17 and 18 in Appendix E.1.

balance across judge and case characteristics, in this estimation and all others in this section I report results controlling for potentially relevant panel-level variables: number of unique law schools and number of women on the panel, as well as the Bloomberg Law topic of the case. (Results are unchanged with and without these controls.) I estimate the model as a conditional logit (since the outcome variable is a probability of dissent, and the fixed effects stratify the data) and cluster robust standard errors at the circuit $\times$ year level. The estimate $\hat{\alpha}$, the causal effect of switching from a party-unanimous panel to a bipartisan one, appears in the first column of Table 2.

Adding a minority-party member to a party-unanimous panel causes a 23.5% increase in dissents, equal to 2.0 percentage points (p.p.), i.e., from 8.4% to 10.4%, or roughly 800 more cases with dissents each year under the modern appellate caseload.¹² This effect is in line with estimates in earlier literature.

My second specification allows for finer variation in judges' political leanings by using ideology scores rather than the bipartisan indicator:

$$\begin{aligned} \text{dissent}_p = & \beta_1 \text{extremal-dist}_p + \beta_2 \text{2nd-extremal-dist}_p + \beta_3 |\text{median}|_p \\ & + \text{controls}_p + \gamma_{ct} + \varepsilon_{pct}. \end{aligned} \tag{3}$$

I again estimate the specification as a conditional logit and cluster robust standard errors at the circuit $\times$ year level. The new variables of interest are the three distance measures, which replace the bipartisan panel indicator in Equation (2). For each panel, I compute the median score, and label the judge who is furthest from the median as extremal (the second-furthest judge is thus second-extremal). Both of those variables appear on the right-hand side, as does the location of the median.¹³ I also include panel-level controls as in Equation (2). The estimated coefficient $\hat{\beta}_1$ then measures the causal effect of moving the extremal

¹²If the single indicator for bipartisan panels in Equation (2) is replaced with two separate indicators for Democratic-majority bipartisan panels and Republican-majority bipartisan panels, the respective coefficients are 0.255 (0.026) and 0.223 (0.024), both significantly different from 0 at a 1% level and not significantly different from each other at a 10% level.

¹³The location of the median judge influences the possible distances, e.g., a median judge located at 0 has a maximum distance of 1 from any other judge, while a median judge located at -0.5 has a distance of 1.5 to a judge located at 1.

judge one unit further from the median judge (holding fixed second-extremal distance and median location), while $\hat{\beta}_2$ measures the effect of moving the second-extremal judge one unit further from the median judge (holding fixed an extremal judge who remains further from the median). The results are in columns (2)–(7) of Table 2.

Focusing first on columns (2)–(4), which use JCS scores, $\hat{\beta}_1$ corresponds to a 24.1% increase in dissents from a one-standard-deviation increase in the extremal judge’s score. $\hat{\beta}_2$ is both much smaller—only a 3.2% increase—and not significantly different from 0. This result is driven by bipartisan panels; changing either distance has no effect on dissent rates of party-unanimous panels. This result matches the intuition that JCS scores capture precisely the legal settings most aligned with left-right ideology. The increased influence of the extremal judge is less clear when using DIME-CF scores, seen in columns (5)–(7). In the full sample, $\hat{\beta}_1$ corresponds to a 13.6% increase in dissents from a one-standard-deviation increase in extremal judge polarization, while $\hat{\beta}_2$ corresponds to a statistically significant 9.7% increase in dissents from the same increase in second-extremal judge polarization. It is once again true that the effect is driven by bipartisan panels, but on that sub-sample $\hat{\beta}_1$ is only significant at the 10% level and is approximately half the size of $\hat{\beta}_2$. The latter result suggests that DIME-CF scores may capture a broader cross-section of legal ideology and thus be a noisier measure on bipartisan panels than JCS scores.

These results provide some support for the median-voter-like predictions of the model in Section 5.1 (and the extension in Appendix C.2), where the presence of noise (or strategic considerations) could lead the second-extremal judge to have a lesser effect on total dissents than the extremal judge. However, that approach predicts that the median judge will dissent markedly less than their colleagues, even in the presence of very large noise (as shown in Section 5.2). To directly test this conclusion, and explore more general departures from classical median voter frameworks, in the next section I separate out dissents according to the “ideological role”—median, second-extremal, or extremal—of the dissenting judge. This approach clarifies whether the increased distance of a judge from the median affects only their own probability of dissent or also the behavior of their fellow panel members.

6.2. Minority Partisans Are Not Dissenting More. To show that partisanship fails to predict *who* dissents, I replace the binary dissent indicator on the left-hand side of equations

(2) and (3) with a four-outcome categorical variable: no dissent (omitted), dissent by the median judge, dissent by the second-extremal judge, and dissent by the extremal judge:

$$\text{dissent}_p = \alpha \text{bipartisan}_p + \text{controls}_p + \gamma_{ct} + \varepsilon_{pct}, \quad (4)$$

$$\begin{aligned} \text{dissent}_p = & \beta_1 \text{extremal-dist}_p + \beta_2 \text{2nd-extremal-dist}_p + \beta_3 |\text{median}|_p \\ & + \text{controls}_p + \gamma_{ct} + \varepsilon_{pct}. \end{aligned} \quad (5)$$

Equation (4) is the analogue of Equation (2), and Equation (5) is the analogue of Equation (3). I estimate both new specifications as conditional multinomial logits—i.e., estimating the change in each type of dissent separately—and cluster robust standard errors at the circuit $\times$ year level. Estimates for these specifications are in Table 3.¹⁴ Columns (1)–(4) use JCS scores to determine the median, second-extremal, and extremal judge and as variables of interest on the right-hand side. Columns (5)–(8) use DIME-CF scores instead.

Contra the median voter prediction, and new to the literature on how judicial polarization affects outcomes, dissents from all three judges increase equally. The differences between the coefficients are not significant even at the 20% level in any of the full-sample specifications of columns (1), (2), (5), and (6). The magnitude of the differences is no more than a tenth the size of the overall increase in dissent rates by each judge. As with Table 2, the effect is more clearly visible for JCS scores, where it is driven entirely by bipartisan panels. Using DIME-CF scores gives the same roughly equal increase in dissent rates, but the effect is weak enough on bipartisan panels to be marginally significant only at the 10% level.

These results offer a sharp contrast to existing models of judicial politics. Both the theoretical literature discussed in Section 2 and the model I specialize to my setting in Section 5.1 emphasize that median judges are much less likely to dissent, whether because they strategically join majority coalitions or because noise may place them further to the left or right than the data implies. Using the point estimates, dissents by the median judge increase by 1.7% less than dissents by the second-extremal judge and by 1.2% more than dissents by the extremal judge, for an average of 0.25% less than their more extreme colleagues overall. This value represents a dissent share of 33.2% compared to the proportional share of

¹⁴As with the earlier results on total dissents, these results are robust to considering Democratic- and Republican-majority bipartisan panels separately; see tables 19–21 in Appendix E.1.

33.3%. Re-computing this difference in 1,000 Bayesian bootstrap draws of the data gives a 95% confidence interval for the median dissent share ranging from 27.2% to 39.1%. Even at the lower bound of the confidence interval, on over 1,000 cases each year the two politically extreme judges form a coalition which excludes the median judge. That lower bound still only accounts for about one-fourth of the causal increase in dissents on bipartisan panels: when partisanship causes judges to fail to reach consensus, that failure is non-partisan in nature almost 75% of the time. To conclusively establish equal dissent rates, in the next section I provide two robustness checks using alternative estimation strategies, followed by an investigation of the effect’s strength across time and case topics.

6.3. Equal Dissent Rates Are Robust. The models presented in Section 5 provide some theoretical foundations for believing that dissent rates may truly be equal across majority and minority partisans, as well as across median and extremal judges. However, one clear possible alternative is that ideology scores are merely a low-quality, noisy measure of partisan polarization. To rule out this explanation, recall that Table 2 shows ideology scores are just as predictive of increased dissents as party, and the magnitude of that increase is practically significant. In addition, increased dissents are driven exclusively by the extremal judge’s distance. If noise alone explained similarities in dissent rates, this distance should not affect the second-extremal judge’s dissent rate, which should instead be driven by their own distance from the median. However, the second-extremal judge’s distance has no greater effect on their own dissent rate than on the other judges, and that effect is generally not significantly different from zero.

To further alleviate the concern that noise drives the results, I provide two additional tests that dissents are indeed equal across majority-party/median judges and minority-party/extremal judges. The first is a binary-outcome specification at the judge level:

$$\text{Dissent}_{jp} = \alpha \text{judge-char}_{jp} + \text{controls}_p + \gamma_j + \gamma_{ct} + \varepsilon_{jpct}, \quad (6)$$

which I can estimate as a conditional logit with standard errors clustered at the judge level. In this specification, the additional subscript j indicates that I consider specifically dissents by, and characteristics of, judge j on panel p of circuit $\times$ year ct . That is, each observation of a panel becomes three observations of individual judges, and “judge-char” captures, e.g.,

whether or not judge j is a minority partisan on panel p . I use judge fixed effects γ_j to control for variation in individual judge characteristics. Results from this specification are in columns (1)–(3) of Table 4.¹⁵

Before interpreting the results, I comment on a challenging issue of statistical power. To a first approximation, there are 200-500 cases observed on average per judge. Given that most judges serve for several decades, I can conservatively upper-bound each judge at 50 cases observed per circuit $\times$ year. Since dissent rates are only around 10%, this means a judge dissents fewer than five times in each random sample, too small to separate out noise in dissents from the effects of panel assignments. In contrast, the panel-level specification has an average of 300-500 cases per circuit $\times$ year and can thus detect more notable differences in the authorship of the corresponding ≈ 50 dissents.

With this caveat in mind, I can use the confidence interval to rule out any differences in dissents between minority- and majority-partisan judges larger than 0.1 percentage points, or about one-twentieth the size of the increase in dissents caused by switching a party-unanimous panel to a bipartisan one. The difference in dissents between median and extremal judges is similar, at most 0.15 percentage points. Clearly, the increase in dissents on bipartisan panels cannot be driven by markedly higher dissent rates by judges when they are in the partisan minority, nor is it driven by dissents from the ideological extremes.

Constructing a panel-level test of equal dissent rates by minority partisans is more challenging and requires carefully defining the counterfactual. In Table 3, I take advantage of the fact that ideology scores are defined on both party-unanimous and bipartisan panels, i.e., extremal and median judges exist regardless of whether all three judges are Democrats or one of them is a Republican. Thus, I can ask how making the extremal judge more extreme affects the dissent rates of all three judges, using the full sample of panels where ideology scores exist in the data. I can also pose a well-defined question relating party to ideology scores: if one of the judges on a three-Democrat panel were made to be a Republican, would

¹⁵In Table 22 (Appendix E.2), I use controls for party, age, and gender instead of judge fixed effects; results are qualitatively unchanged.

the extremal judge dissent more? However, a minority-party judge is (clearly) only well-defined on a bipartisan panel, so I cannot test whether a minority-party judge dissents more on bipartisan panels than party-unanimous ones.

Since the dissent rate of majority-party judges is well-defined on both party-unanimous panels and bipartisan panels, I can estimate it and impute the dissent shares of majority- and minority-party judges. Let d be the dissent rate on bipartisan panels, which I can derive from the dissent rate u on party-unanimous panels and the logit coefficient estimated in Equation (2). I estimate a modified version of that specification where I replace the left-hand-side indicator for dissents with an indicator for majority-party dissents. Of course, on all party-unanimous panels the indicators will have the same value, so u will be unchanged. However, I obtain a new estimate d_{maj} of the dissent rate on bipartisan panels using this new indicator, reflecting dissents by majority partisans only. If $d_{\text{maj}}/d = 2/3$, then the dissent rate of majority partisans is exactly as would be expected if party has no effect on dissent rates; if instead $d_{\text{maj}}/d < 2/3$ (resp. $> 2/3$), then minority partisans are making a larger (resp. smaller) than proportional contribution to the change in dissent rates. I can perform a similar test to see whether the median judge (measured using either JCS or DIME-CF scores) makes up a greater-than-expected share of dissents on bipartisan panels compared to party-unanimous ones, but must compare the ratio d_{maj}/d to 1 instead of $2/3$ because median judges exist on both party-unanimous and bipartisan panels. To form a confidence interval and perform statistical significance tests for these ratios, I compute them for 1,000 Bayesian bootstrap draws of the data; results are in columns (1)–(3) of Table 5.

In all cases, the ratios are qualitatively close to the proportional values expected if party or score polarization did not affect dissent rates, though in five of the six cases I reject the null of exact equality. In particular, the ratio suggests that minority partisans make up a share of dissents about 0.4% larger than proportional, while median judges' share of dissents is about 0.4% less. This difference is minute; scaling by the approximately 10% of cases with dissents, it translates to dissenting 0.04 percentage points more often, essentially insignificant compared to the 2.0 p.p. increase in dissents on bipartisan panels.

Having established that, in the full sample, majority partisans and ideological-median judges dissent at essentially the same rate as minority partisans and ideological-extreme judges, I

now confirm that this effect persists across various key subsets of the data. The share of dissents by each judge role (minority-party, median, second-extremal, or extremal) is approximately one-third in each year from 1970 to 2013, the full time period for which scores are available. This pattern is illustrated in figures 18 and 19 (Appendix E.2), which aggregate results across circuits for ease of presentation. The results across different legal topics are summarized in Table 6, which presents coefficients equivalent to column (1) of Table 3 estimated within each of the Bloomberg legal topics, i.e., the increase in dissent rates of the median, second-extremal, and extremal judge (as measured using JCS scores) when switching from party-unanimous to bipartisan panels. In particular, equality continues to hold for the three most common topics—criminal, due process, and economics and regulation—as well as for miscellaneous cases. The median judge dissents far less for civil rights cases, but far more for labor relations cases. Split-party panels have no more dissents than unanimous-party panels on First Amendment cases, but the sample is an order of magnitude smaller and the baseline dissent rate is 50% higher than for any other topic.

Returning to the comparison with judge-level specifications, the panel-level result clarifies an important subtlety about how a judge’s individual distance from the median affects dissents. For panels where that judge is not extremal, their distance from the median has no effect on any judge’s dissent probability. When plotting individual dissent probability as a function of distance from the median, this result will manifest as a positive slope which is nearly zero for small distances from the median, where the individual judge is highly unlikely to be extremal, and increases with that distance—not because the effect is quadratic, but because the probability of being extremal and having any effect at all increases with distance. The effect on the overall probability of dissent is in fact three times larger than the effect on a judge’s own probability of dissent. Both of these results are confirmed qualitatively in Figure 2, which uses local polynomial regression to show dissent probability as a function of individual or extremal distance from median.

7. JUDGE IDENTITY INFLUENCES DISSENTS

Prior literature has found that non-partisan demographic characteristics can influence outcomes and dissents (Kastellec, 2013; Ash et al., 2024). In this section, I explore whether my

two main conclusions about partisanship—partisan polarization causes more dissents, but those dissents are not driven by the most-polarized outlier judge—hold for ideologically relevant non-partisan characteristics as well. For example, does taking a panel where all three judges share a law school (the equivalent of a party-unanimous panel) and replacing one member with a judge from a different law school (the equivalent of switching to a bipartisan panel) cause more dissents? If so, is the increase driven by a judge’s increased likelihood of dissenting when they are an outlier, i.e., in the law school minority?

To answer this question, I replicate the four types of analysis from Section 6: the effect of mixed panels on total dissents (Table 2 in Section 6, Table 7 here), the effect of mixed panels on dissents by outlier judges (Table 3 in Section 6, Table 8 here), the effect of mixed panels on judge-level dissents (Table 4), and the share of dissents by majority judges (Table 5). Across three important non-partisan characteristics—law school, gender, and age—I find marginally statistically significant increases from the presence of outlier judges, but the magnitude of the point estimates is smaller than the increase caused by partisanship. I also provide suggestive evidence that dissents are somewhat concentrated among “outlier judges,” e.g., the law school minority judge as described above. This combination of empirical facts is more consistent with both a classical median-voter-style framework where outliers directly cause dissents and the model of sincere dissents with noise I presented in Section 5.1.

7.1. Non-Partisan Characteristics Cause Some Dissents. I begin by following the approach of Equation (2) to test whether the presence of outlier judges (e.g., replacing a man on an all-men panel with a woman, or an oldest judge whose age is close to their colleagues with a much older judge) causally increases total dissents. In particular, I use specifications of the form

$$\text{dissent}_p = \alpha \text{panel-char.}_p + \text{controls}_p + \gamma_{ct} + \varepsilon_{pct}, \quad (7)$$

restricting the sample to contain only panels where the selected characteristic is relevant. As with Equation (2) in Section 6.1, I estimate each specification as a conditional logit and cluster robust standard errors at the circuit $\times$ year level. Results are in Table 7.

Only gender causes a statistically significant change in dissent rates; however, all signs are as expected, suggesting that the presence of an outlier on a panel (whether that means mixed

parties, law schools, or genders, or ages) increases dissents.¹⁶ Effects are roughly one-third to one-fourth as large as the effect of partisanship. When estimating a single conditional logit model including indicators for all of these possible outlier effects (Table 26 in Appendix E.3), the coefficient for party is essentially unchanged. All other coefficients, with the exception of those for gender outliers (e.g., replacing a man on an all-male panel with a woman), are slightly reduced in magnitude and become not statistically significant. These results suggest that outlier effects are present for non-partisan characteristics, but their effect on judges’ tendency to disagree is small in comparison to the effect of partisanship.

7.2. Dissents are Somewhat Concentrated on Outlier Judges. The clearest test that the increase in dissents caused by partisan polarization is distributed equally across judges is Table 3 (Section 6.2). That result separates the change in dissents on bipartisan panels according to a judge’s role as median, second-extremal, and extremal using a partisan ideology score. To replicate this test for, e.g., law school, I need a continuous measure of law school similarity, paralleling how JCS and DIME-CF ideology scores measure the partisan similarity of two judges. Such a measure would allow me to define a “law-school-extremal” judge on each panel, analogous to the ideology-score-extremal judge.

As a first step in this direction, I use the semantic embedding vectors for each opinion to construct a simple measure of similarity. Each law school is represented by the average vector across the semantic embeddings for all opinions whose authors went to that law school, and the similarity between two law schools is simply the cosine similarity of their representing vectors. Because cosine similarity is symmetric, a panel therefore includes three similarity measures: the similarities of judges 1 and 2, 2 and 3, and 1 and 3. (If two judges share a law school, their similarity is 1.) The intuitive analogue of the ideology-score-extremal judge in this setting is therefore the judge who is *not* a member of the most similar pair.¹⁷ Restricting

¹⁶Particularly striking is the coefficient for panels with exactly two unique law schools; it flips sign from column (2), where these panels are more mixed than those with one unique law school, to column (3), where those panels are less mixed than those with three unique law schools.

¹⁷I can analogously define the second-extremal judge as the one who is not a member of the second-most similar pair. The remaining judge, who is a member of both the most and second-most similar pairs, is the median judge.

to panels with either two or three unique law schools—since on panels with one unique law school all judges have the same similarity—I can thus estimate the specification,

$$\begin{aligned} \text{dissent}_p = & \text{has-three-schools}_p \\ & + \text{controls}_p + \gamma_{ct} + \varepsilon_{pct}, \end{aligned} \tag{8}$$

where “dissent” on the left-hand side is a categorical variable capturing no dissent, dissent by the (law-school-) extremal judge, dissent by the second-extremal judge, and dissent by the median judge.¹⁸ As with Table 3, I estimate the specification as a conditional multinomial logit and cluster standard errors at the circuit $\times$ year level. Results from this estimation are in column (1) of Table 8.

Before discussing those results, I briefly describe the parallel estimation approach for my other two variables of interest, gender and age. To reproduce these results for gender, I could use (or re-construct) the measure of gender slant used in [Ash et al. \(2024\)](#); I could then test whether the increase in dissents on mixed-gender panels is concentrated among more gender-slanted judges. Though this exercise is straightforward, it requires additional data and extensive computations, so I leave it for future work. However, effects by age are easily computed using existing data: there are always youngest, median, and oldest judges on each panel, so I can test how the (not statistically significant) increase in dissents from the presence of an age outlier is distributed across judges. In particular, I estimate the specification

$$\begin{aligned} \text{dissent}_p = & \text{has-age-outlier}_p \\ & + \text{controls}_p + \gamma_{ct} + \varepsilon_{pct}, \end{aligned} \tag{9}$$

where “dissent” on the left-hand side is a categorical variable capturing no dissent, dissent by the youngest judge, dissent by the age-median judge, and dissent by the oldest judge. As with the other parallel specifications, I estimate Equation (9) as a conditional multinomial logit and cluster standard errors at the circuit $\times$ year level. Results from this estimation are in columns (2)–(4) of Table 8.

¹⁸Note that with two unique law schools, both the median and second-extremal judge are in the law school majority; which is which is determined by their similarity to the law school minority judge.

While the difference in coefficients for law school in column (1) of Table 8 is not significant, the difference in point estimates is meaningful given the magnitude of the overall effect (columns (2) and (3) of Table 7). In particular, the increase in dissents is 4.1% for the (law-school-) median judge compared to 2.0% and 2.3% for the second-extremal and extremal judges respectively. That is, dissents are reduced for outlier judges. However, this reversal is less surprising when considering that on a panel with two unique law schools, there is a sharply defined law school majority and minority; when that division is eliminated, the members of the majority are more affected by the removal of their likely coalition party than the already-isolated minority.

Effects for age are also mostly suggestive. The oldest judge sees a clear increase in dissents and the median judge appears to dissent less than the others, but the differences between the coefficients are again not statistically significant. Furthermore, the oldest judge actually dissents most often when there is a young outlier, rather than (as expected) when they are the old outlier themselves. Given the role of the senior judge in assigning opinion authorship, further analysis—and potentially more data—is needed to untangle the endogenous effects of opinion assignment and dissenting behavior.

I next return to the judge-level specification of Equation (6), which produced the result that switching an individual judge from the partisan majority to the partisan minority did not increase their dissent rate (column (1) of Table 4). Here, I test the effect of switching a judge from the majority to the minority in a non-partisan characteristic—e.g., comparing the dissent rate of Judge A, a woman, on panels where she is the only woman to her dissent rate on panels where she is one of two women.¹⁹ As a reminder, the relevant specification is a conditional logit,

$$\text{dissent}_{jp} = \alpha \text{judge-char}_{jp} + \text{controls}_p + \gamma_j + \gamma_{ct} + \varepsilon_{jpct},$$

where “judge-char” is, e.g., an indicator for whether judge j is in the panel gender minority. To ensure that this comparison is not capturing a difference between, e.g., mixed-gender and gender-homogeneous panels, akin to the difference between bipartisan and party-unanimous

¹⁹For the effect of age, I separately test the effect of being the youngest, median, and oldest judge on the panel. Unlike ideology scores, where the direction of distance from the median is likely irrelevant, young and old judges may behave differently.

panels, the sample in each column contains only mixed-characteristic panels for the appropriate characteristic.²⁰ Results are in columns (4)–(10) of Table 4.

Noise is even more of an issue in this setting than in the partisan one, as shown by the dramatically smaller sample sizes; this issue is driven by the rarity of panels where all three judges share a law school (about 3.5% of the sample), as well as panels with two women (about 2.7% of the sample). Still, in both the version with controls for judge characteristics and the version with judge fixed effects, I can rule out large (5-10%) differences in dissent rates at the judge level for all non-partisan characteristics except for gender, where there is a 13.4% increase in dissents for women judges from being in the gender minority.

Lastly, aggregating back to the panel level, I repeat the comparison of panel-majority judges’ behavior on unanimous and mixed panels, following the approach in Section 6.2. Specifically, I estimate the conditional logit specification

$$\text{dissent}_p = \alpha \text{mixed-char}_p + \text{controls}_p + \gamma_{ct} + \varepsilon_{pct}, \quad (10)$$

where the left-hand side is an indicator for the presence of a dissent by any judge on panel p and “mixed-char” is an indicator for, e.g., panel p being mixed-gender rather than gender-unanimous.²¹ The estimated coefficient $\hat{\alpha}$ and baseline dissent rate on characteristic-unanimous panels u allow me to compute the dissent rate on mixed-characteristic panels, d . I then re-estimate the specification replacing the left-hand side with a new indicator “dissent_maj” which only tracks dissents by judges in the characteristic majority on panel p , and obtain a new dissent rate d_{maj} . If the ratio d_{maj}/d is less than (resp. greater than) the proportional level of 2/3—or 1, for characteristics present on both mixed and unanimous panels—then the characteristic majority contributes fewer (resp. more) dissents than is proportional. Results from this procedure for law school, gender, and age are in columns (4)–(10) of Table 5. As in Section 6.2, standard errors and statistical significance are obtained by re-computing the ratio using 1,000 Bayesian bootstrap draws of the data.

²⁰Recall that I replicate the results using judge-level controls in Table 22 (Appendix E.3).

²¹For age, I use the indicator for a young outlier when looking at dissents by youngest judges, the indicator for an old outlier when looking at dissents by oldest judges, and the presence of any age outlier when looking at dissents by median judges.

Recall that majority partisans account for an almost-exactly-proportional 66.3% of the increase in dissents on bipartisan panels (column (1) of Table 5. Here, for all characteristics except law school I am able to reject the null that dissent shares are proportional, i.e., as expected if that characteristic played no role in dissenting behavior. Indeed, for all characteristics except law school and women judges (where the sample size is extremely small, driven by the rarity of majority-women panels), both the difference from proportionality of the point estimate and the level of rejection of the null are larger than for any of the analogous partisan measures. In particular, gender-minority judges' share of dissents is 3.5% larger than proportional; youngest judges' share is 1.7% smaller; median-age judges' share is 1.8% smaller; and oldest judges' share is 1.7% larger. Converting these ratios into total dissents, the gender-minority judge contributes 0.36 percentage points more dissents than the gender-majority judge, more than half of the 0.64 p.p. increase in dissents caused by mixed-gender panels. Oldest judges contribute about 0.15 p.p. more dissents, almost half of the 0.35 p.p. increase in dissents caused by having an old outlier. While these effects seem small in percentage terms, they represent hundreds of additional dissents per year and may affect judges' ability to reach consensus on highly impactful issues.

Overall, analyzing non-partisan judge characteristics in the same manner as partisanship shows two clear differences from the results presented in Section 6. First, the effect of non-partisan characteristics on total dissents is much more limited. Only the effect of gender is significantly different from zero at the 5% level, and while the signs of all other effects are as expected, the point estimates are only about one-fourth the size of the highly statistically significant effect of party. It therefore seems that partisanship is uniquely powerful in causing disagreements. The second difference is in which judges are affected by the increase in dissents. While that the effect of party is robustly distributed across not only majority and minority partisans but also extremal and median judges, the increase in dissents driven by mixed-characteristic panels is more concentrated among outlier judges. This effect is often weak, but given the reduced increase in total dissents, milder effects are anticipated. These results are consistent with classical models of judicial politics, e.g., the framework in Section 5.1. Even if the increase in total dissents and the localized effects on outlier judges were both approximately zero—which I cannot rule out in some cases—that pair of results would be consistent with a simple decision-making process which is not affected by these non-partisan

traits. In contrast, the presence of a strong positive effect of partisanship on total dissents rules out the explanation that partisanship is simply irrelevant to appellate decisions.

7.3. Partisan and Non-Partisan Characteristics Have Limited Interactions. The results of the previous section make clear that the effect of partisanship on dissents is both stronger and more evenly distributed than that of other non-partisan judge characteristics. However, these effects may interact; a “double-minority” judge who is, e.g., a minority partisan and the only woman on a panel with two men, may behave differently than either a minority partisan in the panel gender majority or a majority partisan who is in the panel gender minority. In Table 25 (Appendix E.3), I interact the outlier effects of Table 7 with whether the outlier judge is a majority or minority partisan. Double-minority law school judges do seem to dissent more than their single-minority outlier counterparts, but double-minority age and gender judges do not.

To more directly analyze these relationships, I follow the approach of Equation (2) and consider the “cross-effect” of increased dissents from panel partisanship on characteristic-majority and characteristic-minority judges.²² I estimate a conditional multinomial logit,

$$\text{dissent}_p = \alpha \text{bipartisan}_p + \text{controls}_p + \gamma_{ct} + \varepsilon_{pct}, \quad (11)$$

where the left-hand side is a categorical variable with outcomes no dissent, dissent by the characteristic-minority judge, and dissent by the characteristic-majority judge. (As with the estimation of Equation (2), I include controls for panel-level variables and cluster robust standard errors at the circuit $\times$ year level.) Specifically, I compare the law school minority and majority on panels with two unique law schools, the gender minority and majority (as well as separately considering a gender-minority man and gender-minority woman) on mixed-gender panels, and all three possible age roles (youngest, age-median, and oldest). Results are in Table 9.

Beginning with law school, the difference in increased dissents between minority and majority judges is not statistically significant, but the qualitative results are sharply different

²²Given the lack of power in the judge-level specification when looking at each characteristic individually, with this data it is infeasible to estimate, e.g., the dissent rate of double-minority party and law school judges compared to single-minority judges along each dimension.

from those of Table 3, where the point estimates across median and extremal judges were essentially identical. Here, I find a 10.9 percentage point (p.p.) larger increase in the dissent rate for the law school minority judge (36.5% compared to 25.6%), which represents 0.9 p.p. more cases with dissents. Of course, this effect only applies on about 25% of bipartisan panels (those with two unique law schools), and thus could account for about one-eighth of the total increase in dissents caused by partisanship. That is, if bipartisan panels merely activated latent disagreements across law schools, causing differential increases in dissents by law school minority and law school majority judges—with no other effects—bipartisan panels would have about 0.25 p.p. more dissents than party-unanimous ones (one-eighth of the result from Section 6.1).

Turning next to gender, the difference between majority and minority judges is not statistically significant when combining men and women judges, but is statistically significant when separately examining the comparison between majority and minority men and majority and minority women. The increase in dissent rate is 125.7 p.p. larger for men in the gender minority (151.6% compared to 25.9% in the gender majority), representing 11.2 p.p. more cases with dissents. The increase in dissent rate is actually 32.3% smaller for women in the gender minority (13.5% compared to 45.7% in the gender majority), representing 3.1 p.p. more cases with dissents. That is, women are more affected by partisan polarization when in the gender majority than when in the gender minority. Since panels with exactly two women make up only about one-eighth of mixed-gender panels, the pooled effect of being in the gender minority is slightly larger than the pooled effect of being in the gender majority. Benchmarking these effects against the 2.0 p.p. more cases with dissents on bipartisan panels, panels with men in the gender minority are about 2% of bipartisan panels and could thus account for about one-tenth of the total increase in dissents caused by partisanship; panels with women in the gender minority are about 17% of bipartisan panels and could thus account for one-quarter of the total increase in dissents caused by partisanship.

Finally, examining age, the difference between youngest, age-median, and oldest judges is again not statistically significant; however, the point-estimated increase in dissent rate for youngest judges is 4.0 p.p. larger than for median judges (36.6% compared to 32.6%); the difference between median and oldest judges is minimal (32.6% compared to 33.0%). The

former represents 2.7 p.p. more cases with dissents, and the latter only 0.4 p.p. more cases with dissents. Since the sample for this specification includes all bipartisan panels, the effect of age could in fact cover the whole effect of partisanship. However, caution is needed in interpreting this result because of the senior judge’s role in assigning opinion authorship. This assignment may be endogenous, e.g., youngest judges know that a senior judge of the opposite party is unlikely to give them authorship, and thus dissent more on bipartisan panels where such a judge can be present. Still, it supports a framework where non-partisan disagreements make up a larger share of total dissents than partisan ones.

As seen in this section, any study of the interaction between partisanship and other judge characteristics is limited by the dramatically reduced number of “doubly-mixed” panels. As such, these results merely suggest that partisanship activates other latent disagreements rather than showing it conclusively. However, I can provide stronger evidence that these non-partisan characteristics have quantitatively important effects on opinion content—and thus justify disagreement by outlier judges—through a closer analysis of opinion text. This complementary approach, which I lay out in the next section also allows me to work with the full sample of opinions, increasing statistical power and more robustly establishing the influence of law school and gender in particular.

8. USING NATURAL LANGUAGE PROCESSING TO UNDERSTAND OPINION TEXT

Dissents are rare, but judges may have reason to be dissatisfied with an opinion even if they are willing to sign it—or may even sign an opinion they disagree with for strategic reasons (as in the model extension of Appendix C.2). Because these disagreements do not produce differences in observed outcomes, they cannot be measured using standard metadata. However, by analyzing opinion text using natural language processing (NLP) tools, I can quantify differences in opinions that suggest potential areas of disagreement. This approach completes the picture of whether partisanship is directly motivating disagreements or simply contributing to an environment where judges who differ along non-partisan dimensions are less likely to reach consensus.

I focus on one important way in which an outlier judge might affect a panel’s discussion: bringing new legal topics to the table, or contributing to a far-ranging discussion which

covers more topics in total. Focusing on broad topics abstracts away from other dimensions along which judges might disagree—such as the presence of emotive language, or the valence with which particular arguments are discussed—and may not fully capture the richness of legal decision-making. However, these topics are still an important feature of decisions, determining which areas of law they are most likely to influence through precedent and citations. They are also an input into the decision to dissent: most dissents include an additional topic not present in the lead opinion, suggesting that the dissenting judge felt a particular issue was not given sufficient consideration. Thus, by looking at the distribution of topics across panel types and individual characteristics, I can obtain a picture of when an opinion is likely to be unsatisfying to a given judge without having to rely on the presence of an actual dissent.

Using state-of-the-art NLP transformer models to identify and measure the prevalence of legal topics in the full sample of opinion text, I find evidence that a panel’s partisan composition has little effect on the topics discussed in lead opinions.²³ Author party has some influence, but is lesser in both range and magnitude to the effect of gender, while author age is largely irrelevant. The effect of law school on opinion content is more challenging to measure and requires a less supervised approach using local similarity measures, but appears to be comparable in magnitude to the effect of gender.

To obtain these results, I follow a standard fine-tuning pipeline using a transformer model pre-trained on legal documents (LegalBERT from [Chalkidis et al. 2020](#)) and predict the legal topics hand-coded for each case by Bloomberg Law.²⁴ In Section 8.1, I follow in spirit [Gentzkow et al. \(2019\)](#), who use predicted author party as a measure of polarization in Congressional speech, and compare predicted topic distributions across various panel compositions and author characteristics. I supplement analysis of the distribution of individual topics with another important dimension of potential heterogeneity: the breadth of topics covered, which I measure as the number of “uniform major topics,” with probability larger

²³In this section, I focus on the time period 1970–2013, when Democratic and Republican positions on these topics—civil rights in particular—are more consistent, and data is fully available for all judge characteristics of interest.

²⁴For details of the prediction task and performance metrics, see Appendix D.

than 1/8 (i.e., larger than if drawn uniformly at random) and the number of “Bayesian major topics” with higher predicted probability than their prior probability.²⁵ In Section 8.2, I move beyond predicted topics and use novel measures based directly on semantic embeddings to study whether author characteristics affect the overall similarity of opinions.

8.1. Analyzing Predicted Legal Topics. A partisan-minority judge is less likely to be chosen as lead opinion author under both random author selection and author selection by similarity to the senior judge, whose identity is itself quasi-random. Thus, if there is a large difference in opinion topics across parties, a partisan-minority judge will often disagree with the content of the lead opinion. The sharpest comparison across parties is of all-Democratic to all-Republican panels; these panels are more likely to be ideologically homogeneous and ensure there can be no moderating effect of author assignment (e.g., a Republican senior judge writes an opinion even though the panel is majority Democratic). On the other hand, the comparison of Democratic-majority bipartisan panels to Republican-majority bipartisan panels is especially crucial for a minority partisan’s decision to dissent. In the analysis that follows, I present both comparisons, as well as a comparison of the transition from party-unanimous to bipartisan panels.

Focusing first on party-unanimous panels, I use the following specification:

$$\begin{aligned} \mathbb{P}(\text{topic})_p = & \beta_1 \text{panel-type}_p \\ & + \text{controls}_p + \gamma_{ct} + \varepsilon_{pct}, \end{aligned} \tag{12}$$

where “panel-type” captures whether the panel is all-Democratic or all-Republican. I estimate a conditional logit model for each individual topic probability (since they are bounded between 0 and 1) and a conditional Poisson model for uniform and Bayesian major topics (since they are integer counts), and cluster robust standard errors at the circuit $\times$ year level. Because author assignment may be endogenous, I focus on panel partisanship, which is randomly assigned. However, to allow qualitative comparisons of party to gender and age,

²⁵Tables 28 and 29 in Appendix F.1 show that the latter two computations are robust to scaling those thresholds. Figures 20–22 show the full distributions of both kinds of major topics across panels with different numbers of Democrats, women, and unique law schools.

in Appendix F.2 I estimate this specification with controls for author age and gender (Table 30). Results from the estimation of Equation (12) are in the first row of Table 10.

Recall that criminal, due process, and economics and regulation make up about 85% of the Bloomberg-coded ground truth topic labels. On the first two of those, there are no significant differences across parties. However, Democratic panels are less likely to discuss economics and regulation. Among the remaining topics, Democratic panels are more likely to discuss both First Amendment and miscellaneous issues. They are also about 15% more likely to include an additional major (Bayesian) topic. This result fits well with Democrats' increased tendency to discuss the broad miscellaneous topic.

These effects represent changes in the average share of text devoted to each topic, rather than shifts from, e.g., an opinion entirely about economics and regulation to one about civil rights. However, interpreting them in the latter way allows me to benchmark them against the overall share of dissents. Democratic panels cover the First Amendment 0.17 percentage points (p.p.) more often, economics and regulation 1.13 p.p. less often, and miscellaneous topics 0.88 p.p. more often. These estimates are generally smaller than, but are comparable in magnitude to, the 2.0 p.p. increase in dissents on bipartisan panels compared to party-unanimous ones.

I next break down the effect of a more Democratic bipartisan panel using three related comparisons. First, I look only at Democratic-majority panels, and compare unanimous ones to bipartisan ones; I then do the same for Republican-majority panels, and finally compare topics across all bipartisan panels separated by majority party. For each comparison, I follow the specification in Equation (12) and again estimate a conditional logit model for all individual topic probabilities and a conditional Poisson model for uniform and Bayesian major topics, clustering robust standard errors at the circuit $\times$ year level.²⁶

The effects of partisanship on Democratic- and Republican-dominated panels qualitatively match those in Table 30, with the additional Democratic judge having a stronger effect on

²⁶Tables 31–32 in Appendix F.2 replicate the analysis with controls for author party, age, and gender. Table 34 in the same appendix replicates the analysis with author fixed effects, essentially estimating the difference in, e.g., opinions written by Republican Judge A when on a panel with two Democrats to opinions written by Judge A when on a panel with one more Republican and one Democrat.

majority-Democratic panels than majority-Republican panels (where they have essentially no effect). Strikingly, but perhaps not surprisingly given the lack of influence from partisanship on observable disagreement, the effect of switching from a Republican-majority bipartisan panel to a Democratic-majority one is essentially null. The only statistically significant coefficient is a marginal increase in discussion of First Amendment issues (0.06 p.p.), and the other point estimates are generally smaller than in any of the other comparisons.

More speculatively, I can use the results in tables 30–33 (Appendix F.2) to compare the influence of partisanship and other judge characteristics. I caveat this comparison by noting that if author assignment is endogenous to these characteristics, then the results may be biased; however, most of the literature on author assignment suggests that it is driven primarily by idiosyncratic features like judge workloads and domain-specific knowledge. Across the four comparisons, the effect of gender is almost universally larger topic-by-topic than the effect of party, while the effect of age is largely null. In particular, women judges tend to discuss civil rights and miscellaneous topics more frequently, but labor relations and economics and regulation topics less frequently. The magnitudes of these effects are generally between 0.5 p.p. and 2 p.p., as substantial as the increase in dissents on bipartisan panels. Also of interest in this setting with controls is the ability to compare panel effects (through the coefficient on panel characteristics) to author effects (through the coefficient on author party). In Tables 31–33, the effect of author partisanship is generally stronger than, but in the same direction as, the effect of panel partisanship. However, the effect of author partisanship is almost always smaller than the effect of author gender.

To establish conclusively that authorship is more important than panel peer effects, or that gender is definitively more important than party, further research is needed to establish when author assignment is quasi-random. However, these results strongly support the idea that party has little effect on topic choice. Additionally, the effect of gender compared to party is especially clear on bipartisan panels (Table 33 in Appendix F.2), where control of the majority has no effect on any individual topics or the number of major topics, while the effect of author party is dominated by that of gender in all categories but one (where it is essentially equal). While these results do not speak directly to the *existence* of dissents, they suggest that a woman judge in the gender minority on a panel is more likely to disagree

with the content of the lead opinion than a panel-minority partisan, providing evidence for the influence of non-partisan characteristics on the *motivation* to dissent. A missing variable in this analysis is law school training, which is challenging to compare under this approach because it takes many unordered values. In the next section, I address this gap by directly using the learned text embeddings rather than the transformer model’s already-powerful reduced-form predictions.

8.2. Analyzing Opinion Similarity. A major benefit of modern transformer approaches to NLP is that these models encode a complete embedding representation of any text input as a high-dimensional vector, which is then used for classification and prediction tasks as in the previous section. Since the model I use is fine-tuned to represent opinions in topic space, closeness of these embeddings reflects similarity in legal topics, but at a much finer-grained level than the predictions can convey. To gain sharper insight into which judge characteristics have the most influence on legal topics in opinion text, I select the 1,000 nearest neighbors for each opinion—about 0.4% of the full sample—by cosine similarity (the standard measure of closeness in embedding space) and assess the properties of these neighbors. This measure provides a more nonparametric approach to the motivating question of the previous section: should minority partisans be worried that an opinion written by a majority partisan will be different from the opinion they would have liked to write?

Using author party as an example, I describe below the steps for constructing my measure of by-characteristic similarity:

- (1) For each opinion, compute the share of neighboring opinions for which the author’s party matches the neighbor’s author’s party.
- (2) For each party in the sample, compute the overall share of opinions authored by judges of that party.
- (3) Normalize the share of matches by subtracting the appropriate mean and dividing by the entropy.²⁷ E.g., for a Republican-authored opinion, subtract the share of

²⁷I use entropy rather than variance as in a standard *z*-score because several of the variables, e.g., law school, are categorical rather than numeric.

Republican-authored opinions in the sample from the share of neighboring Republican-authored opinions and divide by the entropy of authorship party.

- (4) Compute the distribution of normalized scores and the average normalized score across all opinions.

The normalized match scores allow me to compare similarity along dimensions like party, where there are few categories and the sample is balanced, to dimensions like gender, where there are still few categories, but the sample is highly unbalanced, and dimensions like law school, where there are many categories and the sample is unbalanced. I present the results, averaging across opinions in Table 11; full distributions of normalized and de-meanned match scores are in figures 23–28 (Appendix F.3).²⁸

The score for legal topic serves as a useful benchmark; not only was this variable the target for fine-tuning the model, it also displays the clearest clustering patterns when viewing a low-dimensional representation of the embedding vectors. Figure 3 compares this by-topic visualization to one which colors opinions by the number of Democrats on the panel; the difference in visible clusters is immediately apparent. While these visualizations necessarily sacrifice some of the information contained in the full embedding vectors, they provide a first suggestion that partisanship will not markedly alter the pattern of nearby opinions.

The normalized match score for legal topic is an order of magnitude larger than the scores for law school or gender, both of which are roughly of the same magnitude as the match scores for author party and number of Democrats, but opposite in sign. For the three partisan characteristics, the match scores are negative, indicating that along the chosen dimension opinions are less similar to actual neighbors than if those neighbors were randomly selected. For the three non-partisan characteristics, the match scores are positive, indicating higher similarity than under a random choice of neighbors. While not directly comparable due to the differing numbers of outcomes for each variable, the de-meanned coefficients have the most straightforward interpretation: they represent the percentage of similar neighbors in excess of the population average. For example, in column (5) of Table 11 the average opinion is

²⁸Results on average match scores are robust to choosing the 100 or 10,000 nearest neighbors (tables 35 and 36, Appendix F.4) instead of the 1,000 nearest neighbors, as well as to considering only neighbors with different authors (tables 37 – 39, Appendix F.4).

2.7% more likely to have a neighbor authored by a judge from the same law school than if that neighbor had been drawn by random chance. Focusing on this column in particular, the most-represented law school in the sample makes up only 15.7% of the observations, so this effect represents at least a 20% increase over the base rate, and much more for less-frequent law schools. Also striking are the share of opinions with de-meaned match scores greater than 0, i.e., the share of opinions whose neighbors are more similar to them than is the sample as a whole. Slightly less than half of opinions written by Democratic judges have more Democratic-authored neighbors than Republican-authored neighbors. In contrast, almost 70% of opinions authored by a judge from a given law school or of a given gender have more shared-characteristic neighbors than random chance would predict. All results hold across different numbers of neighbors—including in the 10,000-neighbor sample, where the neighborhood is about 4% of the full sample—and regardless of whether or not those neighbors include other opinions by the same author.

The results on legal topic clearly show that local similarity in topic space is well-captured by semantic embeddings, further validating the NLP approach to this question. Using those embeddings to compare partisan and non-partisan judge characteristics shows that opinions' neighbors are essentially random with respect to party, corroborating the reduced-form results on topic distributions in the previous section. Importantly, this new approach allows me to quantify the degree to which shared law schools affect local similarity. While the effect is an order of magnitude smaller than that of legal topic, it is still comparable in magnitude to the effect of gender. Given the strong reduced-form findings for the effect of gender on opinion text, these results suggest that law school training is an important driver of how opinions are written. Thus, judges in the law school minority are likely to find themselves disagreeing with the lead opinion even when they are unwilling to dissent.

9. CONCLUSION

Appellate court rulings are in many ways an ideal setting for research in political economy: they are high-volume, high-impact, and richly documented, but present important challenges due to the technical nature of many appellate rulings. In this work, I complement existing work on partisan outcomes of appellate panels by documenting and explaining a key empirical

pattern: partisanship causes an increase in dissents, but that increase is distributed evenly across all judges on a panel, not just the partisan minority. This combination of facts adds a new dimension to the well-documented claims of increased judicial polarization in the 21st century. I also show that it is unique to partisanship; other important judge characteristics have a more limited effect on total dissents and concentrate that effect among outlier judges. While the behavior along those non-partisan dimensions can be rationalized by classical median-voter or sincere dissent frameworks with sufficient noise in preferences, I show that even with very large levels of noise the effect of partisanship cannot be explained by either of those two approaches.

I instead describe a framework where multidimensional judges have preferences over majority coalitions based on the similarity of the judges in the coalition. To explain the results, this framework requires non-partisan effects to dominate partisan ones. I provide suggestive evidence that this dominance does indeed occur by showing that dissents on bipartisan panels are higher among law school and gender outliers, though reduced sample size limits the strength of these findings. Moving beyond observed dissents, I contribute novel applications of NLP tools to capture the legal topics discussed in appellate opinions, and show that the effect of gender is indeed larger than that of party. The flexibility and informativeness of text embeddings also allow me to compare the full content of opinions, not just reduced-form measures like predicted topics or partisanship. Local similarity metrics in topic space show that judges from the same law school cover similar topics, and benchmarks the magnitude of that effect favorably against those of other demographic variables. Supported by both classical metadata and novel text data, I conclude that non-partisan characteristics do indeed have a strong influence on opinions, and thus on judges' decision to dissent.

This work focused on appellate courts largely in isolation; a key direction for future work is to consider them as part of the hierarchical U.S. court system and combine appellate court data with information about federal trial courts and the Supreme Court. In line with recent work by [Cohen and Dehejia \(2024\)](#), I find that bipartisan panels are more likely to uphold a decision by a 2-1 vote than to reverse it by the same margin, and that unanimous reversals are slightly decreased.²⁹ That work finds that appellate judges are more deferential to lower

²⁹See Table 27 in Appendix E.3 for details.

court judges of their own party; combined with the intuitive assumption that reversing a decision requires greater agreement and a stronger argument than upholding one, I obtain my empirical result. An interesting test of this reasoning would be to combine the appellate court data with information about the party (and possibly ideology score) of the lower-court judges to test whether the dissenting judge in a 2-1 reversal is more likely to be of the same party as the lower court judge. I could also use text similarity measures combined with automatic NLP-based sentiment analysis to assess the degree to which an appellate ruling favorably discusses and extends the lower court’s reasoning. In general, the rapidly advancing state of the art in NLP offers a variety of avenues along which to extend my results. One immediately promising methodological approach is to directly augment text with metadata. This can be done either by concatenating a BERT-like transformer-generated embedding with encoded information about panel and judge characteristics and jointly optimizing the resulting vector for classification tasks, or by including those characteristics in few-shot prompts to a large language model. Beyond these richer embeddings, approaches like sentiment analysis and automated argument detection can provide more explainable descriptions of how opinions differ across judge characteristics and bring insights into the intricacy of their reasoning. These innovations, and surely many others, represent not only important contributions to research in political economy, but also key steps in advancing public understanding and monitoring of a well-functioning judiciary.

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TABLES AND FIGURES FROM THE MAIN TEXT

TABLE 1. Summary Statistics for Opinion Text

Type	Total	Has Text	Known Author	> 500 Words	> 1000 Words	> 5000 Words
Lead/Majority	387,898	313,745	274,416	251,720	224,270	24,842
Dissenting	31,433	31,116	30,734	22,431	15,746	1,694
Concurring	21,361	16,005	16,005	8,499	5,329	560

My sample contains the universe of published cases from U.S. federal courts of appeals, 1890–2013. Of particular interest is the period 1970–2013, when ideology scores are available for all judges and partisan legal positions are relatively stable; this time period comprises approximately 70% of cases for which I have metadata.

TABLE 2. Polarization Causes Total Dissents to Increase

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Party	JCS Scores			DIME-CF Scores		
	All Panels	All Panels	Party-Unan.	Bip.	All Panels	Party-Unan.	Bip.
bipartisan	0.232*** (0.023)						
extremal dist.		0.610*** (0.049)	0.096 (0.150)	0.381*** (0.084)	0.223*** (0.025)	0.021 (0.062)	0.065* (0.035)
2nd-extremal dist.		0.088 (0.086)	−0.117 (0.271)	0.056 (0.103)	0.169*** (0.055)	−0.106 (0.159)	0.129** (0.063)
<i>p</i> for coeff. diff.	—	0.000	0.527	0.013	0.393	0.487	0.409
Control dissents	0.084	0.077	0.077	0.077	0.079	0.078	0.077
Score std. dev.	—	0.388	0.382	0.386	0.704	0.672	0.697
Observations	257,712	150,841	39,095	89,140	155,129	40,471	91,595

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Results from conditional logit estimation of dissent rates using circuit $\times$ year fixed effects (robust standard errors clustered at the circuit $\times$ year level), following Equation (2) for column (1) and Equation (3) for columns (2)–(7). The latter group of columns is divided into three estimates using JCS scores to measure distances between judges—columns (2)–(4)—and three columns repeating the analysis with DIME-CF scores—columns (5)–(7). *p*-values test for equality between extremal distance and second-extremal distance coefficients using a Wald test. Control dissent rates for columns using scores are computed by averaging panels in the sample where the extremal distance to the median is < 0.25 .

TABLE 3. Polarization Causes Equally Increased Dissents Across Judges

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
		JCS Scores				DIME-CF Scores			
Outcome	Variable	All Panels	All Panels	Party-Unan.	Bip.	All Panels	All Panels	Party-Unan.	Bip.
dissent by median judge	bipartisan	0.320*** (0.035)				0.310*** (0.035)			
	ext. dist.		0.542*** (0.057)	0.329 (0.209)	0.330*** (0.099)		0.229*** (0.031)	0.114 (0.085)	0.067 (0.045)
	2nd-ext. dist.		0.171 (0.119)	-0.279 (0.423)	0.112 (0.144)		0.137** (0.069)	-0.145 (0.206)	0.116 (0.082)
dissent by 2 nd -extremal judge	bipartisan	0.333*** (0.035)				0.322*** (0.035)			
	ext. dist.		0.503*** (0.057)	0.014 (0.213)	0.299*** (0.100)		0.182*** (0.032)	-0.094 (0.095)	0.057 (0.046)
	2nd-ext. dist.		0.040 (0.120)	0.037 (0.414)	-0.029 (0.144)		0.212*** (0.070)	0.189 (0.207)	0.129 (0.082)
dissent by extremal judge	bipartisan	0.311*** (0.035)				0.308*** (0.035)			
	ext. dist.		0.565*** (0.057)	-0.053 (0.206)	0.362*** (0.100)		0.178*** (0.032)	0.020 (0.089)	0.042 (0.046)
	2nd-ext. dist.		0.025 (0.118)	-0.144 (0.408)	0.079 (0.140)		0.106 (0.070)	-0.340 (0.209)	0.100 (0.082)
	bip./ext. = across roles	0.907	0.747	0.398	0.910	0.955	0.463	0.275	0.929
	2nd-ext. = across roles	-	0.644	0.870	0.777	-	0.557	0.203	0.971
	Control dissents	0.042	0.076	0.076	0.096	0.044	0.078	0.073	0.095
	Score std. dev.	-	0.388	0.351	0.397	-	0.704	0.666	0.709
	Observations	248,693	150,900	40,271	89,295	249,043	155,191	41,530	91,768

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Results from conditional multinomial logit estimation of dissent rates using circuit $\times$ year fixed effects (with robust standard errors clustered at the circuit $\times$ year level), following Equation (4) for column (1) and (5) and Equation (5) for columns (2)–(4) and (5)–(8). Columns (1)–(4) use JCS scores to determine median, second-extremal, and extremal judges and to measure distances to the median; columns (5)–(8) repeat that analysis using DIME-CF scores. p -values test for equality between coefficients across judge roles using a Wald test. Control dissent rates for columns using scores are computed as in Table 2, by averaging panels in the sample where the extremal distance to the median is < 0.25 .

TABLE 4. Outliers Cause Minimal Differences in Judge-Level Dissents

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Min. Party	Is JCS Median	Is DIME-CF Median	Min. Law School	Min. Gender	Only Man	Only Woman	Youngest	Median Age	Oldest
judge char.	0.010 (0.022)	0.014 (0.021)	-0.018 (0.023)	0.017 (0.043)	0.126*** (0.041)	-0.037 (0.072)	0.144*** (0.045)	0.020 (0.022)	0.009 (0.017)	0.004 (0.023)
Control dissents	0.031	0.036	0.036	0.031	0.038	0.033	0.032	0.032	0.033	0.033
Observations	520,148	379,597	388,432	175,973	138,036	138,036	138,036	772,230	772,230	772,230
# judges	1,230	829	841	824	648	648	648	1,370	1,370	1,370

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Results from conditional logit estimation of dissent rates at the judge level using circuit $\times$ year and judge fixed effects (with robust standard errors clustered at the judge level), following Equation (6). All specifications include controls for Bloomberg topic, as well as number of Democrats, number of unique law schools, and number of women whenever those are not the characteristic of interest. The sample for each column includes only mixed-characteristic panels (e.g., in column (1), only bipartisan panels). The control dissent rate is computed at the judge level across included panels where the judge does not satisfy the criterion of interest (e.g., in column (1), over judges on bipartisan panels that are not in the partisan minority).

TABLE 5. Majority Dissent Shares Are Nearly Proportional

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Maj. Party	Non-Med. JCS	Non-Med. DIME-CF	Maj. Law School	Maj. Gender	Maj. Man	Maj. Woman	Youngest	Median Age	Oldest
w/ controls	0.663 (0.062) $p = 0.394$	1.003 (0.120) $p = 0.139$	1.003 (0.119) $p = 0.172$	0.666** (0.114) $p = 0.038$	0.631 (0.083) $p = 0.450$	0.621 (0.086) $p = 0.471$	0.682 (0.178) $p = 0.361$	1.017 (0.127) $p = 0.510$	1.018 (0.111) $p = 0.493$	0.983 (0.129) $p = 0.484$
w/o controls	0.663 (0.062) $p = 0.374$	1.004 (0.120) $p = 0.201$	1.001* (0.118) $p = 0.057$	0.667** (0.110) $p = 0.033$	0.637 (0.080) $p = 0.482$	0.630 (0.083) $p = 0.484$	0.657 (0.163) $p = 0.206$	1.011 (0.124) $p = 0.444$	1.018 (0.108) $p = 0.503$	0.985 (0.127) $p = 0.450$
Control dissents	0.083	0.064	0.065	0.066	0.090	0.090	0.132	0.093	0.094	0.093
Observations	261,364	164,593	164,878	76,333	290,447	281,977	6,857	263,097	287,655	260,019

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Ratios of majority characteristic dissents on bi-characteristic panels computed as described in Section 6.3. Coefficients used to compute the level of dissents on bipartisan panels are estimated using a conditional logit following Equation (2) or the analogous Equation (10), with circuit $\times$ year fixed effects and robust standard errors clustered at the circuit $\times$ year level. The row with controls always includes Bloomberg topic, as well as number of Democrats, number of unique law schools, and number of women whenever those are not the characteristic of interest. Standard errors are computed using 1,000 Bayesian bootstrap draws of the data. p -values are for a two-sided test of equality with the proportional value (2/3 for columns (1) and (4)–(7); 1 for the other columns) also conducted using 1,000 Bayesian bootstrap draws of the data.

TABLE 6. Equally Increased Dissents Persist Across Legal Topics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Crim.	Civ. Rights	1st Am.	Due Process	Labor Rel.	Econ. & Reg.	Misc.
median dis.	0.367*** (0.074)	0.291*** (0.108)	0.137 (0.331)	0.311*** (0.065)	0.528*** (0.143)	0.142* (0.076)	0.472*** (0.129)
2nd-ext. dis.	0.444*** (0.072)	0.717*** (0.118)	0.166 (0.389)	0.239*** (0.064)	0.374*** (0.138)	0.160** (0.076)	0.388*** (0.124)
ext. dis.	0.370*** (0.072)	0.466*** (0.116)	-0.056 (0.385)	0.339*** (0.065)	0.155 (0.130)	0.146* (0.075)	0.429*** (0.127)
<i>p</i> for coeff. diff.	0.701	0.032	0.906	0.540	0.156	0.985	0.898
Control dissents	0.046	0.076	0.135	0.054	0.055	0.030	0.065
Observations	56,825	14,744	1,675	61,533	15,766	82,382	15,479

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Results from conditional multinomial logit estimation of dissent rates using circuit $\times$ year fixed effects (with robust standard errors clustered at the circuit $\times$ year level), following Equation (4). Each column restricts the sample to one of the eight Bloomberg Law topics (dropping privacy cases, with a sample of < 500 , because of co-linearity preventing estimation). Coefficients represent the increase in dissents by each judge role on bipartisan panels compared to that same judge role on party-unanimous panels, i.e., the same specification as column (1) of Table 3 under the sample restriction. *p*-values test for equality between coefficients across judge roles using a Wald test.

TABLE 7. Outliers Weakly Increase Total Dissents

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Partisanship	Shared Law School	Any Law Schools	Mixed Gender	Min. Men	Min. Women	Age
bipartisan	0.232*** (0.023)						
one diff. law school		0.056 (0.064)	-0.024 (0.021)				
all same law school			-0.079 (0.066)				
mixed gender				0.076*** (0.026)			
one man					0.083 (0.165)		
one woman						0.067*** (0.025)	
young outlier							0.041 (0.028)
old outlier							0.040 (0.030)
Control dissents	0.084	0.066	0.096	0.094	0.091	0.090	0.090
Observations	257,712	70,875	257,712	257,712	5,454	251,574	257,712

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Results from conditional logit estimation of outlier effects on total dissents using circuit $\times$ year fixed effects (with robust standard errors clustered at the circuit $\times$ year level), following Equation (7). For columns (1), (3), (4), and (7) all panels are included. For column (2), all panels with two or more judges from the same law school are included. For column (5), all majority-women panels are included. For column (6), all majority-men panels are included. On panels with a “young outlier,” the youngest judge is 10 years or more younger than the median judge, who is in turn less than 5 years younger than the oldest judge. Panels with an “old outlier” have the same phenomenon but with the age ordering reversed.

TABLE 8. Outliers Cause Concentrated Increases in Dissents Across Judges

	(1)	(2)	(3)	(4)
	Three Unique Law Schools	Any Age Outlier	Young Outlier	Old Outlier
second-extremal or youngest judge	0.022 (0.028)	0.024 (0.031)	-0.000 (0.040)	0.046 (0.042)
median judge	0.046 (0.028)	0.009 (0.031)	0.016 (0.039)	-0.001 (0.042)
extremal or oldest judge	0.025 (0.028)	0.072** (0.030)	0.076** (0.038)	0.049 (0.042)
p , all equal	0.819	0.322	0.361	0.649
Control dissents	0.099	0.094	0.093	0.093
Observations	250, 595	259, 821	259, 821	259, 821

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Results from conditional multinomial logit estimation of dissent rates using circuit $\times$ year fixed effects (with robust standard errors clustered at the circuit $\times$ year level), following Equation (8) for column (1) and Equation (9) for columns (2)–(4). The sample for column (1) includes only panels with at least two unique law schools; coefficients represent the increase in dissents on panels with three unique law schools compared to two. In columns (2)–(4), coefficients represent the increase in dissents on panels where the specified age outlier is present. The reported p -values test for equality between coefficients within a column using a Wald test.

TABLE 9. Partisan Polarization Slightly Increases Minority-Characteristic Dissents

	(1)	(2)	(3)	(4)	(5)
	Law School	Gender (Pooled)	Gender (Men)	Gender (Women)	Age
minority / youngest	0.343*** (0.083)	0.253*** (0.073)	1.096*** (0.245)	0.141* (0.078)	0.345*** (0.038)
majority / age-median	0.250*** (0.059)	0.285*** (0.056)	0.258*** (0.062)	0.426*** (0.141)	0.312*** (0.039)
oldest	—	—	—	—	0.316*** (0.039)
<i>p</i> for coeff. diff.	0.372	0.740	0.001	0.079	0.806
Control dissents	0.080	0.095	0.095	0.095	0.083
Observations	65,368	48,610	48,610	48,610	259,821

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Results from conditional multinomial logit estimation of dissent rates using circuit $\times$ year fixed effects (with robust standard errors clustered at the circuit $\times$ year level), splitting dissents by panel-majority and panel-minority characteristic, or by age. The sample in column (1) includes only panels with exactly two judges from the same law school, and in columns (2)–(4) includes only mixed-gender panels. *p*-values test for equality between coefficients in the same column using a Wald test.

TABLE 10. Comparing Topic Use Across Panel Types, No Author Controls

Panel type	Variable	(1) Crim.	(2) Civ Rights	(3) 1A	(4) Due Process	(5) Priv.	(6) Lab. Rel.	(7) Econ. & Reg.	(8) Misc.	(9) Unif.	(10) Bayes	Observations
unanimous	Dem.	−0.014 (0.035)	0.031 (0.034)	0.247** (0.120)	−0.004 (0.020)	−0.000 (0.163)	0.031 (0.048)	−0.064** (0.025)	0.166*** (0.049)	0.001 (0.006)	0.013* (0.007)	49,098
	Control mean	0.279	0.071	0.006	0.297	0.001	0.057	0.236	0.052	1.753	1.635	
majority-Dem.	3-0 Dem.	−0.001 (0.023)	0.007 (0.024)	0.113 (0.074)	−0.004 (0.013)	0.011 (0.108)	0.030 (0.033)	−0.043** (0.018)	0.095*** (0.034)	−0.006 (0.004)	0.006 (0.005)	61,538
	Control mean	0.257	0.074	0.009	0.279	0.001	0.067	0.247	0.066	1.782	1.660	
majority-Rep.	2-1 Rep.	−0.018 (0.017)	0.007 (0.016)	0.026 (0.059)	0.001 (0.010)	0.038 (0.079)	0.012 (0.020)	0.000 (0.012)	0.036 (0.024)	0.004 (0.003)	0.005 (0.003)	101,857
	Control mean	0.279	0.071	0.006	0.297	0.001	0.057	0.236	0.052	1.753	1.635	
bipartisan	Dem. majority	−0.001 (0.016)	0.008 (0.015)	0.080* (0.049)	0.005 (0.009)	0.009 (0.069)	−0.011 (0.020)	−0.015 (0.012)	0.024 (0.021)	0.003 (0.003)	0.022 (0.020)	114,301
	Control mean	0.266	0.072	0.007	0.296	0.001	0.06	0.242	0.055	1.771	1.649	

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Results from conditional logit estimation of predicted topic probabilities (columns (1)–(8)) and Poisson estimation of major topic counts (columns (9) and (10)) using circuit $\times$ year fixed effects (with robust standard errors clustered at the circuit $\times$ year level), following Equation (12). Coefficients represent the effect of making a panel “more Democratic” within the given category (e.g., in the first section, all-Democratic as opposed to all-Republican).

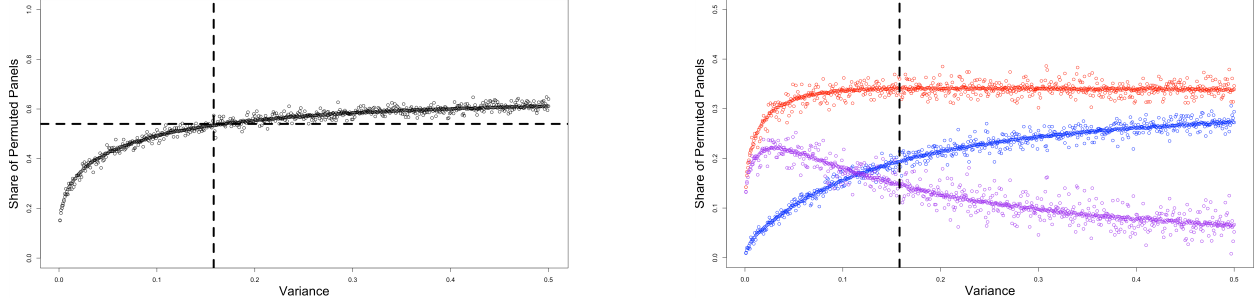
TABLE 11. Non-Partisan Characteristics Influence Opinion Similarity

	(1)	(2)	(3)	(4)	(5)	(6)
	Topic	Author Party	Panel Majority	# Dems.	Law School	Gender
Normalized match score	0.249*** (0.00036)	-0.014*** (0.00022)	-0.199*** (0.00058)	-0.016*** (0.00019)	0.007*** (0.00005)	0.015*** (0.00160)
De-meaned match score	0.401*** (0.00058)	-0.010*** (0.00015)	-0.136*** (0.00037)	-0.021*** (0.00023)	0.027*** (0.00018)	0.004*** (0.00048)
Share de-meaned > 0	0.881	0.438	0.186	0.291	0.682	0.540
Raw match share	0.633	0.495	0.372	0.271	0.074	0.852
Observations	270,451	264,672	246,675	246,675	273,715	274,158

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

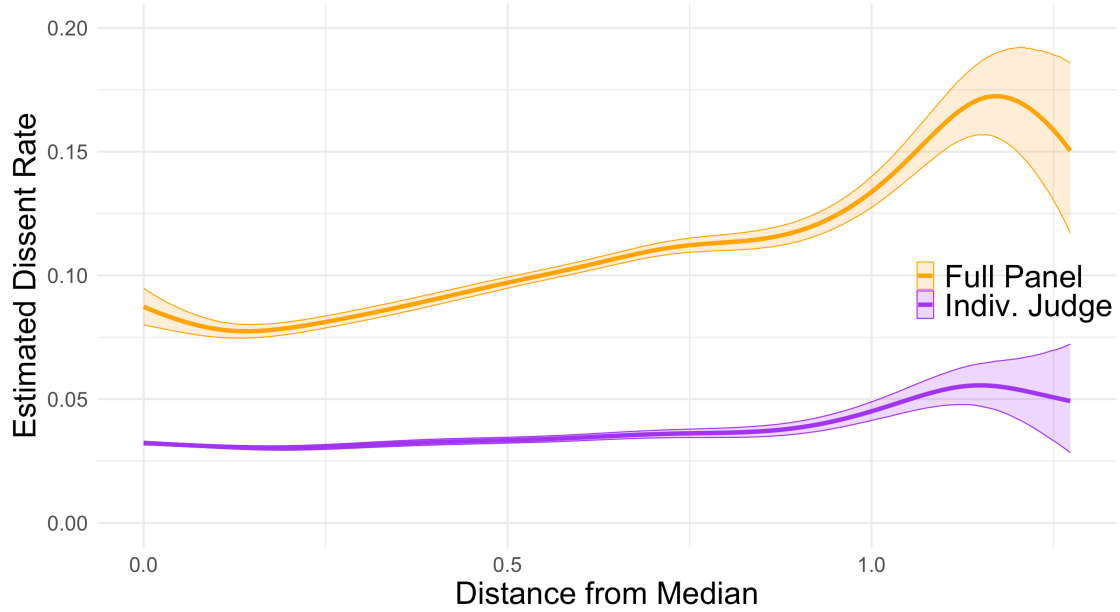
Average shares of 1,000 nearest neighboring opinions matching a given own-opinion characteristic under various normalizations; de-meaning (subtracting the share of opinions matching own characteristics in the full sample) and normalization (de-meaning and division by the entropy of the distribution of characteristics). Standard errors are computed via bootstrap; I draw a new dataset without replacement and re-compute each measure, repeat 10,000 times, and report the standard deviation of the resulting distribution. To calculate statistical difference from 0, I use the share of bootstrap samples with the opposite sign as the observed coefficient, e.g., if 5% of bootstrap samples are weakly positive in column (2), then the coefficient is significant at the $5 \times 2 = 10\%$ level. In fact, no bootstrap draws fit this criterion for any coefficient.

FIGURE 1. Shares of Judge Order Permutations As a Function of Noise Variance



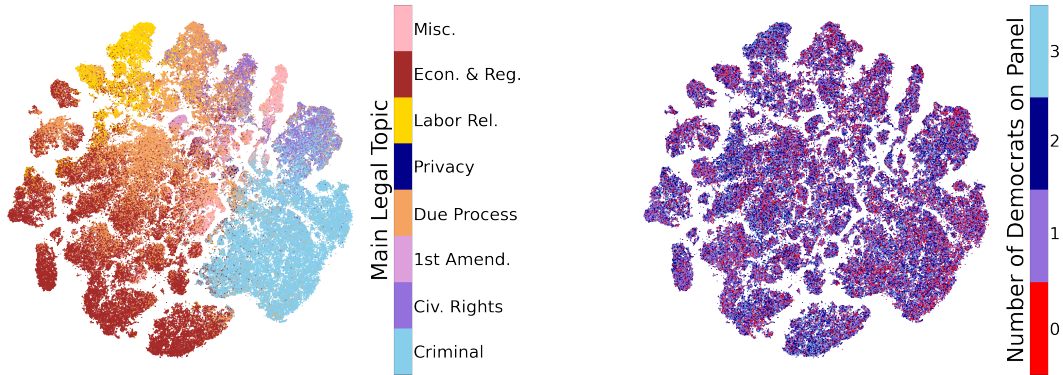
Simulations of judge order with additive i.i.d. $N(0, \sigma^2)$ noise for various values of σ^2 . To produce each point, I draw 100,000 panels at random from my data, add independently drawn realizations of $N(0, \sigma^2)$ to each of the judges' JCS scores, then compute the share of post-noise panel draws where the original median, second-extremal, and extremal judges (using original JCS score) are the post-noise median judge. In the left image, I plot the combined share of panels where the original median judge is *not* the post-noise median judge. The dashed vertical line marks the sample variance of JCS scores; the dashed horizontal line marks the share of permuted panels needed to attain the 95% confidence bound for the median judge's dissent rate. In the right panel, I plot the share of panels where the original second-extremal judge is the post-noise median judge (red); the share of panels where the original extremal judge is the post-noise median judge (blue); and the difference in those two shares (purple). The dashed vertical line again marks the sample variance of JCS scores.

FIGURE 2. Panel and Individual Dissents as a Function of Distance to Panel Median



Local polynomial regression of dissents as a function of median distance using a local fit of degree 1 and a bandwidth of 0.1. The orange line uses a panel-level specification where the presence of a dissent by any judge on the panel is plotted as a function of the extremal judge's distance from the median. The purple line uses a judge-level specification where each judge's own dissent is plotted as a function of their individual distance from the median. 95% confidence intervals, shown by the shaded areas surrounding each line, are constructed by re-estimating the local polynomial on 10,000 bootstrap samples of the data and plotting the pointwise upper and lower quantiles of those bootstrap results.

FIGURE 3. Opinion Embeddings Cluster by Topic, Not by Partisanship

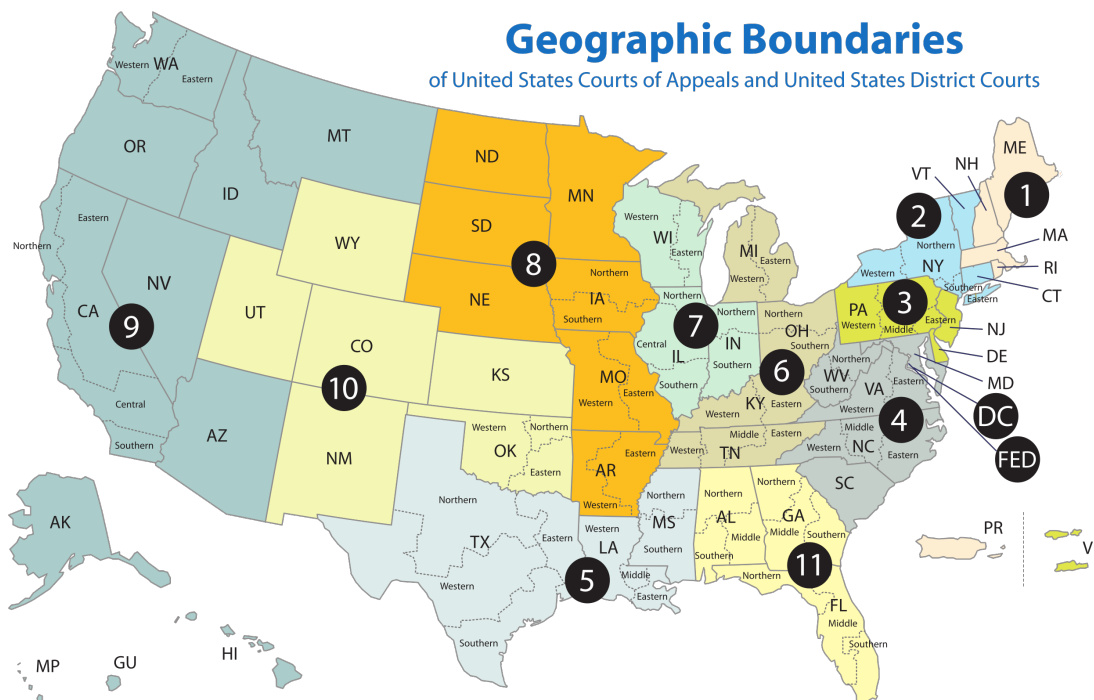


Opinion embeddings visualized using [openTSNE](#) in Python. Following package guidelines I use PCA to reduce the 768-dimensional semantic embeddings to 50 dimensions, then apply t-SNE to represent those compressed vectors in two dimensions. I set t-SNE perplexity to 50. The left image colors by main legal topic following the true Bloomberg labels (criminal, civil rights, First Amendment, due process, privacy, labor relations, economic activity and regulation, and miscellaneous). The right image colors by the number of Democrats on the panel (0–3).

APPENDIX A. DATA VISUALIZATIONS AND DESCRIPTIVE STATISTICS

In this appendix, I present additional information about the appellate court data described in Section 4. Appendix A.1 includes information on opinion data, Appendix A.2 includes information on judges and panel compositions, and Appendix A.3 includes information on ideology scores. Before those sections, though, I include a figure showing the geographic locations covered by different appellate court circuits.

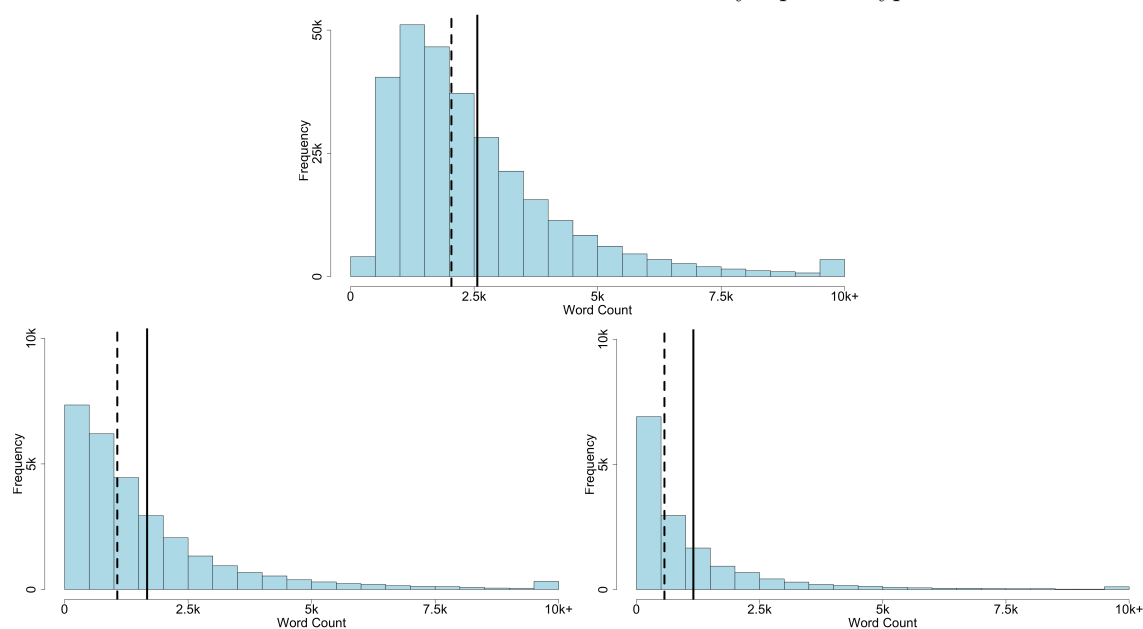
FIGURE 4. Official Map of District and Circuit Courts



A map of the thirteen circuit courts of appeals and the district courts they cover. Source: [US Federal Government](#).

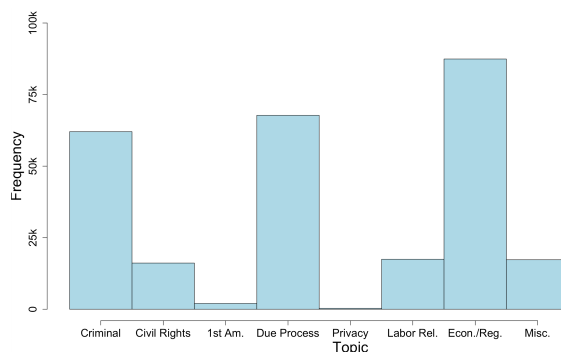
A.1. Opinion Characteristics.

FIGURE 5. Distribution of Word Counts by Opinion Type



In the top center image, word counts for lead/majority opinions where text is available (290,876 opinions); a black solid line marks the mean (2565 words) and a black dashed line marks the median (2401 words). In the bottom left image, the same for dissenting opinions (28,485 opinions; mean 1670 words; median 1067 words). In the bottom right image, the same for concurring opinions (14,879 opinions; mean 1156 words; median 569 words).

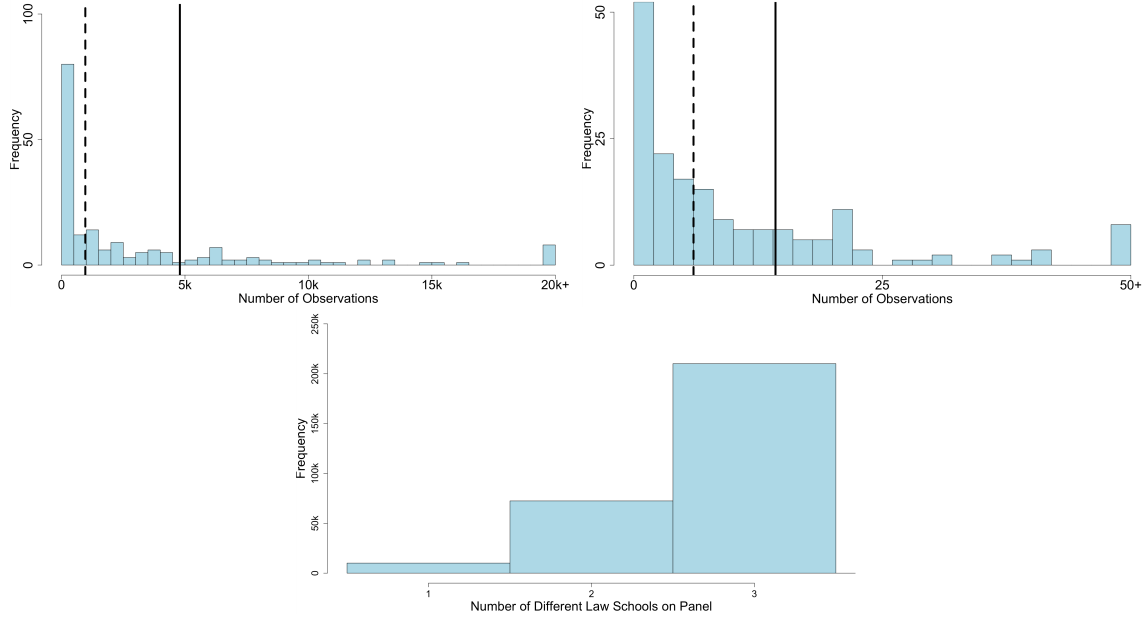
FIGURE 6. Distribution of Legal Topics



Primary legal topics for the 270,451 lead opinions where text and topic are available, as coded by Bloomberg Law. Criminal, due process, and economics and regulation cases dominate.

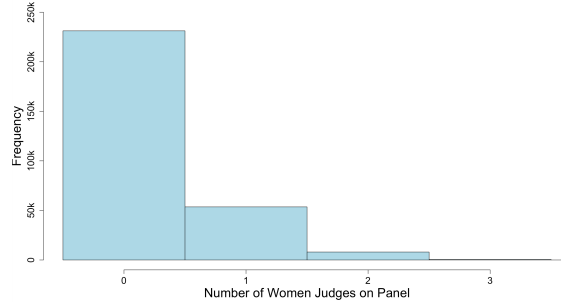
A.2. Judge and Panel Characteristics.

FIGURE 7. Distributions of Judge Law Schools



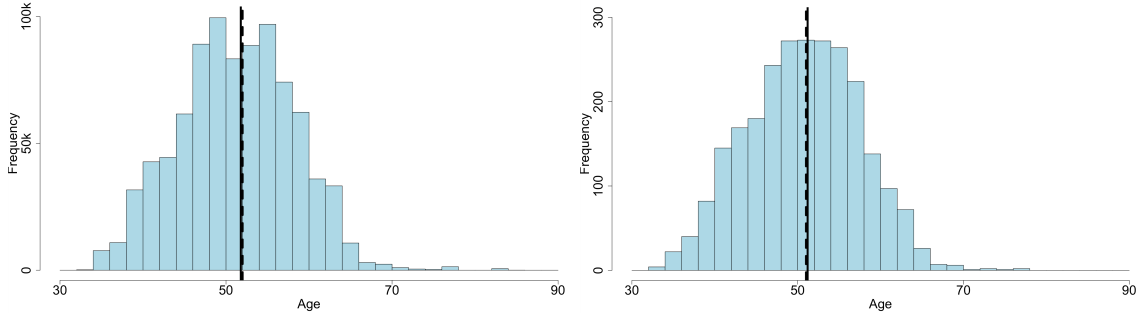
In the top left image, the number of times each of 178 unique law schools appears in the 881,945 observations of law schools on panels with known authors. A black solid line marks the mean (4793), and a black dashed line marks the median (964). In the top right image, the number of times each law school appears across the 2,536 unique judges in that sample (mean 14, median 6). In the bottom center image, the number of panels with known author where one, two, and three different law schools represented (292,636 panels).

FIGURE 8. Distribution of Number of Women on Panels



Number of women for the 293,235 panels where author and gender are available. 8.1% of the 882,554 observations of judges and 11.2% of the 2,539 unique judges are women.

FIGURE 9. Distribution of Judge Ages

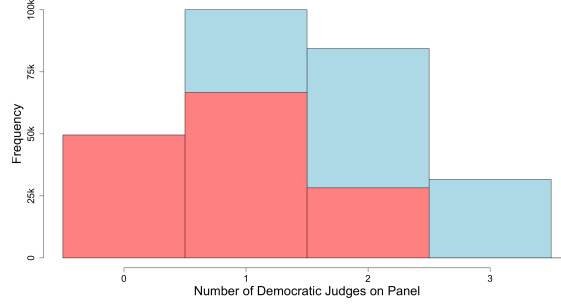


In the left image, judge ages for the 882,554 observations of judges on panels with known authors; a black solid line marks the mean (51.78), and a black dashed line marks the median (52); the standard deviation is 6.95. In the right image, judge ages for the 2,542 unique judges in the sample of panels with known authors (mean 51.20, median 51, standard deviation 6.79).

A.3. Judge Ideology Scores. I first describe in more detail the three measures of judge ideology used in this work, then present visualizations of those measures.

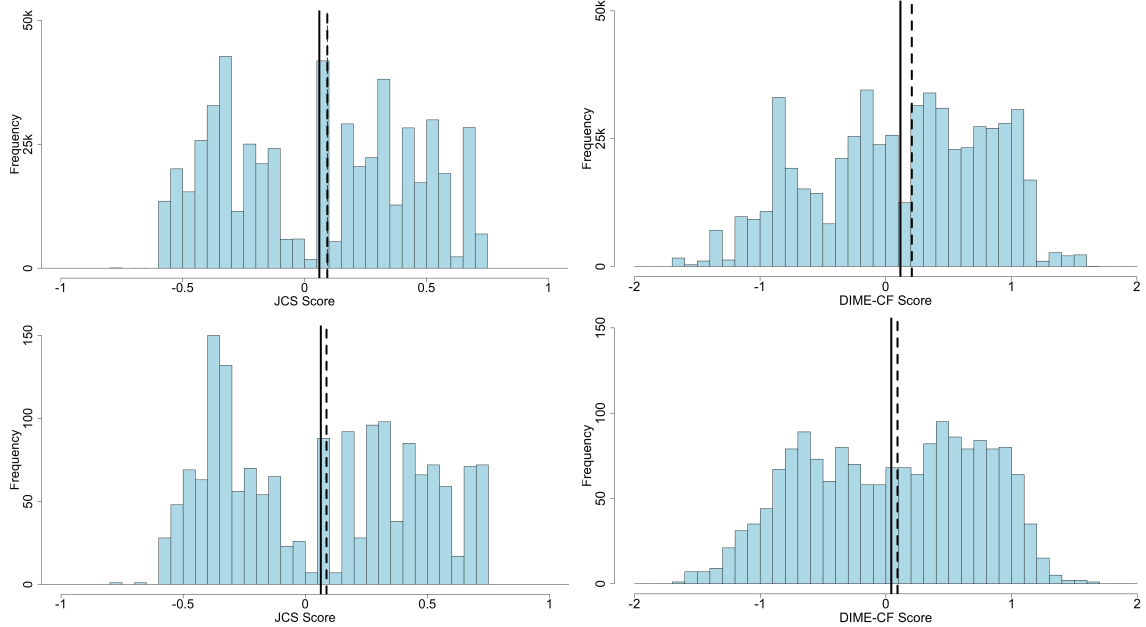
- Self-reported party; available for almost all judges. This value is coarse but model-free. In all but 46 of the over one million observations, this value matches the party of the appointing president (another standard measure in the literature).
- Judicial Common Space (JCS) scores, from [Epstein et al. \(2007\)](#); available from 1970 onward. These scores are based on home-state senator influence in appellate appointments (see Section 3.1). JCS scores average the DW-Nominate score of the appointing president with those of home-state senators in the president’s party. DW-Nominate scores are from [Poole and Rosenthal \(1985\)](#), who estimate a structural model of Congressional votes; legislators who vote similarly have similar ideal points.
- Database on Ideology, Money in Politics, and Elections—Campaign Finance (DIME-CF) scores, from [Bonica and Sen \(2017\)](#); available from 1970 onward. These scores are based on linking judges (and other officials) to their political contributions and applying ideal-point estimation in the style of DW-Nominate. Correlation with JCS and DW-Nominate scores is high, but the authors claim DIME-CF scores better capture judge preferences rather than political deals in the appointment process.

FIGURE 10. Distribution of Number of Democrats on Panels



Number of Democrats on a panel for the 265,592 panels where author and panel parties are available. Stacked blue and red bars represent the share of opinions authored by Democrats and Republicans, respectively. 45.2% of the 834,153 observations of judges, and 45.3% of the 2,407 unique judges, in this sample are Democrats.

FIGURE 11. Distributions of Judge Ideology Scores



In the top images, JCS and DIME-CF scores for the 547,738 and 554,930 observations, respectively, of those scores on panels with known authors. A black solid line marks the mean, and a black dashed line marks the median. Observed JCS scores are roughly bimodal (mean 0.060, variance 0.148), while observed DIME-CF scores are roughly uniform (mean 0.117, variance 0.468). The bottom images replicate the analysis for unique judges (samples of 1,682 and 1,698 respectively); JCS scores are again roughly bimodal (mean 0.064, variance 0.158), while DIME-CF scores now appear bimodal (mean 0.043, variance 0.478).

APPENDIX B. QUASI-RANDOM ASSIGNMENT OF JUDGES

In Appendix B.1, balance tables 12 – 14 aggregate across the full sample to show that the share of cases in each of the eight Bloomberg Law topics is relatively consistent across panel types, and that law school and gender composition is roughly orthogonal to partisanship. Next, in Appendix B.2, I generate—using 1,000 Bayesian bootstrap draws of the set of observed panels—distributions of the shares of panels with 0–3 Democrats; 1–3 unique law schools; and 0–3 women. Those distributions are shown in figures 12–14. Finally, in Appendix B.3, I describe and implement a rigorous procedure to test for random assignment at the individual circuit $\times$ year level; results are in figures 15–17.

B.1. Balance Tables for Topics and Panel Characteristics. Looking first to demographics (tables 12 and 13), the balance tables show a slightly higher probability for women judges to be Democrats, and for panels with judges from three different law schools to have larger numbers of Democrats. Since there are more panels in the sample with a Democratic judge in the minority than with a Republican judge in the minority, these findings translate into a mild negative correlation between bipartisan panels and panels with judges of three different law schools, as well as a mild positive correlation between bipartisan panels and panels with no women. In the framework of Section 5.3, the former slightly weakens the degree to which law school must outweigh party, since double-minority judges across party and law school are more rare than if those characteristics were entirely orthogonal. By the same argument, it slightly strengthens the degree to which gender must outweigh party.

In Table 14, the share of cases in each major topic (criminal, due process, economics and regulation) vary only slightly with panel composition. Panels with more Democrats have slightly fewer cases in the former two topics and slightly more in the latter; the same is true of panels with fewer unique law schools. Panels with more women have more criminal and due process cases and fewer economics and regulation cases. These topic codes take into account both the facts of the case (i.e., pre-trial information) and the ultimate opinion, which may account for some of these trends. Importantly, balance is much stronger when comparing bipartisan to party-unanimous panels, as well as comparing majority-Democrat

to majority-Republican panels. These results suggest that differences in case topics are not driving differences in aggregate panel behavior.

TABLE 12. Distribution of Panels by Number of Law Schools

	Panels w/o Missing Law School	3 Diff. Law Schools	2 Diff. Law Schools	1 Law School
Full sample	292636	0.717	0.248	0.035
Sample w/o missing party	264,993	0.712	0.252	0.036
Panels w/ 0 Dems.	49,323	0.693	0.264	0.043
Panels w/ 1 Dem.	99,684	0.703	0.249	0.048
Panels w/ 2 Dems.	84,365	0.720	0.253	0.027
Panels w/ 3 Dems.	31,621	0.756	0.235	0.009
Bipartisan panels	184,049	0.711	0.251	0.038
Party-unanimous panels	80,944	0.717	0.253	0.030
Majority-Dem. panels	115,986	0.730	0.248	0.022
Majority-Rep. panels	149,007	0.700	0.254	0.046
Sample w/o missing gender	292,636	0.717	0.248	0.035
Panels w/ 0 women	230,580	0.694	0.264	0.042
Panels w/ 1 women	53,550	0.798	0.194	0.008
Panels w/ 2 women	8,037	0.847	0.150	0.003
Panels w/ 3 women	469	0.889	0.111	0

TABLE 13. Distribution of Panels by Number of Women

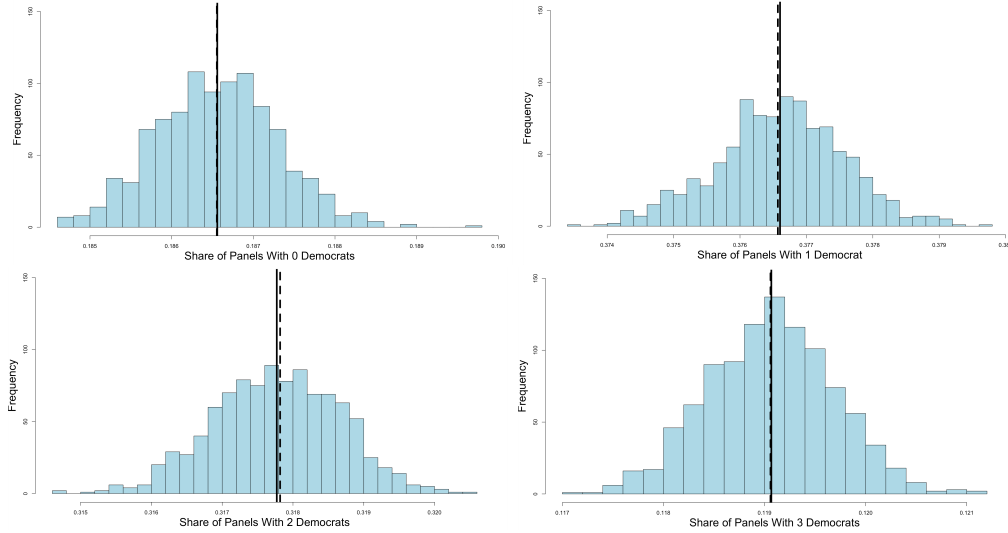
	Panels w/o Missing Gender	0 Women	1 Woman	2 Women	3 Women
Full sample	292,636	0.79	0.183	0.027	0.002
Sample w/o missing party	264,993	0.816	0.163	0.022	0.001
Panels w/ 0 Dems.	49,323	0.827	0.161	0.017	0.001
Panels w/ 1 Dem.	99,684	0.802	0.177	0.023	0.001
Panels w/ 2 Dems.	84,365	0.811	0.164	0.025	0.001
Panels w/ 3 Dems.	31,621	0.861	0.118	0.02	0.001
Bipartisan panels	184,049	0.806	0.171	0.024	0.001
Party-unanimous panels	80,944	0.840	0.144	0.018	0.001
Majority-Dem. panels	115,986	0.824	0.151	0.024	0.001
Majority-Rep. panels	149,007	0.810	0.171	0.021	0.001

TABLE 14. Distribution of Case Topics Across Panel Types

	Cases w/o missing topic	Crim.	Civ. Rights	1st Am.	Due Process	Priv.	Labor Rel.	Econ. & Reg.	Misc.
Full sample	289,894	0.231	0.060	0.007	0.249	0.001	0.064	0.323	0.064
Sample w/o missing party	260,829	0.227	0.059	0.007	0.245	0.001	0.064	0.335	0.062
Panels w/ 0 Dems.	48,932	0.251	0.063	0.006	0.262	0.001	0.057	0.297	0.063
Panels w/ 1 Dem.	98,043	0.232	0.059	0.006	0.252	0.001	0.060	0.327	0.062
Panels w/ 2 Dems.	82,953	0.218	0.058	0.007	0.236	0.001	0.068	0.349	0.062
Panels w/ 3 Dems.	30,901	0.198	0.053	0.008	0.221	0.001	0.077	0.379	0.063
Bipartisan panels	180,996	0.226	0.059	0.007	0.245	0.001	0.064	0.337	0.062
Party-unanimous panels	79,833	0.230	0.059	0.007	0.246	0.001	0.065	0.329	0.063
Majority-Dem. panels	113,854	0.213	0.057	0.007	0.232	0.001	0.070	0.357	0.062
Majority-Rep. panels	146,975	0.238	0.060	0.006	0.255	0.001	0.059	0.317	0.062
Sample w/o missing gender	288,404	0.231	0.06	0.007	0.249	0.001	0.064	0.323	0.064
Panels w/ 0 women	226,553	0.224	0.056	0.006	0.235	0.001	0.065	0.353	0.059
Panels w/ 1 woman	53,375	0.257	0.077	0.010	0.300	0.002	0.059	0.219	0.077
Panels w/ 2 women	8,008	0.257	0.077	0.011	0.316	0.003	0.054	0.185	0.097
Panels w/ 3 women	468	0.250	0.085	0.009	0.297	0.002	0.051	0.203	0.103
Sample w/o missing law school	287,823	0.231	0.060	0.007	0.250	0.001	0.064	0.323	0.064
Panels w/ 3 diff. law schools	206,650	0.241	0.065	0.008	0.260	0.001	0.067	0.293	0.065
Panels w/ 2 diff. law schools	71,286	0.214	0.053	0.006	0.229	0.001	0.058	0.378	0.061
Panels w/ 1 law school	9,887	0.144	0.030	0.003	0.188	0.001	0.032	0.548	0.055

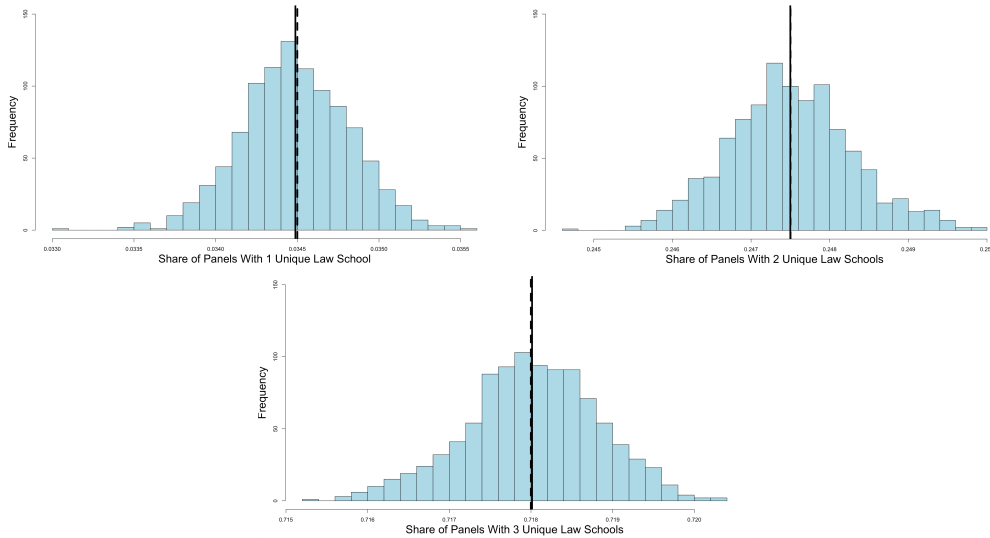
B.2. Panel-Level Simulations with Aggregate Data.

FIGURE 12. Distributions of Aggregate Shares of Panel Partisanship



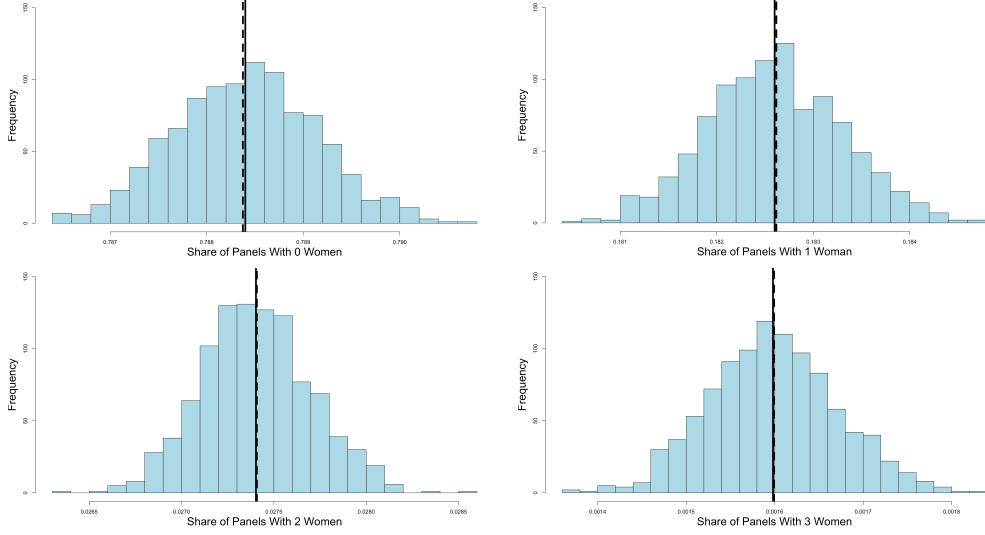
Shares of panels with 0–3 Democrats, sampled at the panel level. The bootstrap means (solid lines) are 0.1866, 0.3766, 0.3178, and 0.1191 respectively; the observed shares (dashed lines) are 0.1866, 0.3766, 0.3178, and 0.1191 respectively.

FIGURE 13. Distributions of Aggregate Shares of Panel Law Schools



Shares of panels with 1–3 unique law schools, sampled at the panel level. The bootstrap means (solid lines) are 0.0345, 0.2475, and 0.7180 respectively; the observed shares (dashed lines) are 0.0345, 0.2475, and 0.7180 respectively.

FIGURE 14. Distributions of Aggregate Shares of Panel Genders



Shares of panels with 0–3 women, sampled at the panel level. The bootstrap means (solid lines) are 0.7884, 0.1826, 0.0274, and 0.0016 respectively; the observed shares (dashed lines) are 0.7884, 0.1826, 0.0274, and 0.0016 respectively.

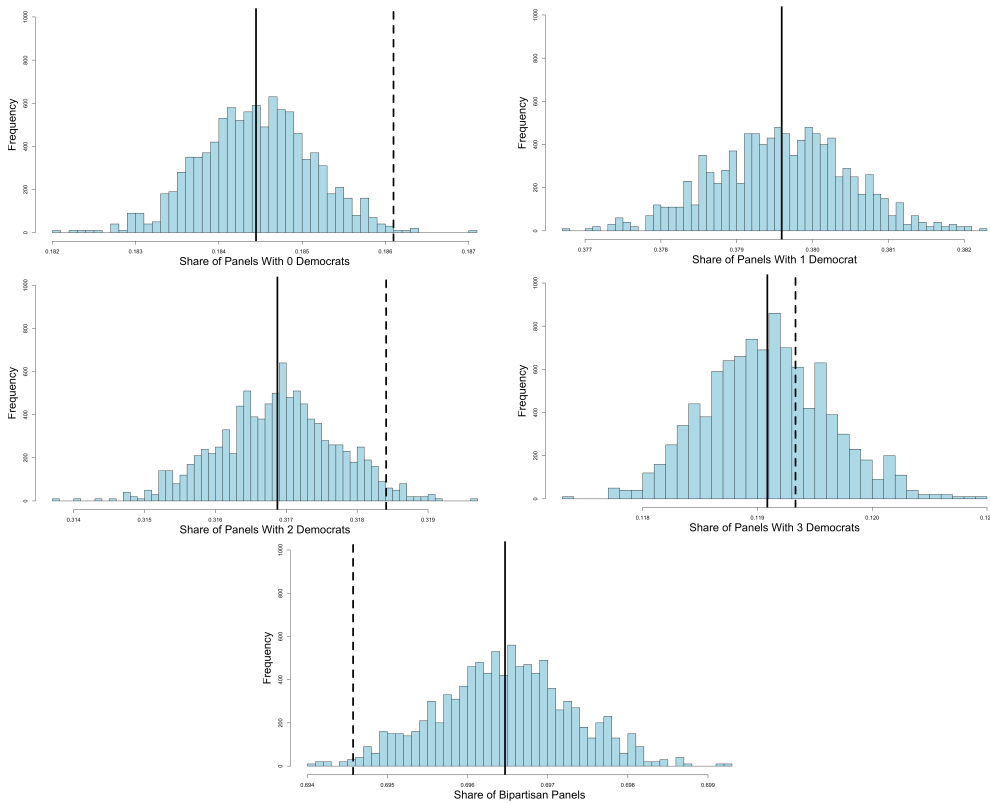
B.3. Judge-Level Simulations by Circuit $\times$ Year. Aggregate balance checks may mask non-random assignment within particular circuit $\times$ year pairs ct . To address this concern, I describe a procedure for sampling from the distribution of judges in each ct , using number of Democrats on a panel as an example. (It is the same for number of unique law schools and women.) For each ct , I do the following:

- (1) Compute the distribution F_{ct} of appearances by unique judges in the N panels on ct .
- (2) Draw a panel p_1 of 3 judges without replacement from F_{ct} . Draw N independent panels $p_1, \dots, p_N$ (replacing judges between panel draws) to form a simulated ct , s_{ct}^1 .
- (3) Draw 1,000 independent simulations $s_{ct}^1, \dots, s_{ct}^{1000}$. For each s_{ct}^i , compute
 - the share of those n panels with 0, 1, 2, and 3 Democratic judges.
 - the share of those n panels with 1, 2, and 3 unique law schools.
 - the share of those n panels with 0, 1, 2, and 3 women.
- (4) For each of those 11 shares, form an empirical distribution G using the s_{ct}^i .

The procedure can be used to return a list of all *ct* where the observed quantile indicates statistically significant deviations from random assignment; below I report the aggregate shares across all *ct*. It requires hundreds of millions of random draws, but does not consider case topics or joint distributions; e.g., panels with 2 Democratic judges, one of whom is a woman and shares a law school with the Republican man. Since these tests are already much more thorough than any previous economics literature and show only small deviations from random assignment, I rely on the aggregate balance checks to test these cases.

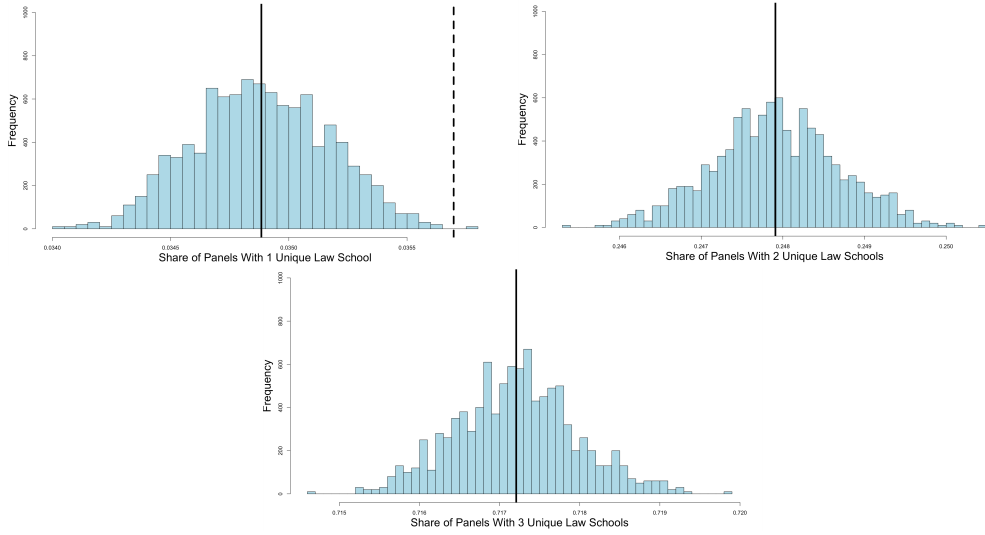
Figures 15–17 show some statistically significant deviations from random assignment, but the magnitudes are small and sensitive to data with missing observations. No share of Democrats or unique law schools differs by more than 0.5 percentage points from the simulated mean; differences for shares of women judges are slightly larger but driven almost entirely by a low observed share of panels with two women judges.

FIGURE 15. Distributions of Aggregate Shares of Panel Partisanship, Judge-Level Simulations



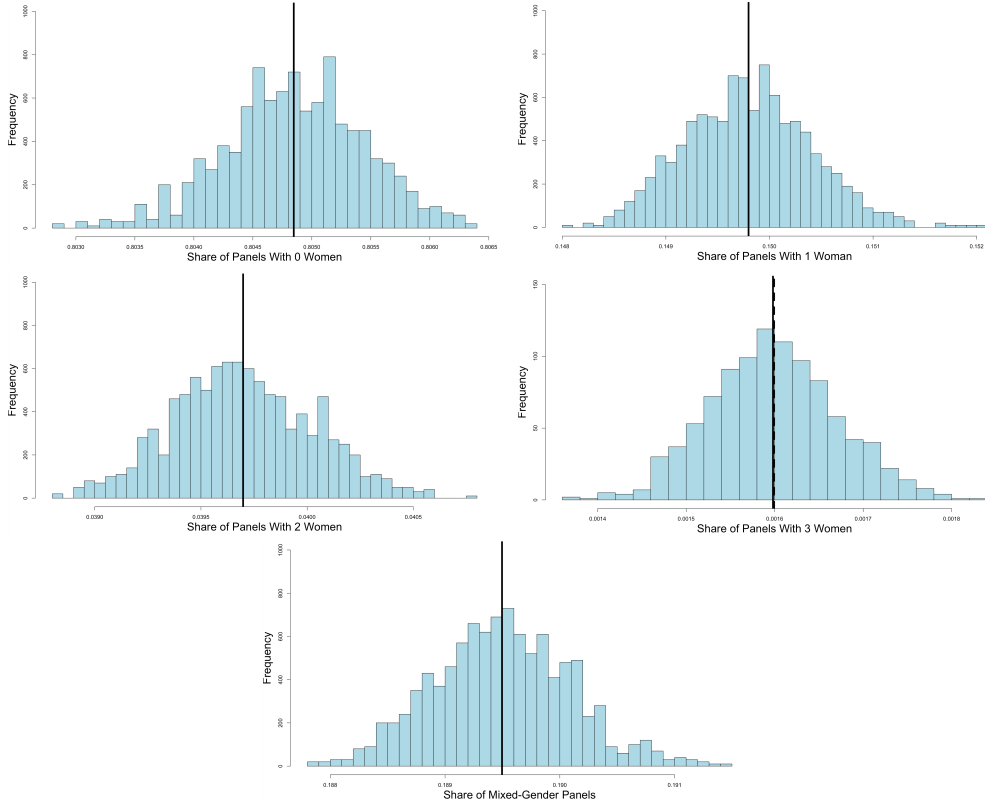
Shares of panels with 0–3 Democrats, sampled at the judge level. The bootstrap means (solid lines) are 0.1845, 0.3780, 0.3169, 0.1191, and 0.0965 respectively; the observed shares (dashed lines, when visible) are 0.1861, 0.3762, 0.3184, 0.1193, and 0.0946 respectively.

FIGURE 16. Distributions of Aggregate Shares of Panel Law Schools, Judge-Level Simulations



Shares of panels with 1–3 unique law schools, sampled at the judge level. The bootstrap means (solid lines) are 0.7172, 0.2479, and 0.0345 respectively; the observed shares (dashed lines, when visible) are 0.7128, 0.2515, and 0.0357 respectively.

FIGURE 17. Distributions of Aggregate Shares of Panel Genders, Judge-Level Simulations



Shares of panels with 0–3 women, sampled at the judge level. The bootstrap means (solid lines) are 0.8048, 0.1498, 0.0397, 0.0057, and 0.1895 respectively; the observed shares (dashed lines, when visible) are 0.8144, 0.1624, 0.0221, 0.0012, and 0.1845 respectively.

APPENDIX C. THEORETICAL APPENDIX

In this section, I describe and prove some additional results for the model in Section 5.1. Appendix C.1 proves results for sincere dissents with noise; Appendix C.2 extends that model by adding a role for opinions as well as dispositional votes; and Appendix C.3 provides details of the coalitional similarity framework with multidimensional judges.

C.1. Additional Results for Sincere Dissents with Noise. I add case-specific noise to the model of Section 5.1, representing the fact that a judge’s cutoff for upholding a ruling may vary depending on the precise facts at hand. For each case c , judges experience i.i.d. preference shocks $\eta_j \sim N(0, \sigma^2)$, and then vote sincerely based on their new cutoffs $x_j + \eta_j$. Each judge’s cutoff is thus a random variable $x'_j \sim N(x_j, \sigma^2)$.

Despite the relative tractability of the normal distribution, it is immediately clear why an exact analytic expression for dissent probabilities is not possible. By additivity, the distance between judges j and k is a normally distributed random variable D_{jk} , as is that distribution conditional on some fixed distance δ between judges i and k , $D_{jk}|\delta$. Judge j may dissent either when $D_{jk} < 0$ and $D_{ik} < 0$, or $D_{jk} > 0$ and $D_{ik} > 0$. Both of these expressions involve the product of a normal distribution with the integral of the normal cdf and have no closed form solution. Furthermore, the resulting expression would not have uniform density over the interval of possible distances. An easier result to establish is the claim that there is a larger increase in total dissents from increasing the extremal distance than the second-extremal distance:

Proposition 1. *In the sincere dissents model with i.i.d. $N(0, \sigma^2)$ noise, there exists a constant K such that for all $\sigma^2 < K$, increasing the second-extremal distance leads to a smaller increase in total dissents than increasing the extremal distance.*

Proof. I prove the proposition in three steps. First, with sufficiently small noise, only three of the six possible permutations of judge orders need be considered. Next, I show that the second-extremal judge is more likely to be permuted to the median than the extremal judge. Last, I show that this effect directly produces a lower effect of varying second-extremal distance.

Consider the permutation $x'_2 < x'_1 < x'_3$, where judge 1 becomes the median judge. To attain it, it is at least necessary that the realization of D_{12} is sufficiently large to produce $x'_2 < x'_1$. By choosing σ^2 small enough, such a realization lies in the tail of the (normally distributed) D_{12} . The probability of a permutation that is twice as large—which would be needed for $x'_3 < x'_1$ —is thus approximately zero in comparison. I can therefore rule out the permutations $x'_2 < x'_3 < x'_1$, $x'_3 < x'_2 < x'_1$, and $x'_3 < x'_1 < x'_2$, which proves the first claim.

Next, I show that of the three remaining permutations, the one which swaps judges 1 and 2 and leaves at least ε distance between them is more likely than the one which swaps judges 2 and 3 and leaves at least ε distance between them. By assumption, the distance between judges 1 and 2 is smaller than the distance between judges 2 and 3. Thus, a realization of D_{12} which makes $x'_2 < x'_1$ is more likely than a realization of D_{23} which makes $x'_3 < x'_2$. Indeed, for σ^2 sufficiently small both realizations are in the tails of their respective distributions (which have the same variances) and thus the probability of the second realization is exponentially smaller than the probability of the first.

Finally, note that when comparing a realized value $x_1 = x$ to a smaller realization $x_1 = x - \varepsilon$, total dissents will increase under the second realization unless $x'_1 - x'_2 > \varepsilon$. Likewise, when comparing $x_3 = y$ to $x_3 = y + \varepsilon$, total dissents will increase unless $x'_2 - x'_3 > \varepsilon$. As established above, the first event is more likely than the second, so total dissents are more likely to increase when x_3 is increased by ε than when x_1 is decreased by ε . $\square$

For $\sigma^2 = 0$ there is no difference in the increase in dissent rates caused by increasing second-extremal and extremal distances. As σ^2 becomes arbitrarily large, the original mean x_j is irrelevant and all permutations of judges become equally likely. Thus there is an important non-monotonicity in the relative importance of extremal distance with respect to noise. In the right image of Figure 1, I show that both the probability of permuting the second-extremal judge to the median and that of permuting the extremal judge to the median rise with σ^2 , but the difference rises for $\sigma^2 \in [0, 0.03]$ before falling for $\sigma^2 \in [0.03, 0.5]$. While a full theoretical characterization of this behavior is beyond the scope of this work, the simulated results in Section 5.2 provide some suggestive evidence that such non-monotonicities are relevant in real-world contexts, even if they do not ultimately affect the fit of this model with my leading empirical results.

C.2. Strategic Dissents. To connect the sincere voting setup to classical median voter models, I consider a simplification of [Parameswaran et al. \(2021\)](#) and introduce a policy utility term that captures not only the disposition of a case, but also the location of the opinion issued by the dispositional majority. Let the location of an opinion o be equal to the median of the x_j among judges j in the dispositional majority. Judges face a policy cost $p(x_j, 0) = p(|x_j - o|) = p(d)$, reflecting the fact that such an opinion shapes future policy in ways that a judge dislikes more the further that opinion is from their ideal point. To introduce a reason for judges to vote strategically on the disposition, I also impose an expressive cost $q(x_j, c) = q(|x_j - c|) = q(d)$ for a judge j who votes against their preferred disposition on case c . By making these costs a function of the distance between x_j and c , I automatically impose location invariance and symmetry, i.e., x_j near zero and x_j near 1 are treated the same, as are insincere upholdings and insincere reversals. I also assume that $p(0) = q(0) = 0$,³⁰ that both costs are continuous, and that both costs are weakly increasing in $|x_j - o|$ and $|x_j - c|$ respectively.

If judge 1 or judge 3 are part of a unanimous opinion and choose to dissent, the opinion will be further from their preferred x_j , since it will be located at the average of the remaining two judges' ideal points instead of being located at the median judge's ideal point. On a fixed panel, this cost does not depend on c , but will differ for the two judges. My next result establishes how the cost of dissent varies with the location of the extremal judges.

Proposition 2. *The cost of dissent to judge $i \in \{1, 3\}$ decreases with the distance of judge i from the median if and only if the policy cost $p(x_i, o)$ has increasing differences in (x_i, o) . The policy cost to that same judge increases with the distance of judge i from the median if and only if it has decreasing differences in (x_i, o) .*

Proof. The definition of increasing differences requires

$$x_i \geq x'_i \text{ and } o \geq o' \Rightarrow p(x_i, o) - p(x'_i, o) \geq p(x_i, o') - p(x'_i, o').$$

The cost of dissent to judge i when their ideal point is x_i is

$$p(x_i, o) - p(x_i, o'),$$

³⁰Generically, $x_1 \neq x_2 \neq x_3$ so that $|x_j - o| = 0$ is never relevant for judges 1 and 3.

where o is the location of the opinion in the coalition with judge i and o' is the location of the opinion in the coalition without judge i .

Letting $i = 1$, I have $o > o'$, since without judge 1 the opinion is set at the median of judges 2 and 3 rather than at the ideal point of judge 2. Moving judge 1 further from the median requires $x'_1 < x_1$, since judge 1 is to the left of the median. Therefore the condition for a decreasing cost of dissent is

$$p(x'_1, o) - p(x'_1, o') \leq p(x_1, o) - p(x_1, o') \Leftrightarrow p(x_1, o') - p(x'_1, o') \leq p(x_1, o) - p(x'_1, o),$$

which is precisely the expression for increasing differences. Likewise, an increasing cost of dissent requires the opposite inequality, which is precisely the condition for decreasing differences.

Letting $i = 3$, I instead have $o < o'$ and $x'_1 > x_1$. Switching both inequalities means that once again the condition for a decreasing cost of dissent is precisely the expression for increasing differences and the condition for a decreasing cost of dissent is precisely the expression for decreasing differences. $\square$

One simple example of a policy cost with increasing differences satisfying the earlier conditions is a logarithmic loss $p(x_i, o) = \ln[(x - y)^2]$; since this expression is twice continuously differentiable away from 0, I can check that its cross-partial derivative is $2/(x - y)^2 > 0$. A simple policy cost with decreasing differences satisfying the earlier conditions is the quadratic loss $p(x_i, o) = (x_i - o)^2$; since this expression is twice continuously differentiable, I can check that its cross-partial is $-2 < 0$.

This result, combined with symmetry of the policy cost, immediately establishes that under increasing (decreasing) differences the cost of dissent is higher (lower) for judge 1 than for judge 3 because judge 1 is closer to the median than judge 3. Letting δ_1 be the cost of dissent to judge 1 and δ_3 be the cost of dissent to judge 3, I can now establish results about the relative dissent rates of the two judges.

Proposition 3. *If $\delta_1 \geq \delta_3$, then extremal judges dissent weakly more often. The difference is strict if extremal judges dissent at all.*

Proof. I begin with the case $\delta_1 = \delta_3 = \delta$. Because $q(d)$ is weakly increasing and symmetric, it attains its maximum on $[0, x_2 - x_1]$ at $d_{12} = x_2 - x_1$ and attains its maximum on $[0, x_3 - x_2]$ at $d_{23} = x_3 - x_2$. Furthermore, $q(d_{12}) \leq q(d_{23})$. If $\delta \geq q(d_{23})$, then the maximal expressive cost of insincerity is lower than the common cost of dissent, so neither judge ever dissents. If $q(d_{12}) \leq \delta < q(d_{23})$, only judge 3 may dissent. The remaining case is therefore where $q(d_{12}) < \delta$.

Since $q(0) = 0 < \delta$, there is a well-defined closed interval $I = [x_\ell, x_u] \subset (0, x_2 - x_1)$ where $d = q(x_2 - x') \forall x' \in I$. (Of course, if $q(d)$ is strictly increasing, $x_\ell = x_u$ and that interval is a point). Judge 1 will thus insincerely join the majority whenever $c \in [x_1, x_1 + x_\ell]$. This interval is not empty, but may be the entirety of $[x_1, x_2]$.

By symmetry and equal costs of dissent, judge 3 will insincerely join the majority on exactly the same measure of cases as judge 1. However, judge 3 faces a strictly larger interval of cases where they sincerely disagree with the majority, and will thus dissent more often than judge 1.

Now consider the case $\delta_1 > \delta_3$. When $q(d_{23}) < \delta_3$, the maximal cost of insincerity is lower than the minimal cost of dissent so neither judge ever dissents. When $\delta_3 < q(d_{23})$ but $q(d_{12}) < \delta_1$ (regardless of the relationship between $q(d_{23})$ and δ_1 , and the relationship between $q(d_{12})$ and δ_3), the maximal cost of insincerity remains lower than the cost of dissent for judge 1, who never dissents, but judge 3 may sometimes dissent. The remaining case is $\delta_1 < q(d_{12}) \leq q(d_{23})$, in which case both judges may dissent.

By the same argument above, there are cutoffs d_{23}^* and d_{12}^* such that judge 3 will insincerely join the majority only for cases closer than d_{23}^* to x_3 and judge 1 will insincerely join the majority only for cases closer than d_{12}^* to x_1 . However, because $\delta_1 > \delta_3$ and both judges face the same expressive cost, it must be that $d_{23}^* < d_{12}^*$, i.e., judge 1 insincerely joins the majority for a larger measure of cases than judge 3.

Since judge 3 faces both a larger measure of cases where dissent minimizes their expressive cost, and a smaller measure of cases where they insincerely join the majority, the extremal judge dissents more often. □

The expected converse result—that when $\delta_1 < \delta_3$, second-extremal judges dissent more often—does not hold because of two offsetting effects. As seen in the proof, a higher cost of dissent means a larger measure of cases where the judge insincerely joins the majority; however, the total measure of cases where judge 3’s expressive preferences predict dissent is larger than the measure of cases where judge 1’s expressive preferences predict dissent. Thus, the key question is whether the cost of dissent grows more than one-to-one with distance from a judge’s ideal point. If this property holds, the higher cost of dissent for judge 3 outweighs the larger share of cases where their expressive preferences predict dissent (since that share grows one-to-one with distance because of the uniform distribution of cases). Thus, judge 3 actually dissents less often than judge 1.

That same property is important in understanding the effect of increasing second-extremal and extremal distances from the median. If the cost of dissent has increasing differences, then the cost of dissent is decreasing with distance. Moving either judge 1 or judge 3 further from the median increases the share of cases where expressive preferences predict a dissent and the share of cases where the cost of dissent is low enough to permit a sincere dissent. Thus the total share of dissents by each judge increases. That share is higher for the extremal judge than the second-extremal judge; whether the increase is more rapid for one judge than the other depends on the curvature of the cost of dissent.

If instead the cost of dissent has decreasing differences, then moving either judge 1 or judge 3 further from the median increases the share of cases where expressive preferences predict a dissent but decreases the share of cases where the cost of dissent is low enough to permit a sincere dissent. In this case, the curvature of the cost of dissent determines both whether the amount of dissents increases and for which judge it changes more quickly. With a convex cost, the increase in insincere agreements outweighs the increase in cases with possible dissents; this effect is stronger for the extremal judge, who dissents less often than the second-extremal judge, and leads to an extremal dissent rate that falls more quickly than the second-extremal dissent rate. With a concave cost, dissents increase for both judges, though more slowly than if the cost of dissent had increasing differences because the increase in cases with possible dissents must compete with the increase in insincere agreements. Since cost is concave, the

extremal judge dissents more often than the second-extremal judge, and the growth in their dissent rate is faster than that of the second-extremal judge.

This model shows the complex possibilities of a fully strategic framework modeling both case dispositions and opinion locations, but falls short of capturing the empirical facts observed in the appellate court data. In particular, dissents increase when the extremal judge moves further from the median, suggesting a world in which either the cost of dissent has increasing differences and is convex or the cost of dissent has decreasing differences and is concave. But in both of these settings, the extremal judge dissents more often than the second-extremal judge, and the median judge does not dissent at all. As shown in the sincere dissents framework, the presence of noise can dramatically complicate results, and is beyond the scope of this analysis. Even still, the simulation results in Section 5.2 suggest that noise alone is insufficient to induce equal dissent rates by median and extremal judges; more exploration of this rich framework is needed to arrive at a strategic justification for the empirical fact.

C.3. Multidimensional Judges and Coalition Formation. Let each judge have an N -dimensional type $t_i = (x_i^1, \dots, x_i^N)$, where the first dimension captures partisanship and the remaining dimensions capture other characteristics, e.g., demographic features such as gender and age or non-partisan ideological features such as law school training or previous clerkships.

An *opinion* is a partition of the judges into two groups called *coalitions*,³¹ one of which may be empty. For example, $(\{1, 2, 3\}, \{\emptyset\})$ represents a unanimous opinion, while $(\{1, 2\}, \{3\})$ represents a split opinion with judges 1 and 2 in the majority and judge 3 dissenting. The value of any coalition C with at least two judges is given by

$$\frac{1}{|C|} \sum_{\{i,j\} \in C} \langle t_i, t_j \rangle,$$

the average weighted dot product of the types of its members. That is, the more similar a coalition's members are (where one dimension can have greater weight than the other in computing similarity) the higher its value. Additionally, I scale the value of a three-judge

³¹Because I focus on dissents, I ignore the role of concurrences in potentially creating a partition into three groups.

coalition by α —representing the fact that unanimous opinions are more valuable than split ones—and let $d \geq 0$ (for *dissent*) be the value of a one-judge coalition.

I now impose equilibrium conditions for an opinion to be stable. Intuitively, an opinion must distribute all available surplus to the judges and be immune to deviations from judges within the coalition. More formally, let C be the majority coalition in an opinion. For that opinion to be stable, I require

$$\sum_{i \in C} u_i = v(C) \quad \text{and} \quad u_i \geq d \forall i \in C,$$

so that all value generated by the majority coalition is shared among its members, each of whom get at least the payoff from their outside option of dissenting. To ensure a unanimous coalition is not vulnerable to a deviation by a pair of judges, I require

$$\text{If } |C| = 3 : \sum_{i \in C} u_i \geq v(D) \forall D \subseteq C.$$

Majority coalitions of three, two, or one judge(s) are all possible depending on the values of d and α . To check whether the unanimous majority coalition forms, it suffices to check whether the value $v(C)$ of that coalition and the value $v(D)$ of the coalition D of the two most similar judges satisfy $v(C) \geq v(D) + d$. While D is the key coalition to check, more than one two-judge coalition can be stable. For example, with two-dimensional judge types that have equal weights, judge types $t_1 = (0, 0)$, $t_2 = (0, 1)$ and $t_3 = (1, 0)$ produce coalitions $\{1, 2\}$ and $\{1, 3\}$ which give the same value to their members.

In Section 5.3, I describe two ways of equalizing dissent rates among majority and minority partisans when the weight on party is smaller than the weight on law school, but not infinitely so. The first is to add party-level noise. To show this result, consider observed party to be $x_i \in \{0, 1\}$ (representing D and R respectively) but true party to be $x_i + \varepsilon$ where $\varepsilon \sim N(0, \sigma^2)$. Then in some cases a Democratic judge will be the partisan outlier, reducing the share of dissents by Republican judges. More generally, if there is more noise in the partisan dimension than the law school dimension, this effect will still apply in expectation.

The second way of equalizing dissent rates is to introduce anti-correlation between party and law school. Then double-minority judges will become less common, and the increased dissent rate by law-school-minority judges compared to minority partisans will have a larger

effect on total dissents. In the balance tables of [B.1](#), I show that there is a slight anti-correlation between bipartisan panels and panels with a law school minority judge, though the correlation between bipartisan panels and panels with a single woman is slightly positive.

In the analysis of [Section 5.3](#), I do not explicitly select among the potential multiplicity of stable opinions, but informally consider opinions that deliver greater total value to the judges to be more likely than ones that deliver lower total value, in the style of random utility models. Establishing an equilibrium selection rule is an important direction for future theoretical work. Another important next step is to include ideology scores like JCS and DIME-CF, e.g., by considering them as a one-dimensional embedding of a richer type space with more dimensions for possible judges characteristic.

APPENDIX D. EVALUATION AND VALIDATION OF MACHINE LEARNING APPROACHES

Below I describe a typical fine-tuning pipeline:

- (1) Remove names of people and locations using a standard named-entity recognizer, and manually remove the names of all judges in the sample (including those not associated with a given opinion). This step ensures that embeddings reflect the content of the text and do not “cheat” by learning and using, e.g., the author name to predict party, or the geographic location to predict political polarization.
- (2) Divide opinions into overlapping chunks based on the maximum length of the relevant model (in the case of transformer models, this length is usually 512 words).
- (3) Randomly select 25% of the opinion chunks (approximately 60,000 full opinions or 350,000 512-word chunks) to use for training. I refer to this sample as the fine-tuning sample, and the remaining 75% of the opinions as the validation sample.
- (4) Run a standard fine-tuning loop (for transformer models, done in Pytorch) or few-shot prediction task (for LLMs, done using the appropriate API).
- (5) Merge opinion chunks by averaging embeddings across all validation chunks in an opinion and re-classifying based on the new opinion-level embedding.
- (6) Perform additional validation (see discussion below).
- (7) Select a model for use based on the appropriateness of the training task and the performance in validation tasks.

Throughout this work, when using NLP measures I report results holding out opinion chunks from the fine-tuning sample. I am still able to obtain results for almost all opinions in this case (since only opinions where all chunks were used for fine-tuning are omitted) and all results are robust to including fine-tuning opinion chunks.

Model validation and selection is done in four main ways. First, for transformer models, the fine-tuning loop automatically splits the fine-tuning sample into training, testing, and validation subsets. A successful fine-tuning run produces a model that achieves good results (under a metric like accuracy or F1 score) on the validation subset of the fine-tuning sample.

Second, I can evaluate those same metrics on the large validation sample. Third, I can perform permutation inference by drawing random labels for the fine-tuning sample from the empirical label distribution, fine-tuning new “permuted models” on those relabeled opinions, then evaluating the permuted models on the validation sample (with true labels). This process is extremely computationally intensive, since it requires fine-tuning hundreds or even thousands of models to produce a distribution of permuted model results. Finally, I can train a simple dense neural network (or other statistical model) to perform cross-task predictions—for example, after fine-tuning a model to predict legal topics, I can use those embeddings to predict author party. This approach does not validate model performance on the target task, but instead assesses generalizability of the learned embeddings. Below I focus on the second approach, presenting out-of-sample performance on a variety of standard metrics using the validation sample.

TABLE 15. Confusion Matrix for Predicted Legal Topic

True Label	Predicted Label								Sum
	Crim.	Civ. Rights	1st Am.	Due Process	Priv.	Labor	Econ./Reg.	Misc.	
Crim.	64,156	3,968	16	1,471	0	15	1,101	438	71,165
Civ. Rights	2,902	9,255	24	5,444	0	133	586	649	18,993
1st Am.	103	22	1,226	577	0	16	333	39	2,316
Due Process	4,196	3,615	321	41,535	0	2,805	22,828	2,126	77,426
Priv.	13	4	0	349	0	0	23	3	392
Labor	48	36	1	3,473	0	13,846	2,181	47	19,632
Econ./Reg.	1,418	437	67	8,869	0	1,289	87,041	413	99,534
Misc.	1,195	889	55	3,922	1	199	4,622	8,967	19,850
Sum	74,031	18,226	1,710	65,640	1	18,303	118,715	12,286	308,912

Results from the best-performing transformer model for topic prediction, fine-tuned Legal-BERT from [Chalkidis et al. \(2020\)](#). Overall, 226,026 out of 308,912 opinions (73.2%) were classified correctly.

TABLE 16. Performance Metrics by Topic

True Label	Accuracy	Precision	Recall	F1 Score
Crim.	0.945	0.867	0.902	0.884
Civ. Rights	0.940	0.508	0.487	0.497
1st Am.	0.995	0.717	0.529	0.609
Due Process	0.806	0.633	0.536	0.581
Priv.	0.999	0.000	0.000	0.000
Labor	0.967	0.756	0.705	0.730
Econ./Reg.	0.857	0.733	0.874	0.798
Misc.	0.953	0.707	0.452	0.551
Weighted avg.	0.884	0.724	0.731	0.722

Performance metrics by topic for the best-performing transformer model for topic prediction, fine-tuned LegalBERT from [Chalkidis et al. \(2020\)](#). Of the three main topics—criminal, due process, and economics/regulation—the first and last show strong performance, and the weighted averages reflects an overall successful classifier. Weaker performance on due process cases is unsurprising since that topic often overlaps with others, and reflect realistically higher weights for other topics in predictions for due process cases.

APPENDIX E. ROBUSTNESS CHECKS FOR DISSENTS

In this appendix, I provide robustness checks for results about dissent rates and dissent shares. In Appendix E.1, I separate out results by Democratic and Republican judges, replicating tables 2 (Section 6.1) and 3 (Section 6.2) from the main text. In Appendix E.2, I estimate judge-level dissents as in Table 4 (Section 6.3) using alternative specifications; I also plot the amount of minority-party and median dissents over time. In Appendix E.3, I provide additional analysis comparing outlier effects in the style of Table 7.

E.1. Separating Partisan Dissents for Democrats and Republicans.

TABLE 17. Replicating Table 2 with Democratic-Majority Panels

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Party	JCS Scores			DIME-CF Scores		
	All Panels	All Panels	Party-Unan.	Bip.	All Panels	Party-Unan.	Bip.
bipartisan	0.128*** (0.031)						
ext. dist.		0.538*** (0.071)	0.494** (0.246)	0.296** (0.117)	0.147*** (0.044)	0.018 (0.111)	0.062 (0.056)
2nd-ext. dist.		0.165 (0.212)	1.582*** (0.609)	−0.075 (0.232)	0.078 (0.088)	0.154 (0.297)	0.029 (0.090)
<i>p</i> for coeff. diff.	—	0.121	0.130	0.178	0.489	0.697	0.762
Control dissents	0.099	0.093	0.088	0.119	0.097	0.104	0.100
Score std. dev.	—	0.359	0.341	0.353	0.663	0.628	0.655
Observations	111, 142	45, 339	8, 660	35, 601	47, 288	9, 348	36, 890

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

The effects of Table 2 persist when considering Democratic-majority panels only. All estimation is performed as in that table with an appropriate modification of the sample.

TABLE 18. Replicating Table 2 with Republican-Majority Panels

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Party	JCS Scores			DIME-CF Scores		
	All Panels	All Panels	Party-Unan.	Bip.	All Panels	Party-Unan.	Bip.
bipartisan	0.323*** (0.029)						
ext. dist.		0.722*** (0.069)	−0.090 (0.188)	0.411*** (0.124)	0.261*** (0.031)	0.016 (0.075)	0.032 (0.048)
2nd-ext. dist.		0.094 (0.113)	−0.334 (0.313)	0.152 (0.129)	0.186** (0.076)	−0.175 (0.192)	0.229*** (0.087)
<i>p</i> for coeff. diff.	—	0.000	0.547	0.156	0.399	0.393	0.071
Control dissents	0.075	0.072	0.071	0.106	0.073	0.064	0.099
Score std. dev.	—	0.355	0.355	0.354	0.671	0.659	0.665
Observations	143,450	82,951	29,183	52,502	84,741	29,844	53,702

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

The effects of Table 2 persist when considering Republican-majority panels only. All estimation is performed as in that table with an appropriate modification of the sample.

TABLE 19. Replicating Table 3 with Democratic-Majority Panels

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
		JCS Scores				DIME-CF Scores			
Outcome	Variable	All Panels	All Panels	Party-Unan.	Bip.	All Panels	All Panels	Party-Unan.	Bip.
dissent by median judge	bipartisan	0.193*** (0.064)				0.270*** (0.062)			
	ext. dist.		0.419*** (0.091)	0.612 (0.404)	0.315** (0.137)		0.130** (0.057)	0.117 (0.147)	0.027 (0.074)
	2nd-ext. dist. med		0.287 (0.306)	1.638* (0.879)	0.138 (0.332)		0.109 (0.119)	0.016 (0.393)	0.063 (0.128)
dissent by 2 nd -extremal judge	bipartisan	0.251*** (0.062)				0.188*** (0.061)			
	ext. dist.		0.449*** (0.087)	0.704** (0.314)	0.201 (0.134)		0.084 (0.057)	−0.122 (0.166)	0.018 (0.072)
	2nd-ext. dist.		0.075 (0.289)	1.288 (0.881)	−0.194 (0.317)		0.036 (0.117)	0.285 (0.372)	−0.010 (0.126)
dissent by extremal judge	bipartisan	0.350*** (0.066)				0.240*** (0.061)			
	ext. dist.		0.500*** (0.089)	−0.187 (0.409)	0.238* (0.133)		0.158*** (0.058)	0.011 (0.154)	0.112 (0.074)
	2nd-ext. dist.		0.140 (0.291)	1.083 (1.088)	−0.057 (0.312)		0.042 (0.120)	0.136 (0.365)	0.003 (0.128)
	bip./ext. = across roles	0.238	0.819	0.213	0.841	0.644	0.668	0.574	0.623
	2nd-ext. = across roles	−	0.881	0.922	0.778	−	0.894	0.888	0.915
	Control dissents	0.032	0.090	0.089	0.096	0.034	0.096	0.100	0.090
	Score std. dev.	−	0.359	0.195	0.380	−	0.662	0.574	0.678
	Observations	107,498	46,187	9,775	36,412	107,684	48,149	10,464	37,685

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Replication of Table 3 restricting to Democratic-majority panels only. All estimation is performed as in that table with an appropriate modification of the sample. In column (1), using JCS scores, the increase in dissents on bipartisan panels compared to party-unanimous panels is largest for extremal judges and smallest for median judges. However, this difference is not in evidence for any other columns, and is only significant at the 10% level when comparing the extremal and median judges.

TABLE 20. Replicating Table 3 with Republican-Majority Panels

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
		JCS Scores				DIME-CF Scores			
Outcome	Variable	All Panels	All Panels	Party-Unan.	Bip.	All Panels	All Panels	Party-Unan.	Bip.
dissent by median judge	bipartisan	0.392*** (0.043)				0.336*** (0.043)			
	ext. dist.		0.683*** (0.086)	0.235 (0.270)	0.312** (0.158)		0.262*** (0.041)	0.105 (0.111)	0.047 (0.059)
	2nd-ext. dist.		0.077 (0.157)	−0.613 (0.500)	0.073 (0.177)		0.129 (0.098)	−0.137 (0.247)	0.175 (0.111)
dissent by 2 nd -extremal judge	bipartisan	0.378*** (0.043)				0.392*** (0.043)			
	ext. dist.		0.630*** (0.089)	−0.386 (0.291)	0.356** (0.171)		0.243*** (0.042)	−0.084 (0.125)	0.051 (0.060)
	2nd-ext. dist.		0.062 (0.160)	−0.027 (0.508)	0.056 (0.180)		0.222** (0.100)	0.260 (0.263)	0.224** (0.111)
dissent by extremal judge	bipartisan	0.296*** (0.043)				0.342*** (0.043)			
	ext. dist.		0.587*** (0.088)	−0.074 (0.253)	0.390** (0.164)		0.181*** (0.042)	0.029 (0.113)	−0.028 (0.060)
	2nd-ext. dist.		0.108 (0.155)	−0.343 (0.452)	0.297* (0.177)		0.159 (0.098)	−0.611** (0.266)	0.224** (0.109)
	bip./ext. = across roles	0.238	0.746	0.303	0.944	0.601	0.367	0.537	0.586
	2nd-ext. = across roles	−	0.979	0.720	0.579	−	0.806	0.072	0.939
	Control dissents	0.048	0.071	0.069	0.095	0.049	0.072	0.064	0.099
	Score std. dev.	−	0.355	0.240	0.387	−	0.671	0.543	0.703
	Observations	141, 195	83, 379	30, 496	52, 883	141, 359	85, 149	31, 066	54, 083

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

The effects of Table 3 persist when considering Republican-majority panels only. All estimation is performed as in that table with an appropriate modification of the sample.

TABLE 21. Replicating Table 3 with Separate Democratic-Majority and Republican-Majority Bipartisan Panels

		(1) JCS	(2) DIME-CF
dissent by median judge	Rep-maj.	0.313*** (0.038)	0.285*** (0.037)
	Dem-maj.	0.333*** (0.041)	0.350*** (0.041)
dissent by 2nd-extremal judge	Rep-maj.	0.288*** (0.038)	0.295*** (0.037)
	Dem-maj.	0.403*** (0.041)	0.364*** (0.041)
dissent by extremal judge	Rep-maj.	0.268*** (0.038)	0.268*** (0.037)
	Dem-maj.	0.380*** (0.041)	0.369*** (0.040)
	Control dissents	0.042	0.044
	Observations	248,693	249,043

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Columns (1) and (2) replicate columns (1) and (5), respectively, of Table 3 with two separate indicators for Democratic-majority and Republican-majority bipartisan panels. The estimation method and sample are the same as in those columns. I perform a Wald test for equality of the three coefficients within each of the four score $\times$ party categories, then for corresponding Democratic and Republican coefficients across all six judge role $\times$ score pairs. These results show that dissent rates are equal across judge roles within party, but differ within judge role across parties.

	JCS	DIME-CF
Rep.	0.710	0.879
Dem.	0.484	0.944

	JCS	DIME-CF
median	0.572	0.001
2nd-extremal	0.001	0.059
extremal	0.047	0.004

E.2. Dissents at the Judge Level.

TABLE 22. Judge-Level Dissents with Controls

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Min. Party	Is JCS Median	Is DIME-CF Median	Min. Law School	Min. Gender	Only Man	Only Woman	Youngest	Median Age	Oldest
judge char.	0.001 (0.017)	−0.002 (0.019)	−0.005 (0.019)	−0.005 (0.029)	−0.005 (0.044)	−0.036 (0.074)	0.145*** (0.038)	0.009 (0.014)	−0.009 (0.014)	−0.001 (0.013)
is Dem.	0.008 (0.016)	−0.018 (0.022)	−0.018 (0.021)	−0.016 (0.034)	−0.022 (0.034)	−0.022 (0.034)	−0.021 (0.034)	−0.008 (0.016)	−0.008 (0.016)	−0.008 (0.016)
age	−0.027** (0.012)	−0.068*** (0.019)	−0.064*** (0.019)	−0.005 (0.026)	−0.034 (0.022)	−0.034 (0.022)	−0.033 (0.022)	— —	— —	— —
age ²	0.000** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	— —	— —	— —
is woman	0.165*** (0.044)	0.147*** (0.042)	0.147*** (0.042)	0.128 (0.083)	0.181*** (0.050)	— —	— —	0.171*** (0.040)	0.171*** (0.040)	0.171*** (0.040)
Control dissents	0.031	0.035	0.035	0.031	0.030	0.031	0.030	0.032	0.032	0.032
Observations	533, 214	389, 766	398, 745	184, 767	144, 555	144, 555	144, 555	772, 230	772, 230	772, 230

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Replicating Table 4 with controls for judge characteristics instead of judge fixed effects. Controls for panel characteristics—number of Democrats, number of unique law schools, number of women, and Bloomberg topic, as appropriate—are also included.

TABLE 23. Judge-Level Dissents with Fixed Effects, Bipartisan Panels

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Min. Party	Is JCS Median	Is DIME-CF Median	Min. Law School	Min. Gender	Only Man	Only Woman	Youngest	Median Age	Oldest
judge char.	0.010 (0.022)	0.019 (0.026)	0.002 (0.026)	0.009 (0.050)	0.122*** (0.046)	−0.013 (0.084)	0.136*** (0.050)	−0.024 (0.025)	0.009 (0.020)	0.011 (0.026)
Control dissents	0.035	0.039	0.039	0.034	0.033	0.033	0.033	0.034	0.035	0.035
Observations	533, 214	389, 766	398, 745	184, 767	144, 555	144, 555	144, 555	772, 230	772, 230	772, 230
# judges	1, 230	756	771	710	592	592	592	1, 230	1, 230	1, 230

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Replicating Table 4, restricted to bipartisan panels only. Controls for panel characteristics—number of Democrats, number of unique law schools, number of women, and Bloomberg topic, as appropriate—are also included.

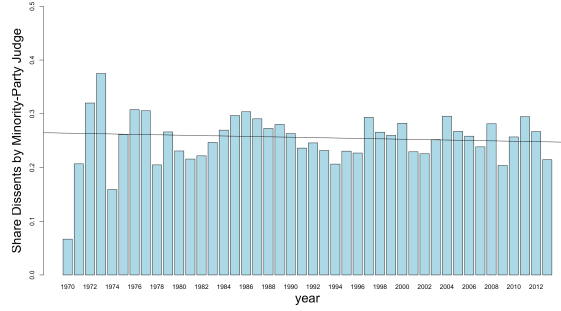
TABLE 24. Judge-Level Dissents with Controls, Bipartisan Panels

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Min. Party	Is JCS Median	Is DIME-CF Median	Min. Law School	Min. Gender	Only Man	Only Woman	Youngest	Median Age	Oldest
judge char.	0.001 (0.017)	−0.010 (0.021)	−0.005 (0.021)	0.014 (0.035)	−0.007 (0.051)	−0.004 (0.080)	0.135*** (0.044)	0.006 (0.016)	−0.009 (0.016)	0.003 (0.016)
is Dem.	0.008 (0.016)	−0.017 (0.022)	−0.017 (0.022)	−0.017 (0.034)	−0.022 (0.034)	−0.022 (0.034)	−0.021 (0.034)	−0.008 (0.016)	−0.008 (0.016)	−0.008 (0.016)
age	−0.027** (0.012)	−0.052** (0.021)	−0.047** (0.021)	0.002 (0.032)	−0.034 (0.025)	−0.035 (0.025)	−0.034 (0.025)	— —	— —	— —
age ²	0.000** (0.000)	0.000** (0.000)	0.000** (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	— —	— —	— —
is woman	0.165*** (0.044)	0.122*** (0.046)	0.122*** (0.046)	0.194** (0.093)	0.180*** (0.054)	— —	— —	0.165*** (0.044)	0.165*** (0.044)	0.165*** (0.044)
Control dissents	0.034	0.038	0.038	0.034	0.033	0.033	0.033	0.034	0.034	0.034
Observations	533, 214	267, 420	274, 416	125, 121	105, 492	105, 492	105, 492	533, 214	533, 214	533, 214

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Replicating Table 4 with controls for judge characteristics instead of judge fixed effects, and restricted to bipartisan panels only. Controls for panel characteristics—number of Democrats, number of unique law schools, number of women, and Bloomberg topic, as appropriate—are also included.

FIGURE 18. Minority-Party Dissents Over Time

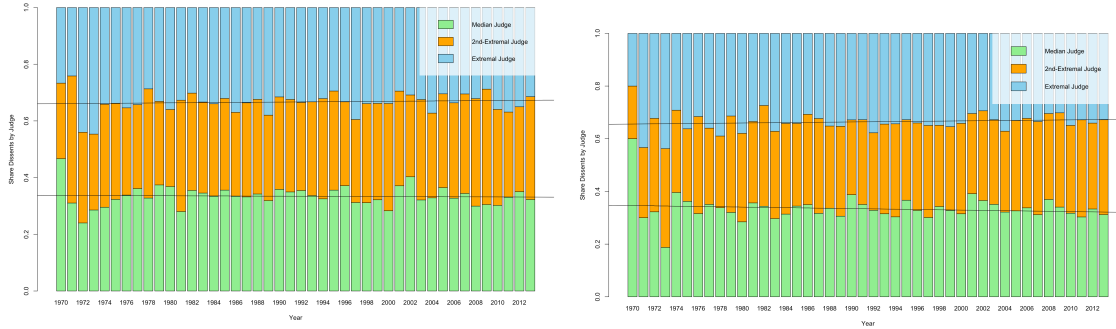


Share of dissents on bipartisan panels authored by the minority-party judge, 1970–2013. The line is a linear regression of minority-party dissent share on year,

$$\text{dissent} = 0.264 - 0.0003 \text{ yr.}$$

The slope is not significantly different from 0 at the 10% level.

FIGURE 19. Dissents by Score Role Over Time



Share of dissents on bipartisan panels authored by each score role—median, second-extremal, extremal—using JCS scores (left) and DIME-CF scores (right), 1970–2013. The lines are linear regressions of dissent share on year. For JCS scores (left),

$$\text{dissent} = 0.339 - 0.0002 \text{ yr} \quad \text{and} \quad \text{dissent} = 0.338 - 0.0001 \text{ yr,}$$

for the extremal and median judges, respectively. For DIME-CF scores (right),

$$\text{dissent} = 0.344 - 0.0003 \text{ yr} \quad \text{and} \quad \text{dissent} = 0.347 - 0.0004 \text{ yr,}$$

for the extremal and median judges, respectively. No slope is significantly different from 0 the 10% level.

E.3. Comparing Different Outlier Effects.

TABLE 25. Comparison of Outlier Effects Interacted with Party

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Partisanship	Shared Law School	Any Law Schools	Mixed Gender	Min. Men	Min. Women	Age
bipartisan	0.232*** (0.023)						
one diff. law school, panel-maj. party		0.078 (0.080)	-0.092*** (0.031)				
one diff. law school panel-min. party		0.200** (0.084)	0.036 (0.036)				
all same law school			-0.177** (0.079)				
panel-min. gender, panel-maj. party				0.051 (0.032)			
panel-min. gender, panel-min. party				0.036 (0.039)			
one man, panel-maj. party					0.146 (0.181)		
one man, panel-min. party					0.098 (0.199)		
one woman, panel-maj. party						0.032 (0.033)	
one woman, panel-min. party						0.024 (0.040)	
young outlier, panel-maj. party							-0.010 (0.041)
young outlier, panel-min. party							0.099* (0.051)
old outlier, panel-maj. party							0.000 (0.042)
old outlier, panel-min. party							0.113* (0.058)
<i>p</i> for coeff. diff.	—	0.012	0.004	0.695	0.604	0.854	0.432 (young); 0.159 (old)
Control dissents	0.084	0.067	0.107	0.098	0.161	0.098	0.102
Observations	178,037	48,720	178,037	178,037	3,848	173,426	178,037

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Effect of panel composition on total dissents, considering bipartisan panels only and splitting each explanatory variable by whether the outlier judge is a majority or minority partisan. Column (1) shows the effect of partisanship for reference. Columns (2)–(6) use the subset of corresponding panels from Table 7 which are bipartisan. All columns follow the same estimation method as in that table.

To directly compare effect sizes (as much as is possible), I estimate a conditional logit model including all coefficients together:

$$\begin{aligned}
\text{Dissent}_p = & \beta_1 \text{bipartisan}_p \\
& + \beta_2 \text{3 same law school}_p + \beta_3 \text{2 same law school}_p + \\
& \beta_4 \text{young outlier}_p + \beta_5 \text{old outlier}_p \\
& + \beta_6 \text{one man}_p + \beta_7 \text{one woman}_p \\
& + \text{legal-topic}_p + \gamma_{ct} + \varepsilon_{pct}.
\end{aligned} \tag{13}$$

Results are in Table 26.

TABLE 26. Direct Comparison of Outlier Effects

Bipartisan	One Unique Law School.	2 Unique Law Schools	One Man	One Woman	Young Outlier	Old Outlier
0.233***	-0.087	-0.027	0.202***	0.080**	0.042	0.044
(0.023)	(0.066)	(0.021)	(0.052)	(0.025)	(0.030)	(0.030)

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

There are 257,712 panels in the sample, and the baseline dissent rate is 8.2%. Party and gender have the strongest effects, with the other coefficients not significant at the 10% level. Averaging across categories for a back-of-the-envelope calculation, shared law school training reduces dissents by 5.1%, a mixed-gender panel increases dissents by 13.8%, and an age outlier increases dissents by 4.0%. These effects sum to 23.1%, comparable to the 23.5% for party but with a much wider confidence interval.

TABLE 27. Dissents and Deference to Lower-Court Rulings

Outcome	Variable	Party	JCS Scores			DIME-CF Scores		
		All Panels	All Panels	Party-Unan.	Bip.	All Panels	Party-Unan.	Bip.
uphold w/ dissent	bipartisan	0.254*** (0.021)						
	ext. dist.		0.537*** (0.045)	0.070 (0.168)	0.195** (0.079)	0.208*** (0.025)	0.014 (0.072)	0.041 (0.036)
	2nd-ext. dist.		0.270*** (0.094)	-0.175 (0.341)	0.252** (0.112)	0.224*** (0.056)	-0.167 (0.174)	0.195*** (0.065)
reverse w/ dissent	bipartisan	0.153*** (0.021)						
	ext. dist.		0.536*** (0.046)	0.103 (0.170)	0.462*** (0.080)	0.186*** (0.026)	0.016 (0.072)	0.072* (0.037)
	2nd-ext. dist.		-0.119 (0.097)	-0.064 (0.328)	-0.165 (0.117)	0.070 (0.057)	-0.038 (0.160)	0.025 (0.067)
reverse w/ dissent	bipartisan	-0.027*** (0.007)						
	ext. dist.		-0.076*** (0.015)	-0.218*** (0.050)	-0.012 (0.029)	-0.025*** (0.009)	0.022 (0.021)	-0.019 (0.013)
	2nd-ext. dist.		-0.155*** (0.032)	-0.156 (0.099)	-0.126*** (0.041)	-0.018 (0.018)	0.013 (0.046)	-0.005 (0.023)
	bip./ext. = across dis.	0.000	0.000	0.115	0.020	0.000	0.926	0.139
	2nd-ext. = across dis.	-	0.000	0.959	0.003	0.000	0.337	0.006
	Control dissents	0.083	0.077	0.076	0.097	0.078	0.073	0.095
	Score std. dev.	-	0.388	0.351	0.398	0.666	0.709	0.709
	Observations	260,123	151,082	40,299	89,417	155,378	41,561	91,891

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Results from conditional multinomial logit estimation of dissent rates using circuit $\times$ year fixed effects (with robust standard errors clustered at the circuit $\times$ year level), in the style of equations (4) and (5) but with the four categories representing uphold or reverse $\times$ unanimously or with dissent. Increased dissents on bipartisan panels come mostly from upholdings; this disparity vanishes or even reverses when looking at the distance between the extremal and median judges.

APPENDIX F. ROBUSTNESS CHECKS FOR NLP RESULTS

In this appendix, I provide robustness checks for the results from Section 8. In Appendix F.1, I show that the differences in uniform and Bayesian major topic use are robust to scaling those thresholds. and provide distributions of major topics using the un-scaled measures. In Appendix F.2, I replicate Table 10 (Section 8.1) with author controls and fixed effects. Finally, in Appendix F.3, I provide distributions of neighbor match scores from Table 11 (Section 8.2), and in Appendix F.4 I show that the results of that table are robust to alternative specifications of the nearest-neighbor measure.

F.1. Robustness of Major Topic Measures.

TABLE 28. Robustness of Uniform Major Topics to Alternative Thresholds

Metric	All-D. vs. All-R.	2D. vs. 3D.	2R. vs. 3R.	Bip. D. vs. R.	Sample Mean
# topics w/ $\mathbb{P} > 0.25$	-0.001 (0.006)	0.003 (0.004)	0.005* (0.003)	0.001 (0.003)	1.388
# topics w/ $\mathbb{P} > 0.125$	0.001 (0.006)	0.006 (0.004)	0.005* (0.003)	0.003 (0.003)	1.772
# topics w/ $\mathbb{P} > 0.1$	0.003 (0.006)	0.005 (0.004)	0.007** (0.003)	0.003 (0.003)	1.906
# topics w/ $\mathbb{P} > 0.05$	0.013** (0.007)	-0.006 (0.004)	0.006** (0.003)	0.004 (0.003)	2.281
# topics w/ $\mathbb{P} > 0.01$	0.015** (0.005)	-0.007* (0.004)	0.006* (0.003)	0.004 (0.002)	3.250
Observations	49,099	61,538	101,862	114,301	189,538

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Robustness to alternative uniform thresholds for major topics. A threshold of 0.125 (reported in the main text, and again here) represents a uniform distribution over all 8 topics. The number of major topics on all-Democratic panels compared to all-Republican panels grows slightly as the threshold decreases, but the difference remains small; the number of major topics is the same on bipartisan and party-unanimous panels.

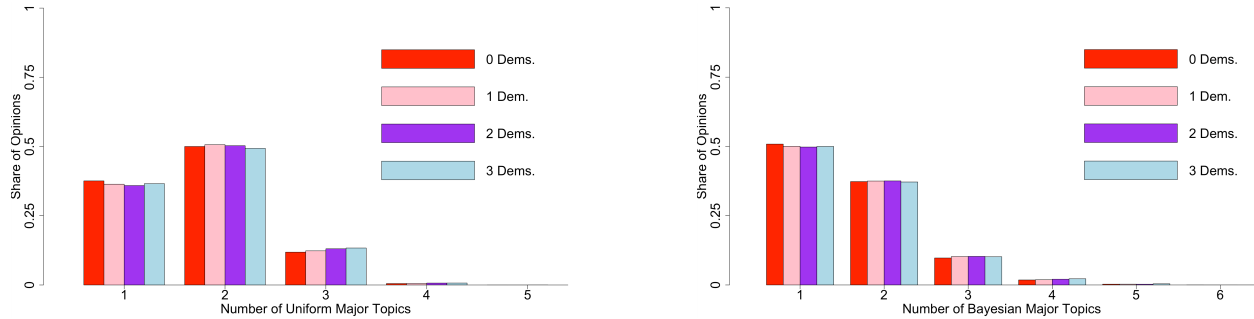
TABLE 29. Robustness of Bayesian Major Topics to Alternative Thresholds

Metric	All-D. vs. All-R.	2D. vs. 3D.	2R. vs. 3R.	Bip. D. vs. R.	Sample Mean
# topics w/ posterior > prior	0.014* (0.007)	-0.007 (0.005)	0.007* (0.004)	0.004 (0.003)	1.651
# topics w/ posterior > 1.25×prior	0.014** (0.007)	-0.002 (0.005)	0.007** (0.003)	0.004 (0.003)	1.480
# topics w/ posterior > 1.5×prior	0.006 (0.006)	0.001 (0.005)	0.005 (0.003)	0.003 (0.003)	1.331
# topics w/ posterior > 2×prior	0.014** (0.005)	-0.008** (0.004)	0.000 (0.003)	0.000 (0.002)	1.010
# topics w/ posterior > 3×prior	0.025** (0.010)	-0.014* (0.008)	-0.001 (0.005)	-0.005 (0.005)	0.735
Observations	49,099	61,535	101,862	114,301	189,538

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

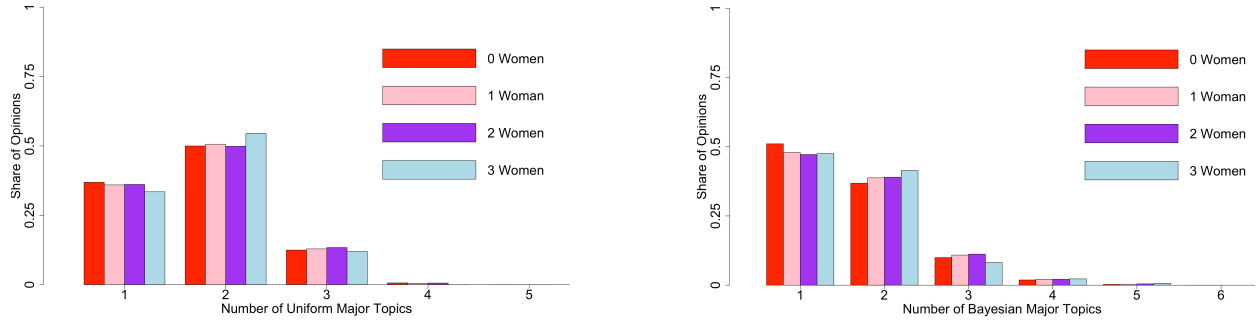
Robustness to alternative Bayesian thresholds (posterior > $\alpha \times$ prior) for major topics. A value of $\alpha = 1$ (reported in the main text, and again here) represents those topics where the posterior distribution places more weight than the prior. Across thresholds, all-Democratic panels have slightly more major topics than all-Republican panels, and bipartisan panels have the same number of topics as party-unanimous panels.

FIGURE 20. Comparing Number of Major Topics By Panel Party Composition



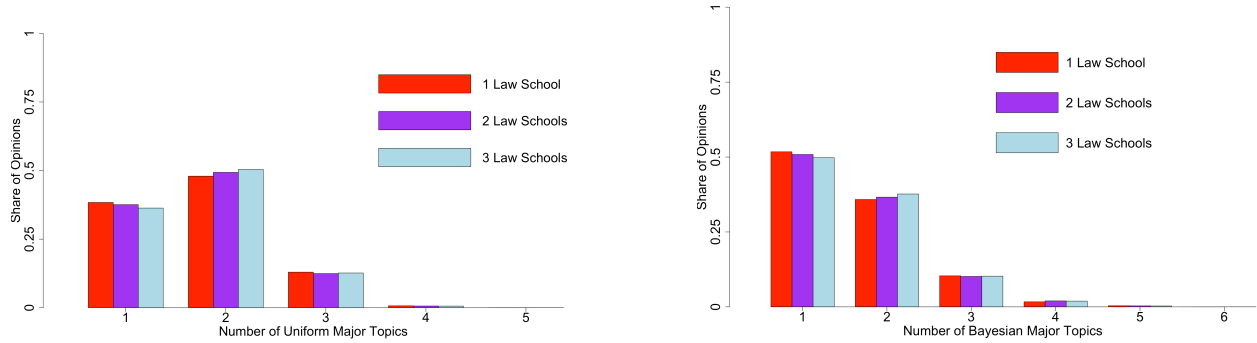
In the left image, distributions of topics with greater-than-uniform probability across panels, 1970–2013 (sample of 163,400) by number of Democrats (means 1.753, 1.771, 1.786, and 1.782 respectively; median 2 for all panel types). In the right image, distributions of topics with greater-than-prior probability across panels, same sample (means 1.635, 1.649, 1.656, and 1.660 respectively; medians 1, 1, 2, and 2 respectively).

FIGURE 21. Comparing Number of Major Topics By Panel Gender Composition



In the left image, distributions of topics with greater-than-uniform probability across panels, 1970–2013 (sample of 188,551) by number of women (means 1.775, 1.761, 1.761, and 1.779 respectively; median 2 for all panel types). In the right image, distributions of topics with greater-than-prior probability across panels, same sample (means 1.651, 1.649, 1.666, and 1.713 respectively; medians 1, 1, 2, 2 respectively).

FIGURE 22. Comparing Number of Major Topics By Panel Law School Composition



In the left image, distributions of topics with greater-than-uniform probability across panels, 1970–2013 (sample of 188,551) by number of unique law schools represented (means 1.631, 1.644, and 1.653 respectively; medians 1, 2, and 2 respectively). In the right image, distributions of topics with greater-than-prior probability across panels, same sample (means 1.631, 1.644 and 1.653 respectively; medians 1, 2, and 2 respectively).

F.2. Robustness of Topic Distributions from Section 8.1 to Alternative Controls.

TABLE 30. Comparing Topics Across All-Democratic and All-Republican Panels, Author Controls

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Crim.	Civ Rights	1A	Due Process	Priv.	Lab. Rel.	Econ. & Reg.	Misc.	Unif.	Bayes
is Dem.	−0.013 (0.035)	0.034 (0.034)	0.252** (0.120)	−0.005 (0.020)	−0.007 (0.165)	0.030 (0.048)	−0.065** (0.025)	0.163*** (0.049)	0.002 (0.006)	0.015*** (0.045)
age	0.022 (0.016)	0.003 (0.018)	−0.028 (0.069)	−0.013* (0.008)	0.119 (0.094)	−0.001 (0.023)	−0.006 (0.012)	0.022 (0.022)	−0.003 (0.003)	0.020 (0.021)
age ²	−0.000 (0.000)	−0.000 (0.000)	0.000 (0.001)	0.000* (0.000)	−0.001 (0.001)	0.000 (0.000)	0.000 (0.000)	−0.000 (0.000)	0.000 (0.000)	−0.000 (0.000)
is woman	0.096** (0.044)	0.165*** (0.044)	0.033 (0.149)	0.007 (0.024)	0.001 (0.177)	−0.164** (0.071)	−0.125*** (0.036)	−0.015 (0.057)	0.009 (0.008)	−0.014 (0.052)
Control mean	0.278	0.070	0.006	0.293	0.001	0.061	0.241	0.050	1.754	1.631

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Replicating the first row of Table 10 with controls for author characteristics; estimation follows that table.

TABLE 31. Comparing Topics Across Democratic-Majority Panels, Author Controls

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Crim.	Civ Rights	1A	Due Process	Priv.	Lab. Rel.	Econ. & Reg.	Misc.	Unif.	Bayes
bipartisan	0.009 (0.024)	−0.021 (0.025)	−0.119 (0.075)	−0.000 (0.014)	−0.070 (0.114)	−0.023 (0.035)	0.043** (0.019)	−0.098*** (0.036)	0.005 (0.004)	−0.008 (0.005)
is Dem.	0.023 (0.019)	−0.038* (0.021)	−0.015 (0.071)	−0.010 (0.012)	−0.176* (0.094)	0.018 (0.029)	0.001 (0.016)	−0.008 (0.028)	−0.001 (0.004)	−0.006 (0.004)
age	0.006 (0.016)	−0.006 (0.016)	−0.045 (0.066)	0.008 (0.007)	0.092 (0.069)	0.004 (0.021)	−0.012 (0.010)	0.007 (0.019)	0.001 (0.003)	−0.002 (0.003)
age ²	−0.000 (0.000)	0.000 (0.000)	0.000 (0.001)	−0.000 (0.000)	−0.001 (0.001)	−0.000 (0.000)	0.000 (0.000)	−0.000 (0.000)	−0.000 (0.000)	0.000 (0.000)
is woman	−0.093*** (0.035)	0.079** (0.034)	0.203** (0.097)	0.061*** (0.021)	0.119 (0.133)	−0.042 (0.053)	−0.063** (0.031)	0.127*** (0.043)	0.021*** (0.007)	0.019** (0.008)
Control mean	0.278	0.071	0.006	0.293	0.001	0.060	0.242	0.050	1.754	1.636

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Replicating the second row of Table 10 with controls for author characteristics; estimation follows that table.

TABLE 32. Comparing Topics Across Republican-Majority Panels, Author Controls

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Crim.	Civ Rights	1A	Due Process	Priv.	Lab. Rel.	Econ. & Reg.	Misc.	Unif.	Bayes
bipartisan	-0.027 (0.017)	0.019 (0.017)	0.029 (0.063)	0.014 (0.010)	0.033 (0.084)	-0.002 (0.022)	-0.000 (0.012)	0.020 (0.026)	0.005* (0.003)	0.007* (0.004)
is Dem.	0.027* (0.016)	-0.035* (0.018)	-0.008 (0.067)	-0.036*** (0.009)	0.029 (0.084)	0.043* (0.026)	-0.000 (0.014)	0.045* (0.024)	-0.003 (0.003)	-0.004 (0.003)
age	0.017 (0.011)	-0.002 (0.011)	0.085* (0.051)	-0.002 (0.005)	-0.016 (0.048)	-0.007 (0.017)	-0.010 (0.007)	0.004 (0.014)	-0.001 (0.002)	0.000 (0.002)
age ²	-0.000 (0.000)	-0.000 (0.000)	-0.001* (0.001)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)
is woman	0.007 (0.030)	0.060** (0.029)	-0.135 (0.101)	0.022 (0.017)	0.071 (0.111)	-0.119*** (0.043)	-0.033 (0.023)	0.039 (0.046)	-0.001 (0.005)	0.003 (0.006)
Control mean	0.279	0.071	0.006	0.295	0.001	0.059	0.238	0.051	1.756	1.636

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Replicating the third row of Table 10 with controls for author characteristics; estimation follows that table.

TABLE 33. Comparing Topics Across Bipartisan Panels, Author Controls

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Crim.	Civ Rights	1A	Due Process	Priv.	Lab. Rel.	Econ. & Reg.	Misc.	Unif.	Bayes
Dem. majority	-0.009 (0.017)	0.019 (0.016)	0.085* (0.052)	0.013 (0.009)	0.030 (0.074)	-0.021 (0.021)	-0.014 (0.013)	0.015 (0.022)	0.004 (0.003)	0.014 (0.020)
is Dem.	0.026** (0.012)	-0.034** (0.014)	-0.020 (0.050)	-0.025*** (0.007)	-0.063 (0.066)	0.032 (0.020)	-0.001 (0.011)	0.023 (0.018)	-0.002 (0.002)	0.022 (0.016)
age	0.011 (0.011)	-0.010 (0.010)	0.051 (0.046)	0.007 (0.006)	-0.014 (0.043)	-0.005 (0.014)	-0.012* (0.007)	0.001 (0.014)	0.000 (0.002)	0.001 (0.013)
age ²	-0.000 (0.000)	0.000 (0.000)	-0.001 (0.000)	-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000* (0.000)	0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)
is woman	-0.073*** (0.027)	0.034 (0.023)	0.025 (0.087)	0.046*** (0.015)	0.115 (0.103)	-0.071* (0.038)	-0.017 (0.022)	0.111*** (0.039)	0.007 (0.005)	0.101*** (0.036)
Control mean	0.272	0.071	0.007	0.295	0.001	0.060	0.242	0.0523	1.763	1.640

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Replicating the fourth row of Table 10 with controls for author characteristics; estimation follows that table.

TABLE 34. Comparing Topic Use Across Panel Types, Author Fixed Effects

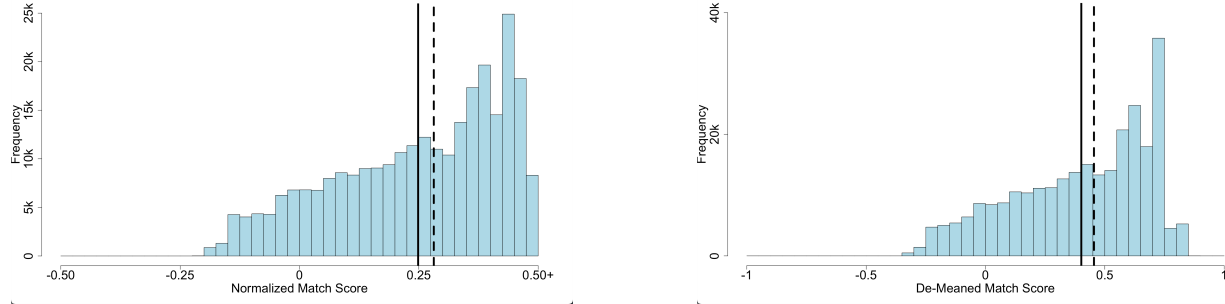
Panel type	Variable	(1) Crim.	(2) Civ Rights	(3) 1A	(4) Due Process	(5) Priv.	(6) Lab. Rel.	(7) Econ. & Reg.	(8) Misc.	(9) Unif.	(10) Bayes	Observations
majority-Dem.	2-1 Dem.	0.019 (0.026)	−0.016 (0.025)	−0.133* (0.076)	−0.002 (0.014)	−0.106 (0.122)	−0.023 (0.037)	0.039* (0.020)	−0.109*** (0.033)	0.003 (0.004)	−0.011** (0.005)	61,494
	Control mean	0.264	0.072	0.008	0.293	0.001	0.062	0.245	0.056	1.774	1.652	
majority-Rep.	2-1 Rep.	−0.026 (0.016)	0.023 (0.017)	−0.005 (0.067)	0.013 (0.010)	0.038 (0.082)	−0.005 (0.024)	0.001 (0.014)	0.016 (0.023)	0.005* (0.003)	0.007* (0.004)	101,773
	Control mean	0.265	0.073	0.008	0.295	0.001	0.061	0.240	0.057	1.773	1.652	
bipartisan	Dem. majority	−0.008 (0.015)	0.025* (0.015)	0.090 (0.057)	0.013 (0.009)	0.032 (0.070)	−0.028 (0.024)	−0.014 (0.011)	0.016 (0.020)	0.004 (0.003)	0.015 (0.018)	114,219
	Control mean	0.265	0.072	0.008	0.295	0.001	0.061	0.241	0.057	1.774	1.654	

Significance is denoted by *** for 1%, ** for 5%, and * for 10%.

Replicating rows 2–4 of Table 10 with author fixed effects; estimation follows that table. Row 1 cannot be replicated with author fixed effects because no judge is an author on both all-Democratic and all-Republican panels.

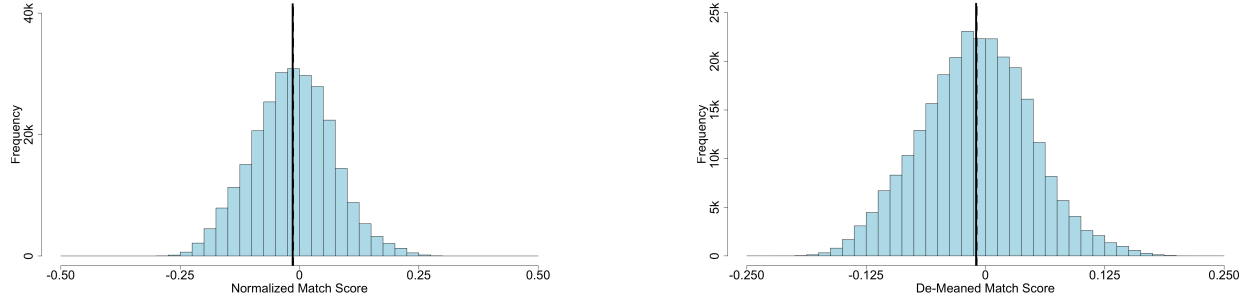
F.3. Distributions of Match Scores from Table 11.

FIGURE 23. Distributions of Category Match Scores



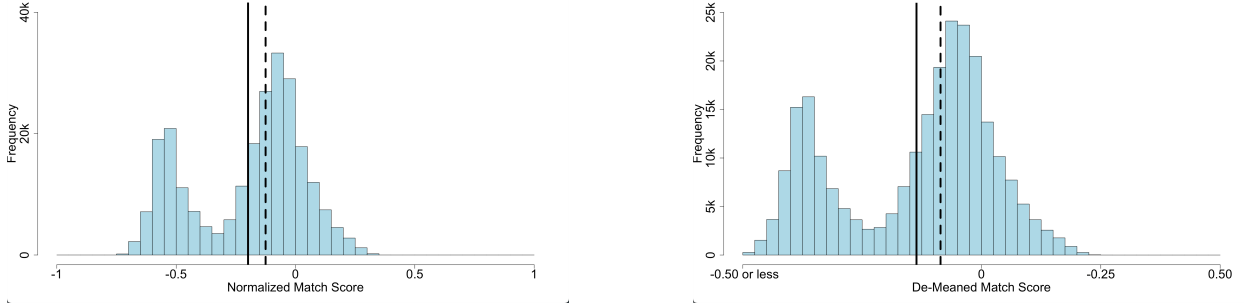
In the left image, the distribution of normalized category match scores; in the right image, the distribution of de-meaned category match scores (without dividing by the entropy of the category distribution); the sample contains 270,451 opinions. Solid lines indicate the mean of each distribution (0.249 and 0.401 respectively) and dashed lines indicate the medians (0.281 and 0.454 respectively).

FIGURE 24. Distributions of Author Party Match Scores



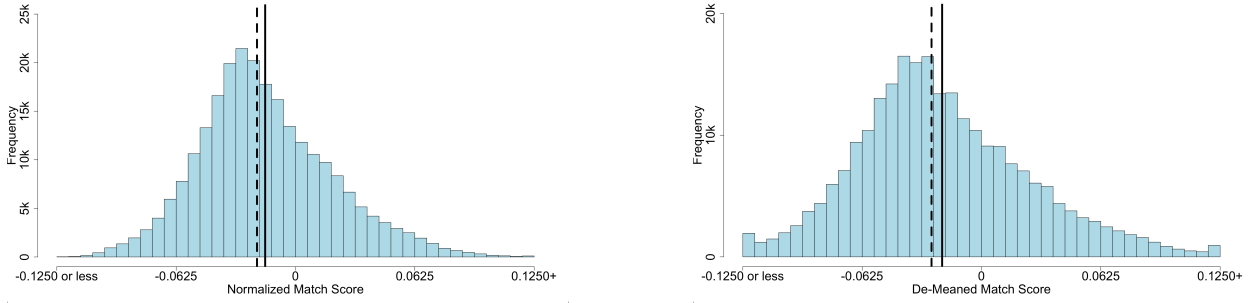
In the left image, the distribution of normalized author party match scores; in the right image, the distribution of de-meaned author-party match scores (without dividing by the entropy of the category distribution); the sample contains 264,672 opinions. Solid lines indicate the mean of each distribution (-0.014 and -0.010 respectively) and dashed lines indicate the medians (-0.014 and -0.009 respectively).

FIGURE 25. Distributions of Panel Majority Match Scores



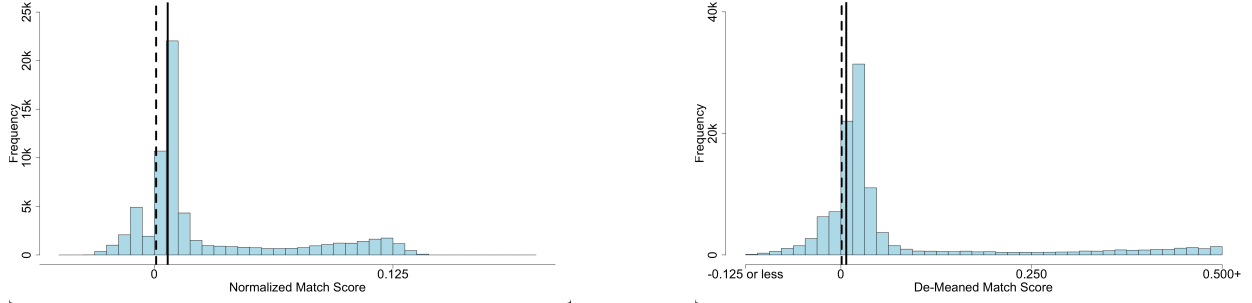
In the left image, the distribution of normalized panel majority match scores; in the right image, the distribution of de-meaned panel majority match scores (without dividing by the entropy of the category distribution); the sample contains 246,675 opinions. Solid lines indicate the mean of each distribution (-0.199 and -0.136 respectively) and dashed lines indicate the medians (-0.125 and -0.086 respectively).

FIGURE 26. Distributions of Number-of-Democrats Match Scores



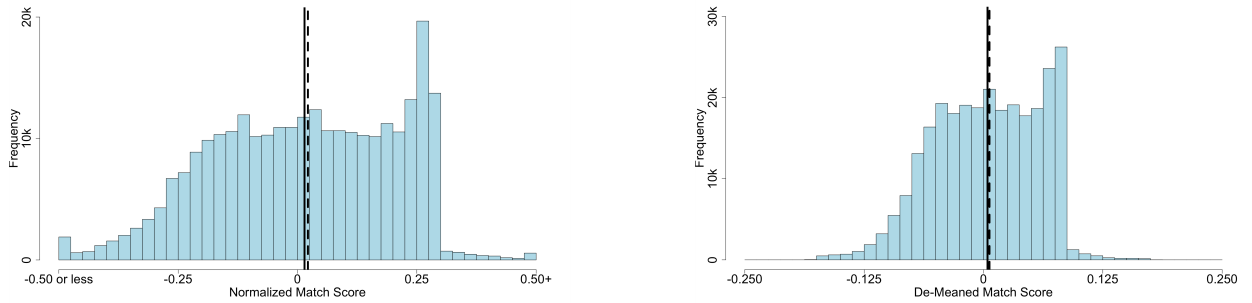
In the left image, the distribution of normalized number-of-Democrats match scores; in the right image, the distribution of de-meaned number-of-Democrats match scores (without dividing by the entropy of the category distribution); the sample contains 246,675 opinions. Solid lines indicate the mean of each distribution (-0.016 and -0.021 respectively) and dashed lines indicate the medians (-0.020 and -0.026 respectively).

FIGURE 27. Distributions of Law School Match Scores



In the left image, the distribution of normalized law school match scores; in the right image, the distribution of de-meaned law school match scores (without dividing by the entropy of the category distribution). Solid lines indicate the mean of each distribution (0.007 and 0.027 respectively) and dashed lines indicate the medians (0.001 and 0.004 respectively). A majority of opinions have normalized match scores in $(-0.005, 0.005)$ and de-meaned match scores in $(-0.01, 0.01)$. For legibility, those opinions are omitted from the figure; the remaining samples contain 66,833 and 108,375 opinions respectively. 76,849 opinions have a normalized match score in $(-0.005, 0)$, and 130,033 opinions have a normalized match score in $(0, 0.005)$. 67,486 opinions have a de-meaned match score in $(-0.01, 0)$, and 97,854 opinions have a de-meaned match score in $(0, 0.01)$.

FIGURE 28. Distributions of Gender Match Scores



In the left image, the distribution of normalized gender match scores; in the right image, the distribution of de-meaned gender match scores (without dividing by the entropy of the category distribution); the sample contains 274,158 opinions. Solid lines indicate the mean of each distribution (0.015 and 0.004 respectively) and dashed lines indicate the medians (0.022 and 0.006 respectively).

F.4. Robustness of Nearest-Neighbor Measures from Section 8.2.

TABLE 35. Opinion Similarity With 100 Neighbors

	(1)	(2)	(3)	(4)	(5)	(6)
	Topic	Author Party	Panel Majority	# Dems.	Law School	Gender
Normalized match score	0.251 (0.00036)	-0.009 (0.00028)	-0.189 (0.00061)	-0.012 (0.00021)	0.009 (0.00005)	0.016 (0.00164)
De-meaned match score	0.406 (0.00059)	-0.006 (0.00019)	-0.130 (0.00004)	-0.015 (0.00026)	0.033 (0.00002)	0.005 (0.00049)
Share de-meaned > 0	0.879	0.464	0.241	0.397	0.596	0.572
Raw match share	0.638	0.499	0.378	0.277	0.080	0.852
Observations	270,451	264,672	246,675	246,675	273,715	274,158

All coefficients are significant at the 1% level.

Replicating Table 11 using the 100 nearest neighbors (about 0.04% of the sample of opinions), instead of the 1,000 nearest neighbors as in that table. All analysis is conducted in the same way as in that table.

TABLE 36. Opinion Similarity With 10,000 Neighbors

	(1)	(2)	(3)	(4)	(5)	(6)
	Topic	Author Party	Panel Majority	# Dems.	Law School	Gender
Normalized match score	0.239 (0.00036)	-0.020 (0.00019)	-0.208 (0.00057)	-0.020 (0.00018)	0.005 (0.00004)	0.018 (0.00150)
De-meaned match score	0.386 (0.00056)	-0.014 (0.00013)	-0.143 (0.00036)	-0.025 (0.00022)	0.018 (0.00015)	0.005 (0.00045)
Share de-meaned > 0	0.883	0.358	0.133	0.201	0.687	0.546
Raw match share	0.618	0.491	0.365	0.267	0.066	0.853
Observations	270,451	264,672	246,675	246,675	273,715	274,158

All coefficients are significant at the 1% level.

Replicating Table 11 using the 10,000 nearest neighbors (about 3.6% of the sample of opinions), instead of the 100 nearest neighbors as in that table. All analysis is conducted in the same way as in that table.

TABLE 37. Opinion Similarity With 100 Neighbors, Excluding Own Author

	(1)	(2)	(3)	(4)	(5)	(6)
	Topic	Author Party	Panel Majority	# Dems.	Law School	Gender
Normalized match score	0.251 (0.00036)	-0.010 (0.00027)	-0.191 (0.00061)	-0.013 (0.00021)	0.008 (0.00005)	0.015 (0.00162)
De-meaned match score	0.405 (0.00006)	-0.007 (0.00019)	-0.131 (0.00039)	-0.017 (0.00026)	0.032 (0.00002)	0.004 (0.00048)
Share de-meaned > 0	0.879	0.462	0.236	0.391	0.586	0.570
Raw match share	0.637	0.498	0.377	0.275	0.079	0.852
Observations	270,451	264,672	246,675	246,675	273,715	274,158

All coefficients are significant at the 1% level.

Replicating Table 11, only considering neighbors with different authors. All analysis is conducted in the same way as in that table.

TABLE 38. Opinion Similarity With 1,000 Neighbors, Excluding Own Author

	(1)	(2)	(3)	(4)	(5)	(6)
	Topic	Author Party	Panel Majority	# Dems.	Law School	Gender
Normalized match score	0.248 (0.00035)	-0.015 (0.00022)	-0.201 (0.00057)	-0.017 (0.00019)	0.007 (0.00004)	0.014 (0.00158)
De-meaned match score	0.400 (0.00058)	-0.011 (0.00015)	-0.138 (0.00037)	-0.022 (0.00023)	0.025 (0.00018)	0.004 (0.00047)
Share de-meaned > 0	0.881	0.428	0.179	0.278	0.672	0.536
Raw match share	0.632	0.495	0.371	0.27	0.073	0.851
Observations	270,451	264,672	246,675	246,675	273,715	274,158

All coefficients are significant at the 1% level.

Replicating Table 11 using the 1,000 nearest neighbors (about 0.4% of the sample of opinions), instead of the 100 nearest neighbors as in that table, and only considering neighbors with different authors. All analysis is conducted in the same way as in that table.

TABLE 39. Opinion Similarity With 10,000 Neighbors, Excluding Own Author

	(1)	(2)	(3)	(4)	(5)	(6)
	Topic	Author Party	Panel Majority	# Dems.	Law School	Gender
Normalized match score	0.251 (0.00035)	-0.009 (0.00018)	-0.189 (0.00056)	-0.012 (0.00018)	0.009 (0.00003)	0.016 (0.00151)
De-meaned match score	0.406 (0.00054)	-0.006 (0.00012)	-0.13 (0.00036)	-0.015 (0.00022)	0.033 (0.00014)	0.005 (0.00046)
Share de-meaned > 0	0.879	0.464	0.241	0.397	0.596	0.572
Raw match share	0.638	0.499	0.378	0.277	0.080	0.852
Observations	270,451	264,672	246,675	246,675	273,715	274,158

All coefficients are significant at the 1% level.

Replicating Table 11 using the 10,000 nearest neighbors (about 3.6% of the sample of opinions), instead of the 100 nearest neighbors as in that table, and only considering neighbors with different authors. All analysis is conducted in the same way as in that table.